GALOIS
Locally constant coefficients, localizers, and relative Galois reconstruction

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We fix Grothendieck universes $\mathbb{U} \in \mathbb{V}$. Categories of shapes and spaces are $\mathbb{U}$-small; categories of coefficients may be $\mathbb{V}$-large. A construction requiring an additional universe says so explicitly. We put $BX = |NX|$, $e = \{*\}$, and $\mathsf{Hot} = \mathsf{Cat}_{\mathbb{U}}[\mathcal{W}_\infty^{-1}]$, where $\mathcal{W}_\infty$ consists of the functors whose nerves are weak homotopy equivalences. We write $\mathcal{W}_n$ for the functors inducing isomorphisms on π_i for $0 \le i \le n$, at all base points; the assertion on π_0 is bijectivity.

Coefficients have the manuscript's presheaf variance. Thus the enhanced example is $\mathbf{D}_{\mathcal{C}}(A) = \operatorname{Ho}\operatorname{Fun}((NA)^{\mathrm{op}}, \mathcal{C})$. In formulas about spaces we identify a groupoid with its opposite by inverting paths. For $u : A \to B$, the comma category A/b has objects $(a, u(a) \to b)$, and $b\backslash A$ has objects $(a, b \to u(a))$. A covariantly final functor has every $b\backslash A$ nonempty and connected. A sieve is closed under sources, and a cosieve under targets.

In this paper a *fundamental localizer* satisfies identities, two-out-of-three and closure under retracts; contains $A \to e$ when A has a terminal object; and satisfies the *relative* Theorem A: if $u : A \to B$ is a morphism over S and every $A/s \to B/s$ belongs to $\mathcal{W}$, then $u \in \mathcal{W}$. In Chapter XVI this last axiom is literally Loc (3 bis), whereas Loc (3) is its absolute special case. This distinction is material below. An object is $\mathcal{W}$-aspherical if its specified map to e is in $\mathcal{W}$.

Every numbered assertion carries one of the requested statuses. *Open* means unresolved by this work, not a claim of literature-wide openness. *Proved* records an argument given here and makes no priority claim. *Known* records a checked source, with the precise version used. Earlier AI-produced companion files are audited inputs, not primary mathematical authorities. The checked CATHOT C.1–C.4 are used as supplied in the prompt. All original arguments and all limitations in this document are attributable only to the credited AI system.

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1 Definitions and first properties

1.1 Axioms, coefficients, and the five comparison classes

A.1. Definitions and the axioms actually required. [Proved]

Let $\mathbf{D} : \mathsf{Cat}_{\mathbb{U}}^{\mathrm{coop}} \to \mathrm{CAT}_{\mathbb{V}}$ be a prederivator. For a small category A, let $\mathrm{LC}_{\mathbf{D}}(A)$ be the full subcategory of $\mathbf{D}(A)$ on objects whose underlying presheaf $A^{\mathrm{op}} \to \mathbf{D}(e)$ sends every arrow to an isomorphism. If $f : A \to B$, restriction gives a functor $f^* : \mathrm{LC}_{\mathbf{D}}(B) \to \mathrm{LC}_{\mathbf{D}}(A)$. We define $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ by this functor being an equivalence and $\mathcal{W}_{\mathbf{D}}^{1}$ by its being fully faithful. Writing $p : A \to e$, $q : B \to e$, we define $\mathcal{W}_{\mathbf{D}}$ by bijectivity of

$$\mathrm{Hom}_{\mathbf{D}(B)}(q^*M, q^*N) \longrightarrow \mathrm{Hom}_{\mathbf{D}(A)}(p^*M, p^*N) \qquad (M, N \in \mathbf{D}(e)). \tag{A.1.1}$$

If all restrictions have adjoints $f_! \dashv f^* \dashv f_*$, we define $\mathcal{W}_{\mathbf{D}}^{\mathrm{coh}}$, respectively $\mathcal{W}_{\mathbf{D}}^{\mathrm{hom}}$, by invertibility, for every $F \in \mathrm{LC}_{\mathbf{D}}(B)$, of the specified comparison

$$q_*F \longrightarrow p_*f^*F, \qquad p_!f^*F \longrightarrow q_!F \quad \text{in } \mathbf{D}(e). \tag{A.1.2}$$

The first map is induced by the unit $F \to f_*f^*F$; the second by the counit $f_!f^*F \to F$. The definitions require no enhancement.

Proof. Evaluation and the underlying presheaf commute with restriction, so restriction preserves the stated full subcategories. Pseudofunctorial coherence identifies qf with p and supplies (A.1.1). The adjunctions and their composition isomorphisms supply exactly (A.1.2). These constructions do not assume that the underlying diagram functor is full or faithful. □

Unless explicitly stated otherwise, a derivator satisfies the usual four axioms in presheaf variance: disjoint unions of shapes become products, including $\mathbf{D}(\varnothing) \simeq e$; point evaluations jointly detect isomorphisms; every restriction has both adjoints; and both Kan extensions are computed by comma categories. In particular $b^*u_!$ is computed on $b\backslash A$ and b^*u_* on A/b, with presheaf variance on those shapes. We do not use the strongness axiom that coherent arrows present all incoherent arrows, unless a statement says so. These conventions agree with the definitions used by Cisinski (2009, §3.2); they are not Grothendieck's original numerical labels Der (1)–(6).

A.2. The inclusion chain and its two separate last arrows. [Proved]

For a prederivator with the adjoints of A.1,

$$\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}} \subseteq \mathcal{W}_{\mathbf{D}}^{1} \subseteq \mathcal{W}_{\mathbf{D}}^{\mathrm{coh}} \cap \mathcal{W}_{\mathbf{D}}^{\mathrm{hom}} \subseteq \mathcal{W}_{\mathbf{D}}. \tag{A.2.1}$$

In fact each of $\mathcal{W}_{\mathbf{D}}^{\mathrm{coh}}$ and $\mathcal{W}_{\mathbf{D}}^{\mathrm{hom}}$ is separately contained in $\mathcal{W}_{\mathbf{D}}$. For a derivator, $\mathcal{W}_{\infty} \subseteq \mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$.

Proof. The first inclusion is the definition of equivalence of categories. If f^* is fully faithful on locally constant objects, then, for $M \in \mathbf{D}(e)$ and $F \in \mathrm{LC}_{\mathbf{D}}(B)$, it gives bijections on $\mathrm{Hom}(q^*M, F)$ and on $\mathrm{Hom}(F, q^*M)$. The adjunctions identify these with the maps obtained by applying $\mathrm{Hom}(M, -)$, respectively $\mathrm{Hom}(-, M)$, to (A.1.2). The ordinary Yoneda lemma in $\mathbf{D}(e)$ makes both comparison maps isomorphisms. Taking $F = q^*N$ proves the last arrows by the same adjunctions. The final assertion is exactly Cisinski (2009, Corollary 3.8), with $A^{\mathrm{op}}, B^{\mathrm{op}}$ if necessary to match variance. Its proof uses Theorem 3.6 and the model-category result Theorem 1.3; it applies to derivators, not merely to derivators known in advance to come from a model category. □

A.3. What happens for a map to the point. [PROVED]

For every prederivator and every A,

$$(A \to e) \in \mathcal{W}_{\mathbf{D}} \quad \Longleftrightarrow \quad (A \to e) \in \mathcal{W}^{1}_{\mathbf{D}}. \tag{A.3.1}$$

Consequently the claim in the CATHOT addendum that an acyclic-group projection $BA \to e$ is a constant-coefficient equivalence but is not 1-complete is false.

Proof. Every object of $\mathbf{D}(e) = \mathrm{LC}_{\mathbf{D}}(e)$ is a coefficient on the target. Full faithfulness of $p^* : \mathbf{D}(e) \to \mathrm{LC}_{\mathbf{D}}(A)$ is precisely (A.1.1) with $B = e$, since $\mathrm{LC}_{\mathbf{D}}(A)$ is full in $\mathbf{D}(A)$. This argument includes $A = \varnothing$ and does not use acyclicity. The correctly oriented strictness examples occur in Part C. □

1.2 Formal closure and coherence

A.4. Closure properties that follow without descent. [PROVED]

For any prederivator, $\mathcal{W}_{\mathbf{D}}$ and $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$ contain identities and are closed under composition, two-out-of-three, and retracts of arrows. The class $\mathcal{W}^{1}_{\mathbf{D}}$ contains identities, is closed under composition and retracts, and satisfies the cancellation rule

$$f \in \mathcal{W}^{1}_{\mathbf{D}}, \quad gf \in \mathcal{W}^{1}_{\mathbf{D}} \quad \Longrightarrow \quad g \in \mathcal{W}^{1}_{\mathbf{D}}. \tag{A.4.1}$$

With adjoints, $\mathcal{W}^{\mathrm{coh}}_{\mathbf{D}}$ and $\mathcal{W}^{\mathrm{hom}}_{\mathbf{D}}$ have these latter closure properties too. The missing two-out-of-three implication for these three intermediate classes is false even for a represented derivator.

Proof. For $\mathcal{W}_{\mathbf{D}}$, (A.1.1) is a natural map of sets for each fixed M, N; bijections have the asserted closure properties. The assignment $A \mapsto \mathrm{LC}_{\mathbf{D}}(A)$ is contravariantly pseudofunctorial, and equivalences of categories have the same properties. A retract of a functor which is an equivalence is fully faithful because its maps on Hom sets are retracts of bijections, and essentially surjective by applying the retraction to an isomorphism exhibiting essential surjectivity. The same Hom-set argument handles fully faithful functors. If f^* and f^*g^* are fully faithful, cancellation of the bijective Hom maps for f^* proves full faithfulness of g^*.

For cohomology, the comparison for gf is the composite of the comparison for g and that for f applied to g^*F; the latter coefficient is locally constant. This proves composition and (A.4.1). For a retract diagram in the arrow category, pull a target coefficient back along the target retraction. Its comparison restricts to the original one, which is therefore a retract of an isomorphism. The homological proof is identical with the arrows reversed. A.5 supplies the stated failure. □

A.5. Two-out-of-three for 1-completeness. [REFUTED]

The assertion that $\mathcal{W}^{1}_{\mathbf{D}}$ is a fundamental localizer for every derivator is false. The same obstruction applies to $\mathcal{W}^{\mathrm{coh}}_{\mathbf{D}}$ and $\mathcal{W}^{\mathrm{hom}}_{\mathbf{D}}$.

Proof. Take the represented derivator of sets. Let $G = C_2 = \{1, s\}$, $s^2 = 1$, and let BG have one object, endomorphism set G, and composition the group multiplication. Let $i : e \to BG$ select the object and $r : BG \to e$ be the projection. Then $ri = \mathrm{id}_e$. The functor $r^* : \mathsf{Set} \to G\text{-}\mathsf{Set}$ gives the trivial action and is fully faithful. The functor $i^* : G\text{-}\mathsf{Set} \to \mathsf{Set}$ forgets the action. On

the regular two-element G-set R, there are two equivariant endomorphisms but four underlying functions, so i^* is not full. Thus $r, ri \in \mathcal{W}^1_{\mathbf{D}}$ while $i \notin \mathcal{W}^1_{\mathbf{D}}$.

For cohomology, the regular action has $R^G = \varnothing$, whereas its pullback along i has two global sections. For homology, R/G is a singleton whereas the pulled-back colimit is the two-element set. Projection r preserves the limits and colimits of every coefficient on e, since BG is nonempty and connected. Thus the same two-out-of-three counterexample applies to both classes. The program `code/A5_finite_counterexample.py` checks the category law, all four functions, and both comparisons; the proof of full faithfulness of trivial actions is independent of a finite size bound. □

A.6. Locally constant coefficient derivators and shifts. [Known]

For every derivator $\mathbf{D}$ and small A, the prescription

$$\mathbb{L}_A(I) = \{F \in \mathbf{D}(A \times I) : F \text{ is locally constant in the } A \text{ direction}\} \tag{A.6.1}$$

defines a derivator. Its inclusion in $\mathbf{D}^A$ has both adjoints as morphisms of derivators. Restriction along $f : A \to B$ gives a morphism $\mathbb{L}_B \to \mathbb{L}_A$ with both adjoints. It is fully faithful, or an equivalence, at every shape I if and only if it is so at e.

Proof. The first two assertions are Cisinski (2009, Theorem 3.13), translated from his covariant notation. Compose the adjoints to the inclusions with the shifted Kan extensions along $f \times \mathrm{id}_I$ to obtain the two adjoints to restriction. These are morphisms of derivators, so their units and counits commute with evaluation at $i \in I$. Full faithfulness of a right adjoint is equivalent to its counit being invertible; full faithfulness of a left adjoint is equivalent to its unit being invertible. At every i, the relevant unit or counit is the one at e. Der 2 therefore proves invertibility at I. Using both unit and counit proves the equivalence assertion. No inference from an arbitrary equivalence of ordinary categories to an enhancement is being made: the specified adjunctions of derivators provide the comparison maps. □

A.7. Finite products, coproducts, opposites, and a limit. [Proved]

For a derivator, all five classes are invariant under natural transformations between the indexing functors and under passage to opposite functors. They are closed under finite products of arrows. The classes $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$, $\mathcal{W}^1_{\mathbf{D}}$, $\mathcal{W}^{\mathrm{coh}}_{\mathbf{D}}$, and $\mathcal{W}^{\mathrm{hom}}_{\mathbf{D}}$ satisfy the disjoint-union equivalence

$$\coprod_{j \in J} f_j \in \mathcal{W} \quad \Longleftrightarrow \quad (\forall j)\ f_j \in \mathcal{W} \tag{A.7.1}$$

for every small family. For $\mathcal{W}_{\mathbf{D}}$ only the right-to-left implication holds in this generality; its converse is false. There is no general closure under infinite products of arrows, even for $\mathcal{W}_\infty$ or the set-valued classes.

Proof. A natural transformation evaluated on a locally constant coefficient is pointwise invertible; Der 2 makes it invertible in the diagram category. The same applies to constants. For opposites, use the twisted-arrow category $\mathrm{Tw}(A)$ with objects $a \to b$ and arrows given by pairs $a' \to a$, $b \to b'$ whose composite is $a' \to b'$. Its two endpoint projections to A^{op} and A induce weak homotopy equivalences: its nerve is the edgewise subdivision of NA, and either endpoint map is homotopic to the subdivision homeomorphism after realization. The natural squares for f compare f and f^{op} by restriction equivalences, by A.2. Such squares preserve full faithfulness, equivalence, and the constant Hom comparisons. For (co)homology they preserve the specified maps by the adjunction characterization in A.2.

For $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ and $\mathcal{W}_{\mathbf{D}}^{1}$, A.6 gives the result for $f \times \mathrm{id}_I$: inside the shift, impose local constancy also in I. For essential surjectivity the preimage is locally constant in I because f^* reflects the invertibility of its I-arrows. For $\mathcal{W}_{\mathbf{D}}$, apply the constant homotopy-colimit characterization from A.2 and Fubini: iterated homotopy colimits of constants over $A \times I$ identify with the homotopy colimit over A of the coefficient $\pi_{I!}\pi_I^* M$. For $\mathcal{W}_{\mathbf{D}}^{\mathrm{hom}}$, first integrate a locally constant coefficient over I; the result is locally constant on B, since Kan extension in one factor is computed pointwise in the other. Fubini then reduces the assertion to f. The dual argument treats $\mathcal{W}_{\mathbf{D}}^{\mathrm{coh}}$. Factoring a finite product of arrows into these products with identities proves the claim.

Der 1 identifies the relevant categories over a disjoint union with products. For $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$, an equivalence of such product restriction functors reflects each factor: fill the other factors with terminal objects and use their unique endomorphisms for full faithfulness; essential surjectivity is tested by filling the other target factors with their terminal objects. This also proves the full-faithfulness case. For cohomology choose the terminal coefficient on every other component, and for homology choose the initial coefficient; limits, respectively colimits, then isolate the desired comparison. For $\mathcal{W}_{\mathbf{D}}$, products of bijections prove the forward closure under sums, but bijectivity of a product can conceal an empty factor. The represented derivator of the complete lattice $\{0 < 1\}$ gives the explicit failure: $\varnothing \to e$ is not in $\mathcal{W}_{\mathbf{D}}$, whereas its sum with id_e is in $\mathcal{W}_{\mathbf{D}}$, since both its source and target are nonempty. The calculation is given in G.2.

For the infinite-product failure, let I_n be the path poset on $\{0, \dots, n\}$ with alternating orientations of consecutive edges. There are no composable nonidentity arrows, and BI_n is an interval. Each $I_n \to e$ is a weak homotopy equivalence. In $\prod_{n\geq 1} I_n$, the objects $(0)_n$ and $(n)_n$ cannot be joined by a finite zigzag: a zigzag of length m changes the nth coordinate by at most m, contradicting $n > m$. Its projection to e fails even bijectivity on components. This explicitly checks the infinite-family edge case rather than transferring an untrue product rule from spaces to nerves of categories. □

A.8. What the ordinary-category comparison does prove with an enhancement. [Proved]

If $\mathcal{C}$ has all small limits and colimits, then

$$\mathrm{LC}_{\mathbf{D}_{\mathcal{C}}}(A) \simeq \mathrm{Ho}\,\mathrm{Fun}(BA, \mathcal{C}). \tag{A.8.1}$$

For the specified restriction map, equivalence or full faithfulness on these homotopy categories is equivalent to the corresponding infinity-categorical property.

Proof. The localization $NA \to \Pi_\infty A$ inverting every edge is universal for functors carrying those edges to equivalences. Its universal property is fully faithful on functor infinity-categories, with exactly this image. Passing to homotopy categories preserves that full image. The groupoid $\Pi_\infty A$ is the space BA; inversion of paths implements the variance convention.

Restriction has both Kan adjoints. Its homotopy-category adjunctions are obtained from these by applying Ho. A functor is fully faithful precisely when the appropriate adjunction unit or counit is an equivalence. This can be tested in the homotopy category. Equivalence is tested by both unit and counit. Equivalently, one may recover all mapping spaces from maps to cotensors by arbitrary small spaces, as in C.1. This argument does not say that a bare ordinary category determines its infinity-category enhancement. □

2 Localizers, descent, and cardinal-controlled Kan–Thurston

2.1 The distinction between absolute and relative Theorem A

B.1. Absolute Theorem A for every derivator. [Proved]

For every derivator $\mathbf{D}$, the class $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ is weakly saturated, contains the categories with a terminal object, and satisfies the absolute Theorem A: if $u : A \to B$ and every $A/b \to e$ belongs to $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$, then $u \in \mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$. The constant-coefficient class $\mathcal{W}_{\mathbf{D}}$ is a fundamental localizer, including the relative Theorem A. The class $\mathcal{W}_{\mathbf{D}}^{1}$ need not be a localizer, by A.5.

Proof. Weak saturation is A.4. A category with a terminal object has contractible nerve, so A.2 applies. To check the first Theorem A, we use covariant coefficients $\mathbb{E}(I) = \mathbf{D}(I^{\mathrm{op}})$. Passage to opposites in A.7 identifies its classes with those of $\mathbf{D}$. Each transition $A/b \to A/b'$ is a $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$-equivalence: its two projections to e are, and two-out-of-three applies. For $F \in \mathrm{LC}_{\mathbb{E}}(A)$, pointwise left Kan extension gives

$$(u_! F)_b = \operatorname*{colim}_{A/b} F|_{A/b}. \tag{B.1.1}$$

Here and below the displayed colimit denotes the derivator Kan extension, an object of $\mathbb{E}(e)$. A transition $b \to b'$ induces an isomorphism between these colimits: A.2 applied to the transition category and the restricted locally constant coefficient is exactly this comparison. Hence $u_! F$ is locally constant.

For each object (a, α) of A/b, restriction from $\mathrm{LC}_{\mathbb{E}}(A/b)$ to $\mathbb{E}(e)$ at that object is an inverse to the constant-diagram equivalence. Its value-to-colimit map is therefore an isomorphism. In particular, at $(a, \mathrm{id}_{u(a)})$, the unit $F \to u^* u_! F$ is an isomorphism. For $G \in \mathrm{LC}_{\mathbb{E}}(B)$ the restriction of $u^* G$ to A/b is canonically isomorphic to the constant value G_b, using the arrows $u(a) \to b$. The counit $u_! u^* G \to G$ is therefore an isomorphism at b. Point evaluations detect both isomorphisms. These are the specified unit and counit, so they prove that u^* is an equivalence on locally constant objects. This is the argument of Cisinski (2009, Proposition 1.27), whose Kan-extension argument applies here using A.6.

For the last assertion we can use Cisinski (2008, Proposition 3.7); its localizer axiom is explicitly the relative one. To specify why it concerns our class, Yoneda identifies (A.1.1) with invertibility of $p_{B*} p_B^* \to p_{A*} p_A^*$. For a triangle $A \xrightarrow{u} B \xrightarrow{\beta} S$, with $\alpha = \beta u$, the pointwise formula identifies the value at s of $\beta_* p_B^* \to \alpha_* p_A^*$ with the constant cohomology comparison for $A/s \to B/s$. If all these comparisons are isomorphisms, Der 2 and then p_{S*} make the comparison on global constant cohomology an isomorphism. This proves relative Theorem A with all maps typed in $\mathbf{D}(S)$ and $\mathbf{D}(e)$. □

B.2. Relative Theorem A with an infinity-category enhancement. [Proved]

Let $\mathcal{C}$ be a complete and cocomplete infinity-category, with the universe bounds of the conventions. Then $\mathcal{W}_{\mathcal{C}}^{\mathrm{lc}}$ is a fundamental localizer. In particular, for a morphism $u : A \to B$ over a small category S, the hypotheses

$$(A/s \longrightarrow B/s) \in \mathcal{W}_{\mathcal{C}}^{\mathrm{lc}} \quad (s \in S) \tag{B.2.1}$$

imply that the specified restriction $\mathrm{Fun}(BB, \mathcal{C}) \to \mathrm{Fun}(BA, \mathcal{C})$ is an equivalence of infinity-categories.

Proof. The hypotheses initially concern homotopy categories, but A.8 upgrades each specified restriction to an infinity-categorical equivalence, using its Kan adjunctions. Consider the covariant diagram $s \mapsto A/s$. Its Grothendieck construction has objects $(s, a, p(a) \to s)$, where $p : A \to S$. The projection $P : \int_S (A/s) \to A$ has the section $J(a) = (p(a), a, \mathrm{id})$, and $J \dashv P$. Thus BP is a weak homotopy equivalence. The Grothendieck-construction formula for homotopy colimits, in the form Cisinski (2008, Theorem 1.13 and Remark 1.14), gives natural equivalences

$$BA \simeq \operatorname*{hocolim}_{s \in S} B(A/s), \qquad BB \simeq \operatorname*{hocolim}_{s \in S} B(B/s). \tag{B.2.2}$$

The space underlying this homotopy colimit is also the colimit of the same diagram in infinity-categories: the inclusion of spaces is left adjoint to the maximal-groupoid functor. Cartesian closure then gives, in the larger universe,

$$\mathrm{Fun}(BA, \mathcal{C}) \simeq \lim_{s \in S^{\mathrm{op}}} \mathrm{Fun}(B(A/s), \mathcal{C}), \tag{B.2.3}$$

and the corresponding formula for B. The diagram of restriction functors is coherently the one induced by u; (B.2.1) makes it an objectwise equivalence. Its limit is an equivalence, proving the claim about the specified map. A.4 and A.2 supply the remaining localizer axioms. If S is empty, both A and B are empty, so the same formula and conclusion hold. □

B.3. Relative Theorem A for an arbitrary derivator. [Proved]

Let $\mathbf{D}$ satisfy Der 1–4 in the presheaf convention of A.1, on all $\mathbb{U}$-small categories. Let $A \xrightarrow{u} B \xrightarrow{q} S$ be functors of $\mathbb{U}$-small categories and put $p = qu$. If, for every $s \in S$, the specified restriction

$$(u/s)^* : \mathrm{LC}_{\mathbf{D}}(B/s) \longrightarrow \mathrm{LC}_{\mathbf{D}}(A/s) \tag{B.3.1}$$

is an equivalence, then $u^* : \mathrm{LC}_{\mathbf{D}}(B) \to \mathrm{LC}_{\mathbf{D}}(A)$ is an equivalence. If every restriction in (B.3.1) is fully faithful, then u^* is fully faithful. Both conclusions also hold for a map of Grothendieck opfibrations preserving cocartesian arrows when the corresponding restriction on every strict fiber is, respectively, an equivalence or fully faithful. Consequently $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$ is a fundamental localizer for every such derivator, while $\mathcal{W}^{1}_{\mathbf{D}}$ satisfies relative Theorem A despite its failure of two-out-of-three in A.5. The statement concerns the ordinary categories $\mathrm{LC}_{\mathbf{D}}(-)$ themselves; neither an infinity-categorical presentation nor strongness of $\mathbf{D}$ is required.

Proof. We first establish the partial localizations needed to make descent coherent. If T is small and $R \subseteq T$ is a subcategory, let $\mathbf{D}(T; R)$ be the full subcategory of $\mathbf{D}(T)$ on objects whose underlying presheaf inverts every arrow of R. Both adjoints to

$$j_{T,R} : \mathbf{D}(T; R) \hookrightarrow \mathbf{D}(T) \tag{B.3.2}$$

exist, and they are the components of adjunctions of shifted derivators. Here and below "shifted" means that, at an auxiliary shape J, the same condition is imposed on the R arrows in $T \times J$.

For completeness, this is the arbitrary-subcategory form of the argument in Cisinski (2009, Theorem 3.6, Lemma 3.9, Proposition 3.10, and Theorem 3.13). Write $C = T^{\mathrm{op}}$ and $R' = R^{\mathrm{op}}$. The universal derivator of simplicial presheaves on C has a localization

$$\gamma : \mathrm{Ho}([C^{\mathrm{op}}, \mathrm{sSet}]) \rightleftarrows \mathrm{Ho}(L_{R'}[C^{\mathrm{op}}, \mathrm{sSet}]) : i, \qquad \gamma \dashv i. \tag{B.3.3}$$

The right adjoint i is fully faithful. It preserves homotopy colimits: a pointwise homotopy colimit of diagrams which invert R' again inverts R', since a homotopy colimit sends an objectwise weak equivalence of diagrams of spaces to a weak equivalence. This argument works at every auxiliary shape. Thus both γ and i are cocontinuous morphisms of derivators. Cisinski's

Theorem 3.6, which is explicitly stated for an arbitrary subcategory R' and every derivator, identifies precomposition by γ on cocontinuous morphisms into $\mathbf{D}$ with (B.3.2). Precomposition by i supplies its left adjoint. The same construction with the opposite derivator and opposite subcategory supplies its right adjoint. The internal-Hom argument of Theorem 3.13 gives these as adjunctions of derivators, because the argument just given applies to every shift. All the categories and sets of arrows used in the localization are $\mathbb{U}$-small; the simplicial-presheaf categories are taken in the ambient larger universe. This uses the universal property of the auxiliary model derivator, not a presentation of $\mathbf{D}$ by a model category. There is no hidden strongness hypothesis in this universal property: the cited foundational paper defines a two-sided derivator using Der 1–4, and Proposition 3.12 makes it eligible for Corollary 3.26 on every small shape (Cisinski, 2008, Definitions 0.1–0.2, Proposition 3.12, and Corollary 3.26).

We use the right adjoint of (B.3.2), denoted $Q_{T,R}$. Suppose that $\pi : T \to S$ is a Grothendieck opfibration and let R consist of its vertical arrows, namely those sent to identities. Write $\mathbf{D}(T)^{\mathrm{vert}} = \mathbf{D}(T; R)$ and $Q_T = Q_{T,R}$. Its objects are exactly the diagrams locally constant on every fiber. For the fiber inclusion $i_s : T_s \hookrightarrow T$, the canonical mate is an isomorphism

$$i_s^* Q_T \;\cong\; Q_{T_s} i_s^*. \tag{B.3.4}$$

We verify the required base change rather than assume descent. The left Kan extension $i_{s!}$ sends locally constant objects on T_s to vertically locally constant objects on T. Indeed, at $x \in T_t$, the comma category $x\backslash T_s$ is the disjoint union, over $\alpha : t \to s$, of categories with initial object the chosen cocartesian lift $x \to \alpha_! x$. Der 4, in presheaf variance, therefore gives

$$x^* i_{s!} F \;\cong\; \coprod_{\alpha \in \mathrm{Hom}_S(t,s)} (\alpha_! x)^* F. \tag{B.3.5}$$

For a vertical arrow $x \to x'$, the comparison in (B.3.5) is the coproduct of the comparisons induced by the vertical arrows $\alpha_! x \to \alpha_! x'$. They are isomorphisms when F is locally constant. The assertion follows, including empty comma categories and infinite coproducts. The equality of the two composites "include, then $i_{s!}$" and "$i_{s!}$ on the local subcategories, then include" now gives (B.3.4) by taking right adjoints. Consequently (B.3.4) is compatible with the units and counits, and the argument also proves it at every shift.

Consider a map $v : T \to T'$ of Grothendieck opfibrations over S preserving cocartesian arrows. For $x' \in T'_s$, the inclusion

$$T_s/x' \hookrightarrow T/x' \tag{B.3.6}$$

has a left adjoint. It sends $(x, \beta : v(x) \to x')$, with $\pi(x) = t$ and $\pi'(\beta) = \alpha : t \to s$, to the object $(\alpha_! x, \overline{\beta})$ obtained by factoring β through the cocartesian transport of $v(x)$. The universal property of a cocartesian arrow gives the adjunction. If $r \dashv i$ is this adjunction, presheaf variance gives $r_* \cong i^*$; composing right Kan extensions to the point shows that restriction along (B.3.6) computes the same homotopy limit. Der 4 therefore gives the canonical isomorphism

$$i_s'^* v_* \;\cong\; v_{s*} i_s^*. \tag{B.3.7}$$

This is the usual base change for a map of opfibrations, proved here with the variance required for $\mathbf{D}$.

Suppose now that every $v_s^* : \mathrm{LC}_{\mathbf{D}}(T'_s) \to \mathrm{LC}_{\mathbf{D}}(T_s)$ is an equivalence. On vertically locally constant diagrams we have an adjunction

$$v^* : \mathbf{D}(T')^{\mathrm{vert}} \rightleftarrows \mathbf{D}(T)^{\mathrm{vert}} : Q_{T'} v_*. \tag{B.3.8}$$

By (B.3.4) and (B.3.7), its unit and counit restrict on every fiber to those of $v_s^* \dashv Q_{T'_s} v_{s*}$. Those units and counits are isomorphisms by hypothesis. Every point lies in a fiber, so Der 2 makes the unit and counit of (B.3.8) isomorphisms. Thus (B.3.8) is an equivalence.

If the fiber restrictions are only fully faithful, the same argument applied to the unit alone proves that v^* in (B.3.8) is fully faithful. Its restriction to the full subcategories of fully locally constant diagrams is therefore fully faithful as well. This proves the asserted fiberwise full-faithfulness conclusion without an essential-surjectivity or reflection argument.

It remains to restrict this equivalence to diagrams locally constant on all arrows. Restriction v^* preserves this condition. It also reflects it among vertically locally constant diagrams. To see this, let $N \in \mathbf{D}(T')^{\mathrm{vert}}$ and suppose v^*N is locally constant. For $\alpha : s \to t$, the cocartesian lifts define a natural transformation between the corresponding fiber inclusions, and hence a morphism

$$\alpha_!^*(i_t'^*N) \longrightarrow i_s'^*N \quad \text{in } \mathrm{LC}_{\mathbf{D}}(T'_s). \tag{B.3.9}$$

Its image under v_s^* is an isomorphism, since it is the transport comparison for v^*N. The equivalence v_s^* is conservative, so (B.3.9) is an isomorphism. Every arrow of T' factors as a cocartesian arrow followed by a vertical arrow. Equation (B.3.9) and vertical local constancy therefore show that N is locally constant. Essential surjectivity in (B.3.8), followed by this reflection, gives essential surjectivity on fully locally constant diagrams; full faithfulness is inherited from (B.3.8). We have proved that $v^* : \mathrm{LC}_{\mathbf{D}}(T') \to \mathrm{LC}_{\mathbf{D}}(T)$ is an equivalence.

Finally replace $p : A \to S$ by the Grothendieck opfibration

$$\widetilde{A} = \int_{s \in S} (A/s) \longrightarrow S. \tag{B.3.10}$$

An object is $(a, s, \alpha : p(a) \to s)$, and a morphism is a pair $(h : a \to a', k : s \to s')$ with $k\alpha = \alpha' p(h)$. Transport along k replaces α by $k\alpha$. The forgetful functor $r_A : \widetilde{A} \to A$ has the left adjoint $a \mapsto (a, p(a), \mathrm{id})$, with $r_A j_A = \mathrm{id}$ and $j_A r_A \Rightarrow \mathrm{id}$. The latter natural transformation induces an isomorphism on every locally constant object, by Der 2. Consequently r_A^* and j_A^* are mutually inverse equivalences on locally constant categories. The same holds for B. The functor u induces a transport-preserving map $\widetilde{u} : \widetilde{A} \to \widetilde{B}$ whose fiber at s is precisely u/s. Apply (B.3.8)–(B.3.9) to $\widetilde{u}$ and then use the forgetful equivalences. For full faithfulness use the unit-only version of (B.3.8) just proved. This proves both relative Theorem A assertions.

The unit, counit, and base-change proof is the ordinary-derivator specialization of the proof of Hörmann (2015, Main Theorem 2.5.5(2), pp. 52–54). The stable compact-generation assumptions in that fibered theorem provide its partial Cartesian projectors. In the present nonfibered situation, (B.3.2)–(B.3.3) provide those projectors for every derivator, so those assumptions are unnecessary. Weak saturation was proved in A.4. A category with a terminal object is $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$-aspherical by the same adjoint-section argument as for r_A, using its projection to the point and the terminal section. Together with the relative theorem just proved, these are precisely the three fundamental-localizer axioms. □

B.4. Accessibility for all computed coefficient theories. [Proved]

For every $\mathbb{U}$-presentable enhanced coefficient category, and in particular for the derivators of sets, groupoids, r-types ($r \geq 0$), spaces, $D(k)$ for a small associative unital ring k, spectra, and $L_E\mathsf{Sp}$ for a small spectrum E, the constant and 2-complete classes are set-generated fundamental localizers. They also satisfy the literal accessibility condition Loc (9) of Chapter XVI ms. 75–79: as full subcategories of $\mathsf{Cat}^{[1]}$, they are generated under sufficiently filtered colimits by a small family of bounded-cardinality arrows. For the listed concrete coefficient theories the intermediate classes are accessible full subcategories of $\mathsf{Cat}^{[1]}$ as well; this does not turn them into fundamental localizers when two-out-of-three fails. No large-cardinal principle is used for these coefficient theories.

Proof. We give the bounded-subfunctor construction, so that the two meanings of accessibility are both verified. First fix $k \neq 0$ and choose an uncountable regular cardinal $\kappa \in \mathbb{U}$ larger than $|k|$ and $\aleph_0$. Size of a category means its number of arrows, including identities. Put $Q = \mathrm{Ex}^\infty$. The functor Q preserves monomorphisms and filtered colimits. Every simplex of QK occurs at a finite Ex-stage and is a map from a finite subdivided simplex; consequently it depends on finitely many simplices of K. If a category has fewer than κ arrows, its Q-nerve has fewer than κ simplices in total. For a vertex a of a Kan complex K, use the based-loop model

$$\Omega_a K = \{(a,a)\} \times_{K\times K} K^{\Delta^1}.$$

The endpoint map is a Kan fibration. Each loop simplex uses only the finite simplicial set $\Delta^n \times \Delta^1$, so this construction also has finite support and commutes with filtered colimits of pointed Kan complexes.

Let $f : A \to B$ belong to $W^{\mathrm{lc}}_{D(k)}$, and start with any subfunctor $g_0 : A_0 \to B_0$ having fewer than κ arrows. Here A_0, B_0 are subcategories and $f(A_0) \subseteq B_0$. We enlarge it inside f using the criterion of C.7. For every vertex $b \in B_0$, choose a vertex $a \in A$ and a finite zigzag from $f(a)$ to b. For every pair $a, a' \in A_0$ whose images are connected in B_0, choose a finite zigzag joining them in A. Ambient bijectivity on components supplies these choices.

For every $a \in A_0$, every $n \geq 0$, and every cycle $z \in C_n(\Omega_{f(a)} QNB_0; k)$, choose an ambient cycle $w \in C_n(\Omega_a QNA; k)$ and an ambient chain b with $f_*(w) - z = db$. Ambient surjectivity on loop homology supplies them. For every cycle $w \in C_n(\Omega_a QNA_0; k)$ and every chain $b \in C_{n+1}(\Omega_{f(a)} QNB_0; k)$ with $db = f_*(w)$, choose an ambient chain c with $dc = w$; ambient injectivity supplies it. Each selected chain and zigzag has finite support. Pull the supporting Q-nerve data back to finitely many original arrows of A and B, add those arrows and their images under f, and generate subcategories by identities and finite composition. There are fewer than κ tests and witnesses: all relevant chain groups have fewer than κ elements because $|k| < \kappa$. Thus the enlarged subfunctor g_1 still has size less than κ.

Repeat this construction countably many times. Its union g_∞ has size less than κ by regular uncountability. Every component relation in the union has a finite zigzag witness and occurs at a finite stage. Every cycle and every bounding chain likewise occurs at a finite stage, because its support is finite and QN and the loop model commute with filtered unions. At the next stage the required lift or bounding chain was added. Thus g_∞ is a component bijection and its based-loop maps are k-homology equivalences. C.7 gives $g_\infty \in W^{\mathrm{lc}}_{D(k)}$.

Let P_f be the poset of all size-$< \kappa$ subfunctors of f lying in this class. Given fewer than κ members, take their union, generate subcategories, and apply the construction just made. This gives an upper bound in P_f, so P_f is κ-filtered. Every original arrow lies in some member, by starting the construction with that arrow. Therefore the specified inclusions give

$$f = \operatorname*{colim}_{g \in P_f} g \quad \text{in } \mathsf{Cat}^{[1]}.$$

This also checks the enlargement form of the manuscript's Loc (9), with inclusions of source into source and target into target. The same finite-support tests show closure of the class under filtered colimits: QN, based loops with a chosen vertex, and homology commute with them; bijectivity on components is detected by finite zigzags. A category with fewer than κ arrows is κ-presentable in Cat, and similarly for a functor in $\mathsf{Cat}^{[1]}$: all its generators and composition equations form a set of size $< \kappa$. These small weak arrows remain κ-presentable in the full subcategory because its filtered colimits are computed in the ambient arrow category. Choosing representatives of their isomorphism classes therefore proves accessibility as a full subcategory.

For set-generation as a fundamental localizer, let S be that small set of representatives and let $\langle S \rangle$ be its generated fundamental localizer. B.2 gives $\langle S \rangle \subseteq W^{\mathrm{lc}}_{D(k)}$. Every $g \in P_f$

belongs to $\langle S\rangle$. Every fundamental localizer is closed under filtered colimits, by Cisinski (2006, Corollary 4.2.22); hence $f \in \langle S\rangle$. This proves equality. The cited corollary alone is not an accessibility theorem; the bounded construction is the indispensable extra argument.

For $L_E\mathsf{Sp}$, represent E by a small cofibrant sequential spectrum and choose κ also larger than the total number of simplices in all its levels. Use C.11 and replace loop-chain cycles and boundaries by representatives and equality witnesses in the stable homotopy groups of $E \wedge \Sigma^\infty_+ \Omega_a QNA$. A stable class is represented at a finite spectrum level by a finite simplicial sphere after levelwise fibrant replacement; an equality or nullity is witnessed at a further finite level by a finite homotopy. Thus each witness uses finitely many simplices of the loop complex. For a size-$< \kappa$ complex there are fewer than κ such representatives and witnesses, over all integer degrees and spectrum levels. The same countable enlargement, filtered-union verification, and generated-localizer argument therefore apply. This requires neither connectivity of E nor a smashing localization. Spectra are the case $E = \mathbb{S}$.

For the constant classes, omit the based-loop construction and use the ordinary chain or suspension-spectrum homology test of C.7 or C.11. Their fundamental-localizer assertion is B.1. For the intermediate classes use the homotopy-fibre test C.2 instead of based loops. A homotopy fibre of a Kan map is constructed from a path object and a finite pullback, so it has the same finite-support and filtered-colimit properties. The resulting full subcategories are accessible, but the set-generation conclusion as a fundamental localizer is not asserted for them.

For sets, groupoids, and r-types, C.4 reduces the tests to finitely many homotopy groups and a surjectivity condition in the next degree. Representatives and homotopies are finite sphere and disk data in Kan replacements, so the same enlargement applies with any uncountable regular κ. For spaces use every homotopy degree, still a countable family of finite-support tests. These prove literal accessibility and, for their constant and 2-complete fundamental localizers, set-generation. Zero coefficients give the whole arrow category and are included by discarding all tests.

All presentable enhanced coefficients. We strengthen the concrete argument to every $\mathbb{U}$-presentable ∞-category $\mathcal{C}$. Choose an uncountable regular $\kappa \in \mathbb{U}$ for which $\mathcal{C}$ is κ-compactly generated. The category $\mathrm{Pr}^{\mathrm{L}}_\kappa$ of κ-compactly generated presentable categories and colimit-preserving functors preserving κ-compact objects is itself $\mathbb{U}$-presentable: Lurie (2009, Proposition 5.5.7.10) identifies it with $\mathsf{Cat}^{\mathrm{Rex}(\kappa)}_\infty$, and Lurie (2017, Lemma 4.8.4.2) applies to the $\mathbb{U}$-small family of representatives of the κ-small simplicial shapes. By Lurie (2009, Proposition 5.5.7.6 and Notation 5.5.7.7), limits in its dual category $\mathrm{Pr}^{\mathrm{R}}_\kappa$ are computed in categories. Consequently

$$F_{\mathcal{C}} : \mathcal{S} \longrightarrow \mathrm{Pr}^{\mathrm{L}}_\kappa, \qquad X \longmapsto \mathrm{Fun}(X, \mathcal{C}), \qquad f \longmapsto f_!,$$

is the copower functor $X \mapsto \mathrm{colim}_X \mathcal{C}$ and preserves all small colimits. The natural direction here is left Kan extension, not restriction; its adjoint is the specified restriction f^*. In particular $F_{\mathcal{C}}$ is accessible. Restriction is an equivalence exactly when its adjoint is, so $\mathcal{W}^{\mathrm{lc}}_{\mathcal{C}}$ is the inverse image under $\mathrm{Arr}(F_{\mathcal{C}})$ of the full subcategory of equivalences. For an accessible category $\mathcal{E}$, that full subcategory of $\mathrm{Arr}(\mathcal{E})$ is accessible: it is equivalent to $\mathcal{E}$ by $e \mapsto \mathrm{id}_e$, and its inclusion preserves filtered colimits. Lurie (2009, Proposition 5.4.6.6), applied to the resulting homotopy pullback, proves accessibility of this arrow class of spaces.

To obtain the literal class of ordinary functors, apply the same pullback theorem to the accessible functor

$$N(\mathrm{Arr}(\mathsf{Cat}_{\mathbb{U}})) \longrightarrow \mathrm{Arr}(\mathcal{S}), \qquad u \longmapsto Bu.$$

It preserves filtered colimits because nerves and classifying spaces do. Since the equivalence condition is full and replete, this pullback is the nerve of the ordinary full subcategory of

arrows in $\mathcal{W}_{\mathcal{C}}^{\mathrm{lc}}$. Accessibility of that nerve is ordinary accessibility: compactness is detected by its discrete mapping spaces, and a filtered infinity-category admits a cofinal ordinary filtered poset (Lurie, 2009, Proposition 5.3.1.18). Thus a $\mathbb{U}$-small set of functors generates the class under sufficiently filtered colimits. It is closed under all filtered colimits by B.2 and Cisinski's Corollary 4.2.22. Its closure under all filtered colimits is therefore exactly the class, proving literal Loc (9). The generated-localizer argument above proves set-generation as well.

For the constant class, choose a $\mathbb{U}$-small set G generating $\mathcal{C}$ under colimits. The colimit-preserving functor $X \mapsto (X \otimes g)_{g \in G}$ from spaces to $\mathcal{C}^G$ inverts precisely $\mathcal{W}_{\mathcal{C}}$ by C.3: the equivalence test is closed under colimits in the coefficient object, so it suffices to test the generators. The identical inverse-image and ordinary-nerve argument applies. This strengthens all the concrete constant and locally constant calculations without a large-cardinal hypothesis. □

The order printed on pairs of subcategories in Chapter XVI ms. 78 reads $X_i \subset Y_i$ and $X_j \subset Y_j$. It must read $X_i \subset X_j$ and $Y_i \subset Y_j$. The former expression is not even typed when source and target categories differ; the proof above uses the latter.

For an arbitrary derivator, accessibility is not proved here. The bounded tests above depend on a fixed small ring, spectrum, or truncation theory. Filtered-colimit closure by itself does not imply set-generation. If one assumes Vopěnka's principle, every fundamental localizer is accessible (Cisinski, 2006, Scholie 4.2.17); that additional set-theoretic hypothesis is not assumed in this paper.

B.5. Sums, components, direct excision, and filtered colimits. [Proved]

For every derivator, $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ satisfies the full disjoint-union axiom Loc (4). It satisfies Loc (5), namely $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}} \subseteq \mathcal{W}_0$, exactly when $\mathbf{D}(e)$ is not equivalent to the terminal category. For every derivator it also satisfies the direct Mayer–Vietoris axiom Loc (6a) and the filtered-colimit axiom Loc (8). The class $\mathcal{W}_{\mathbf{D}}^{1}$ satisfies Loc (4), Loc (6a), and Loc (8) for every derivator, despite its possible failure of two-out-of-three. For arbitrary $\mathbf{D}$, it satisfies Loc (5) exactly when $\mathbf{D}(e)$ is not a preorder up to equivalence. For arbitrary $\mathbf{D}$, the constant class $\mathcal{W}_{\mathbf{D}}$ satisfies Loc (6a) and Loc (8); it satisfies Loc (5) exactly under the nonthinness condition G.2, and need not satisfy the reverse implication in Loc (4).

Proof. A.7 proves Loc (4), and G.1 proves the stated exact component criterion. For Loc (6a), let $A = A_1 \cup A_2$ be a union of two full sieves, with $A_0 = A_1 \cap A_2$, and let $B = B_1 \cup B_2$ be another such union. Let $f : A \to B$ carry each A_i into B_i. Every simplex of NA lies in one of NA_1, NA_2: its last vertex lies in one of the sieves, and all its sources then lie there. Consequently $NA = NA_1 \cup_{NA_0} NA_2$, a homotopy pushout of simplicial sets along monomorphisms. The same holds for B. Applying $\mathrm{Fun}(-, \mathcal{C})$ to their groupoid completions gives pullback squares of infinity-categories. If the three maps for $i = 0, 1, 2$ are equivalences, so is the fourth. This is precisely Loc (6a); it is not either reverse-excision assertion.

For a filtered diagram of arrows in Cat, the nerve commutes with its filtered colimit, since a finite string of composable arrows and every equality between such strings occur at some stage. Filtered colimits of simplicial sets compute the filtered colimit of the represented spaces. Thus the analogue of (B.2.3) identifies restriction at the colimit with a limit of restrictions. Limits preserve equivalences and fully faithful functors. A.8 therefore gives Loc (8) for $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ and for $\mathcal{W}_{\mathbf{D}}^{1}$ in the enhanced case. For the precise component condition on $\mathcal{W}_{\mathbf{D}}^{1}$, G.2 and $\mathcal{W}_{\mathbf{D}}^{1} \subseteq \mathcal{W}_{\mathbf{D}}$ prove sufficiency of nonthinness. If $\mathbf{D}(e)$ is thin up to equivalence, the fold $e \amalg e \to e$ belongs to $\mathcal{W}_{\mathbf{D}}$, and hence to $\mathcal{W}_{\mathbf{D}}^{1}$ by A.3, but does not belong to $\mathcal{W}_0$. This also covers the terminal coefficient category. Thus nontriviality suffices for Loc (5) for $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$, whereas the strictly stronger nonthinness condition is exact for $\mathcal{W}_{\mathbf{D}}^{1}$ and $\mathcal{W}_{\mathbf{D}}$.

For $\mathcal{W}_{\mathbf{D}}$, B.1 and Cisinski (2006, Corollary 4.2.22) give filtered-colimit closure. For direct excision, take the fundamental sublocalizer generated by the three given arrows. It is accessible by definition, is contained in $\mathcal{W}_{\mathbf{D}}$, and corresponds to the simplicial model structure with monomorphisms as cofibrations (Cisinski, 2006, Theorems 1.4.3 and 4.2.15). Its usual cofibrant gluing lemma applies to the two nerve pushouts above. This proves Loc (6a) without an accessibility assumption on $\mathcal{W}_{\mathbf{D}}$ itself. Its sums and component exceptions are the explicit represented-lattice example G.2. By B.3, $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ is also a fundamental localizer for an arbitrary derivator. The same generated-sublocalizer gluing argument and Corollary 4.2.22 give Loc (6a) and Loc (8) without enhancement. For $\mathcal{W}_{\mathbf{D}}^{1}$, use instead the fully faithful version of Grothendieck-construction descent proved in B.3. A natural transformation of diagrams $A_i \to B_i$ whose components are in $\mathcal{W}_{\mathbf{D}}^{1}$ induces a map of their Grothendieck opfibrations with fully faithful restriction on locally constant objects. For a filtered diagram, the comparison from its Grothendieck construction to its ordinary colimit is in $\mathcal{W}_{\infty}$: the homotopy-colimit formula (B.2.2) and preservation of filtered colimits by the nerve identify this comparison with the usual homotopy-colimit-to-colimit equivalence of simplicial sets. For a diagram $A_1 \leftarrow A_0 \to A_2$ of the sieves above, the same comparison is in $\mathcal{W}_{\infty}$ because the nerve union is a pushout along monomorphisms. In both cases these comparisons induce equivalences on local systems by A.2. Conjugating the fully faithful Grothendieck-construction restriction by them proves respectively Loc (8) and Loc (6a) for $\mathcal{W}_{\mathbf{D}}^{1}$, without using two-out-of-three for that class. □

2.2 Reverse excision and the three meanings of fibration

B.6. The reverse Mayer–Vietoris axioms. [Refuted]

For a derivator $\mathbf{D}$ with $\mathbf{D}(e) \not\simeq e$, the class $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ satisfies neither Loc (6b) nor Loc (6c) of Chapter XVI. If $\mathbf{D}(e) \simeq e$, all arrows belong to $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ and both axioms hold. For $\mathcal{W}_{\mathbf{D}}^{1}$ of a nontrivial derivator, Loc (6b) always fails, whereas Loc (6c) holds exactly when $\mathbf{D}(e)$ is a preorder up to equivalence. Thus the unqualified Loc (6), which includes Loc (6b), fails for every nontrivial coefficient theory, even for spaces and ordinary weak homotopy equivalences. In contrast, each constant homology localizer W_E satisfies all three Mayer–Vietoris implications Loc (6a,b,c).

Proof. Write a Mayer–Vietoris square in the order (X_0, X_1, X_2, X_3), where X_1, X_2 are full sieves of X_3, their union is X_3, and $X_0 = X_1 \cap X_2$. For a map of two such squares, the manuscript's precise implications are

$$\begin{aligned} (6b):&\quad f_1, f_2, f_3 \in W \Longrightarrow f_0 \in W,\\ (6c):&\quad f_0, f_3 \in W \Longrightarrow f_1, f_2 \in W. \end{aligned} \tag{B.6.1}$$

We give separate countercubes, so no implication from (6c) to (6b) is being assumed.

For any small category A, let $C_v = A \star \{v\}$ and $C_w = A \star \{w\}$ be its two terminal cones and put $\Sigma A = A \star \{v, w\}$, with no arrows between v and w. Its objects are those of A and v, w. Its arrows are those of A, the two new identities, and exactly one arrow $a \to v$ and one arrow $a \to w$ for every $a \in A$. Composing any arrow of A with either terminal arrow gives the terminal arrow from its source; these and the compositions in A specify the entire composition law. The inclusions $C_v, C_w \subset \Sigma A$ are full sieves with intersection A. Their nerves form a homotopy pushout, and $B\Sigma A \simeq \Sigma BA$ is the unreduced suspension. It suffices to find

$$A \longrightarrow e \notin \mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}, \qquad \Sigma A \longrightarrow e \in \mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}. \tag{B.6.2}$$

First suppose that $\mathbf{D}(e)$ has distinct parallel morphisms $\alpha, \beta : M \to N$. Choose a nontrivial integrally acyclic group G as constructed in B.10, and take $A = BG$, the one-object category

with arrows G and group multiplication as composition. The CW space BG is connected and acyclic. Its suspension is simply connected and acyclic, and therefore weakly contractible by homological Whitehead. Thus the second condition of (B.6.2) holds by A.2.

For the first condition, let $a : e \to BG$ be its object. The coherent object $a_!M \in \mathbf{D}(BG)$ has value $\coprod_{g\in G} M$ at that object; its G-action is regular permutation of the summands. This is a pointwise Kan-extension calculation, and constructs a coherent object rather than merely postulating a diagram in $\mathbf{D}(e)$. A nonidentity permutation is detected by a map to N assigning α to one summand and β to all the others. Consequently its action is nontrivial. It cannot be isomorphic to a constant object, whose action is trivial. Every object of $\mathbf{D}(BG)$ is locally constant since every arrow of BG is invertible. Thus $\mathbf{D}(e) \to \mathrm{LC}_{\mathbf{D}}(BG)$ is not essentially surjective, proving the first condition.

It remains to treat a nontrivial $\mathbf{D}(e)$ which is a preorder up to equivalence. For every nonempty group G, a nonempty coproduct of copies of an object M of this category is isomorphic to M. The counit $a_!a^*F \to F$, for $F \in \mathbf{D}(BG)$, is therefore an isomorphism on its sole point evaluation, and hence an isomorphism by Der 2. Likewise the canonical map $a_!M \to p^*M$, where $p : BG \to e$, is an isomorphism. The adjunction for a now gives

$$\mathrm{Hom}_{\mathbf{D}(BG)}(p^*M, p^*N) \simeq \mathrm{Hom}_{\mathbf{D}(BG)}(a_!M, p^*N) \simeq \mathrm{Hom}_{\mathbf{D}(e)}(M, N).$$

Thus every $BG \to e$ belongs to $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$. Take instead $A = e \amalg e$. Its projection is not in $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$: the diagonal $\mathbf{D}(e) \to \mathbf{D}(e)^2$ is not essentially surjective because $\mathbf{D}(e)$ has two nonisomorphic objects. The category ΣA is the four-object poset $a, b < v, w$. Its nerve is a circle, hence is weakly equivalent to $B\mathbb{Z}$, and the preceding argument proves the second condition of (B.6.2). Explicitly, a weak equivalence $\Sigma A \to B\mathbb{Z}$ sends $a \to v$ to $1 \in \mathbb{Z}$, the other three nonidentity arrows to 0, and every object to the unique object; its map on the unique cycle has degree one.

For Loc (6b), map the square $(A, C_v, C_w, \Sigma A)$ to the identical double-cone construction on e. Each cone has a terminal object, and both total categories are $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$-aspherical by (B.6.2). Thus f_1, f_2, f_3 belong to $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$, whereas $f_0 : A \to e$ does not. For Loc (6c), use the map

$$(A, A, C_w, C_w) \longrightarrow (A, C_v, C_w, \Sigma A). \tag{B.6.3}$$

Both quadruples are Mayer–Vietoris squares, allowing one sieve to equal the whole category. Here $f_0 = \mathrm{id}_A$, $f_2 = \mathrm{id}_{C_w}$, and $f_3 : C_w \to \Sigma A$ belong to $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$, but $f_1 : A \to C_v$ does not, by two-out-of-three with $C_v \to e$. This directly disproves (6c). The finite thin-coefficient instance, including all objects, arrows, compositions, functoriality and sieve checks, is verified by `code/B_verify_reverse_excision.py`; its complete output is included.

For the asserted 1-complete comparison, first assume that $\mathbf{D}(e)$ is nonthin. Reverse the intersection map in the first cube: map the double cone of e to the double cone of the acyclic BG. The three cone and total maps are weak homotopy equivalences. But restriction along $e \to BG$ is not full: with the objects M, N chosen above, it induces the nonbijective diagonal

$$\mathrm{Hom}(M, N) \longrightarrow \mathrm{Hom}(M, N)^G$$

on morphisms from the free coherent G-object $a_!M$ to the constant object N. This refutes (6b).

For (6c), choose a nontrivial acyclic group H and, by B.10, an acyclic simple group G containing H properly. Its larger cardinality guarantees properness of the inclusion. Let Y_1 be the categorical mapping cylinder of $BH \to BG$. It has two objects h, g, endomorphism groups H, G, and $\mathrm{Hom}(h, g) = G$, with $\mathrm{Hom}(g, h) = \varnothing$. Composition on the cross-arrows is left multiplication by G and right multiplication by the specified image of H. This specifies all its arrows and compositions. The full subcategory BH is a sieve; projection $Y_1 \to BG$ is a weak homotopy equivalence, as seen from its terminal-index Grothendieck construction. Let

$C_H = BH \star \{w\}$ and glue it to Y_1 along BH, giving Y_3. Besides the arrows just listed, Y_3 has id_w and a unique $h \to w$, fixed by precomposition with H, and no arrows $g \to w$ or out of w. Its two pieces Y_1, C_H are full sieves. Its classifying space is the homotopy cofiber of $BH \to BG$. Van Kampen gives $\pi_1 BY_3 = G/\langle\langle H \rangle\rangle = 1$ by simplicity; Mayer–Vietoris gives zero reduced integral homology since both groups are acyclic. Thus BY_3 is weakly contractible.

Use the map of squares

$$(BH, BH, C_H, C_H) \longrightarrow (BH, Y_1, C_H, Y_3). \tag{B.6.5}$$

Its f_0 is an identity and its f_3 is a weak homotopy equivalence. Its f_1 is not 1-complete. Indeed, restriction of the free coherent G-object to BH is a coproduct of free coherent H-objects, one for each coset in G/H. To see this at the coherent level, choose coset representatives, use the corresponding arrows of BG to construct the maps from the free H-objects by Kan adjunction, and apply point evaluation: their disjoint regular orbits partition G, so that map is an isomorphism by Der 2. The morphism comparison to the constant object N is consequently

$$\mathrm{Hom}(M, N) \longrightarrow \mathrm{Hom}(M, N)^{G/H},$$

again a nonbijective diagonal. Since $Y_1 \to BG$ is a weak equivalence, the same failure holds for $BH \to Y_1$. This proves failure of (6c) for $\mathcal{W}^1_{\mathbf{D}}$ with nonthin coefficients, including all the nonzero stable coefficient categories.

There is a different answer for thin coefficients. If $P = \mathbf{D}(e)$ is a nontrivial preorder up to equivalence, then every $\mathbf{D}(A)$ is thin as well: its diagonal $F \to F \times F$ is pointwise an isomorphism, hence an isomorphism by Der 2, which forces every Hom set to have at most one element. Let $u : \mathrm{Ob}(A)_{\mathrm{disc}} \to A$ be the inclusion of its discrete object set. The unit $F \to u_* u^* F$ is a pointwise isomorphism. At a the right-hand side is the product of the values F_b over arrows $b \to a$, in presheaf variance. The value F_a maps to every factor by the underlying diagram, and occurs as the identity factor; in a preorder these two facts make the unit an isomorphism. The Kan adjunction therefore makes the underlying-diagram functor fully faithful. Conversely, for a presheaf of values in P, applying u_* to its values has exactly those point values by the same product calculation. Thus $\mathbf{D}(A) \simeq \mathrm{Fun}(A^{\mathrm{op}}, P)$, and

$$\mathrm{LC}_{\mathbf{D}}(A) \simeq P^{\pi_0 A}, \qquad \mathcal{W}^1_{\mathbf{D}} = \{f : \pi_0 f \text{ is surjective}\}. \tag{B.6.6}$$

Here the last equivalence uses nontriviality of the complete preorder P: an omitted target component is detected by its distinct initial and terminal objects. Map the double cone of the empty category to the double cone of e. Its intersection map $\varnothing \to e$ is not in $\mathcal{W}^1_{\mathbf{D}}$; both cone maps are weak equivalences, and the total map from two points to a connected category is surjective on components. This refutes (6b).

To verify (6c) in this case, suppose f_0, f_3 are surjective on components and let C be a component of Y_1. If C meets Y_0, surjectivity of $\pi_0 f_0$ gives an object of $X_0 \subset X_1$ mapping into C. If it does not, the sieve-cover description of the nerve shows that C is a whole component of Y_3 disjoint from Y_2. Surjectivity of $\pi_0 f_3$ then gives an object of X_3 mapping into C; it must lie in X_1, since an object of X_2 maps into Y_2. Thus $\pi_0 f_1$ is surjective, and the symmetric argument proves the same for f_2. This proves the exact thin-coefficient exception for (6c).

If $\mathbf{D}(e) \simeq e$, G.1 shows $\mathbf{D}(A) \simeq e$ for every A, and the exception in the statement follows. Finally, a sieve cover gives a homotopy-pushout square of nerves. Applying $E \wedge \Sigma^\infty_+$ gives the exact triangle

$$E \wedge \Sigma^\infty_+ BX_0 \longrightarrow (E \wedge \Sigma^\infty_+ BX_1) \oplus (E \wedge \Sigma^\infty_+ BX_2) \longrightarrow E \wedge \Sigma^\infty_+ BX_3. \tag{B.6.4}$$

In a map of exact triangles, equivalence of any two terms implies equivalence of the third. An equivalence of the displayed block-diagonal middle map implies equivalence of each block, since

each is a retract. This proves (6a), (6b) and (6c) for W_E, in particular for the constant class of $D(k)$. This stable excision argument does not apply to the 2-complete class of $D(k)$. □

B.7. Classification of the fibration axioms. [PROVED]

Let W be a fundamental localizer with $W \subseteq W_0$. Define $\mathcal{F}_W$ to be the class of Grothendieck fibrations whose transport functors belong to W, and define Fib_W by preservation of all W-equivalences under strict categorical pullback. Then each of Loc (7a), Loc (7b) and Loc (7c) holds if and only if $W = W_\infty$. In particular this applies to $W = \mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ for every nontrivial derivator, by B.3 and G.1. For the trivial derivator all three axioms hold.

If W is accessible, Loc (7F), with the manuscript's class Fib_W, holds if and only if W is proper, equivalently if and only if it is generated as a fundamental localizer by a set of projections $A \to e$. For a nonzero associative unital ring k this specializes to the complete classification

$$\begin{aligned} \mathrm{Loc}(7F) \text{ for } W_{D(k)}^{\mathrm{lc}} &\iff \text{no rational prime is invertible in } k \\ &\iff W_{D(k)}^{\mathrm{lc}} = W_\infty. \end{aligned} \tag{B.7.1}$$

The constant k-homology localizer satisfies none of Loc (7a,b,c,F), for every nonzero k. For $\mathcal{W}_{\mathbf{D}}^{1}$, which need not be a localizer, Loc (7c) nevertheless holds for every derivator. With the enhancement of C.2, Loc (7a) and the factorization assertion of Loc (7F) hold as well; Loc (7b) fails for sets, groupoids, every finite r-type theory, every nonzero $D(k)$, every nonzero $L_E\mathsf{Sp}$, and every nontrivial thin coefficient theory. These facts do not make $\mathcal{W}_{\mathbf{D}}^{1}$ the weak-equivalence class of a Brown category.

Proof. The exact axioms. Chapter XVI, ms. 38–39, states

$$\begin{aligned} &(7a): \quad \mathcal{F}_W \subseteq \mathrm{Fib}_W; \\ &(7b): \quad f : X \to Y \text{ over } S, \quad X \to S, Y \to S \in \mathcal{F}_W, \quad f \in W \\ &\qquad\qquad \Longrightarrow T \times_S f \in W \text{ for every } T \to S; \\ &(7c): \quad \mathcal{F}_W \cap W \subseteq W^{\mathrm{u}}; \\ &(7F): \quad \text{every functor factors as } pi, \quad i \in W, \quad p \in \mathrm{Fib}_W\,. \end{aligned} \tag{B.7.2}$$

The word "cofibrations" in the manuscript's formulation of (7F) is a typographical slip: the surrounding definition is the category-of-fibrant-objects factorization axiom. The class called *weak W-fibrations* in Cisinski (2006, Definition 6.1.13) is precisely Fib_W here. It is not the manuscript's weaker class obtained by testing only contractible bases.

A pullback lemma. Suppose that factorizations in (7F) exist. If $f : X \to Y$ is in W over S and both structural maps belong to Fib_W, then every pullback of f over S is in W. Indeed, factor $t : T \to S$ as $T \xrightarrow{a} Z \xrightarrow{b} S$, with $a \in W$ and $b \in \mathrm{Fib}_W$. The map f_Z is in W by the definition of b. The maps $X_T \to X_Z$ and $Y_T \to Y_Z$ are in W by the definition of the pullbacks of the two structural maps. Two-out-of-three in the resulting square gives $f_T \in W$. In particular

$$\mathrm{Fib}_W \cap W \subseteq W^{\mathrm{u}}. \tag{B.7.3}$$

Why (7a) implies (7b). Every functor $u : A \to S$ factors as a weak homotopy equivalence followed by a member of $\mathcal{F}_{W_\infty}$. For completeness, factor it first as $A \to P \xrightarrow{p} S$ in the ordinary Thomason model structure. Its model fibration p is universally W_∞-preserving, by right properness. Form the Grothendieck fibration $q : \Phi(P) \to S$ of the contravariant diagram $s \mapsto s\backslash P$. An object is $(s, x, \beta : s \to p(x))$; an arrow to (s', x', β') is a pair $(a : s \to s', h : x \to x')$ with $p(h)\beta = \beta' a$. The inclusion $j(x) = (p(x), x, \mathrm{id})$ is a weak homotopy equivalence: forgetting

s, β gives a retraction ϵ, and the arrows (β, id) give $\mathrm{id} \Rightarrow j\epsilon$. For $s \to s'$, the transition $s'\backslash P \to s\backslash P$ is the pullback by p of $s'\backslash S \to s\backslash S$. The latter is a weak equivalence between categories with initial objects, so the transition is a weak equivalence. This proves the asserted factorization.

Since $\mathcal{F}_{W_\infty} \subseteq \mathcal{F}_W$, (7a) supplies (7F), using these factorizations. Applying the pullback lemma then gives (7b). Taking $Y = S$ and $q = \mathrm{id}_S$ gives (7b)⇒(7c).

Why (7c) forces the minimal localizer. The exact hypotheses of Cisinski (2006, Theorem 6.4.22) are: (a) $W \subseteq W_0$; and (b) a W-locally constant functor is a W-equivalence if and only if it is W-aspherical. No accessibility hypothesis occurs. Here "W-locally constant" means that the transitions $A/s \to A/s'$ are in W. Given such a functor $u : A \to S$, replace it by the Grothendieck opfibration

$$q : \int_{s \in S} (A/s) \longrightarrow S.$$

The projection to A is a weak homotopy equivalence, with section $a \mapsto (u(a), a, \mathrm{id})$ and the evident natural comparison; that comparison also gives $q \in W \iff u \in W$. Its fibres are A/s and its transports belong to W. Apply (7c) to its opposite, a Grothendieck fibration. Fundamental localizers are invariant under opposites. We conclude that every A/s is W-aspherical. Thus a W-locally constant W-equivalence is W-aspherical. The converse is Theorem A. The cited theorem proves $W = W_\infty$.

For W_∞, a Grothendieck fibration with weakly equivalent transports computes its strict base changes as homotopy pullbacks, by Quillen's Theorem B and its base-change criterion (Quillen, 1973, §1, Theorem B and its corollary); see also Rezk (1998, Proposition 2.7). It therefore preserves weak equivalences under every base change, giving (7a), and hence (7b,c). This completes their classification. The same result holds for every nontrivial fundamental localizer, without the explicit condition $W \subseteq W_0$, by Cisinski (2006, Remark 6.4.23). The only additional nontrivial case is the gross localizer; the inclusion of one point into two discrete points refutes (7c) there.

Why (7F) is exactly properness. Let W_{asph} denote the fundamental localizer generated by all projections $A \to e$ which belong to W. It is contained in W and has exactly the same aspherical categories and aspherical functors (Cisinski, 2006, §6.1.7). Suppose (7F). For any $f : X \to Y$ in W, factor its graph as

$$X \xrightarrow{i} P \xrightarrow{q} X \times Y, \qquad i \in W, \quad q \in \mathrm{Fib}_W\,.$$

The projections from a product belong to Fib_W, since a fundamental localizer is stable under products. Consequently $r = \mathrm{pr}_X\, q$ and $p = \mathrm{pr}_Y\, q$ belong to Fib_W. Since $ri = \mathrm{id}_X$ and $pi = f$, both r, p are also in W. By (B.7.3) they are universally in W. For each $y \in Y$, pull p back to Y/y. The result $P/y \to Y/y$ is in W, and its target is contractible. Thus P/y is W-aspherical, equivalently W_{asph}-aspherical. Theorem A for W_{asph} gives $p \in W_{\mathrm{asph}}$. The same argument gives $r \in W_{\mathrm{asph}}$. Then $ri = \mathrm{id}_X$ gives $i \in W_{\mathrm{asph}}$, and $f = pi \in W_{\mathrm{asph}}$. We have proved

$$\mathrm{Loc}(7F) \Longrightarrow W = W_{\mathrm{asph}}, \tag{B.7.4}$$

without an accessibility assumption.

If W is accessible, Proposition 6.1.8 and Theorem 6.1.11 of Cisinski (2006) identify this equality with properness, or equivalently generation by a set of aspherical projections. Conversely, for proper W the Thomason model structure is right proper (Cisinski, 2006, Theorem 5.2.15). Its model fibrations belong to Fib_W (Cisinski, 2006, Example 6.1.14), and its factorization axiom supplies (7F). This proves the claimed equivalence. Here properness is right properness of the associated simplicial model structure; the cited theorem gives the equivalent Thomason-model

condition. Literal Loc (9), accessibility of the full subcategory of arrows, must not be confused with the additional assertion $W = W_{\mathrm{asph}}$.

An obstruction to properness. We use the precise statement of Cisinski (2006, Lemma 9.4.5): if $W \subseteq W_n$ and W is proper, every simplicial W-equivalence whose target is W-fibrant is a *strong* n-equivalence. By Scholie 9.2.18 there, this includes surjectivity on π_{n+1}, in addition to isomorphisms through degree n. It is enough below to exhibit a W-equivalence between connected spaces, with W-local target and nonsurjective π_{n+1}. A Kan representative of a W-local space is W-fibrant: restriction against a W-equivalence which is a cofibration is a weak equivalence and a Kan fibration of mapping spaces, hence a trivial fibration. This verifies the target hypothesis in each application, rather than omitting it.

Characteristic zero rings. Let S be the set of rational primes invertible in k, and set $R = \mathbb{Z}[S^{-1}]$. By the arithmetic test of C.9, k-homology equivalences are exactly R-homology equivalences: both are detected by $\mathbb{Q}$ and by $\mathbb{F}_p$ for $p \notin S$. If S is empty, C.9 gives $W^{\mathrm{lc}}_{D(k)} = W_\infty$, which is proper. If S is nonempty, consider

$$j : K(\mathbb{Z}, 2) \longrightarrow K(R, 2). \tag{B.7.5}$$

Its homotopy fibre is $K(R/\mathbb{Z}, 1)$. The group $R/\mathbb{Z}$ is the filtered union of finite groups whose orders are invertible in R. Averaging makes their positive R-homology zero, and the bar construction commutes with filtered unions. Thus this fibre is R-acyclic. The Serre homology spectral sequence proves that j is an R-homology equivalence. Both its spaces are simply connected, so C.7 makes it a $W^{\mathrm{lc}}_{D(k)}$-equivalence.

The target is local for R-homology equivalences. Indeed, the homotopy groups of $\mathrm{Map}(X, K(R, 2))$ are $H^{2-i}(X; R)$ for $0 \leq i \leq 2$; these are invariant under R-homology equivalences, by applying derived R-linear Hom into R to the singular-chain equivalence. It is consequently local for the smaller class $W^{\mathrm{lc}}_{D(k)}$. But $\pi_2(j) : \mathbb{Z} \to R$ is not onto. Since C.7 also gives $W^{\mathrm{lc}}_{D(k)} \subseteq W_1$, the preceding obstruction with $n = 1$ proves that this localizer is not proper.

Positive characteristic rings. Let $d = \mathrm{char}(k) > 0$, and let P be the nonempty finite set of primes dividing d. The nonunits among rational primes are precisely the members of P. C.9 identifies k-homology equivalences with simultaneous $\mathbb{F}_p$-homology equivalences for $p \in P$. Put

$$M = \prod_{p \in P} \mathbb{Z}_p, \qquad j : K(\mathbb{Z}, 2) \longrightarrow K(M, 2), \tag{B.7.6}$$

where $\mathbb{Z} \to M$ is diagonal and $\mathbb{Z}_p$ denotes the additive group of p-adic integers, regarded here as a discrete group. For each $p \in P$, multiplication by p is an automorphism of $A = M/\mathbb{Z}$. For surjectivity, given $m \in M$, choose an integer z congruent to its p-component modulo p. Then $(m - z)/p \in M$: division by p is allowed in the other components. For injectivity, if pm is the diagonal integer z, integrality in the p-component implies $p \mid z$, and then m is itself diagonal.

For clarity, an abelian group uniquely divisible by p is $\mathbb{F}_p$-acyclic as a group. Such a group has no p-torsion. Each finitely generated subgroup H is therefore a sum of a free abelian group and a finite group of order prime to p, so $H_*(BH; \mathbb{F}_p)$ is the exterior algebra on its first homology. Enlarge H to the finitely generated subgroup $p^{-1}H$. Its inclusion induces zero on $H_1(-; \mathbb{F}_p)$, since under the isomorphism $p^{-1}H \simeq H$ it is multiplication by p. It therefore induces zero in every positive homology degree. Filtered colimits of the bar complexes prove the assertion. Hence $K(A, 1)$, the homotopy fibre of (B.7.6), is $\mathbb{F}_p$-acyclic for every $p \in P$. The same Serre argument proves that j is in $W^{\mathrm{lc}}_{D(k)}$.

The target in (B.7.6) is local for k-homology equivalences. To verify this rather than assume it, an $\mathbb{F}_p$-homology equivalence induces all $\mathbb{F}_p$-cohomology isomorphisms by vector-space duality. The coefficient short exact sequences

$$0 \longrightarrow \mathbb{Z}/p^{r-1} \longrightarrow \mathbb{Z}/p^r \longrightarrow \mathbb{F}_p \longrightarrow 0$$

and their long exact cohomology sequences imply the same assertion for $\mathbb{Z}/p^r$. Thus $K(\mathbb{Z}/p^r, 2)$ is $\mathbb{F}_p$-homology local. Local spaces are closed under homotopy limits, and the surjective inverse system of these coefficient groups gives

$$\operatorname{holim}_r K(\mathbb{Z}/p^r, 2) \simeq K(\mathbb{Z}_p, 2);$$

the possible derived-limit group is zero by surjectivity. Taking the finite product over P proves the required locality for the simultaneous homology class, and hence for $W^{\mathrm{lc}}_{D(k)}$. The map $\pi_2(j) : \mathbb{Z} \to M$ is not surjective, since M is uncountable. The $n = 1$ obstruction again proves nonproperness. Together with B.4 and (B.7.4), this proves (B.7.1) for every nonzero ring.

The constant homology class. For every nonzero k, it contains $BG \to e$ for the nontrivial acyclic group used in B.6, so it strictly contains W_∞. It preserves components, and the already proved classification therefore refutes (7a,b,c). It is also never proper. In characteristic zero with S empty it equals the integral homology localizer, whose nonproperness is Cisinski (2006, Scholie 9.4.7). When S is nonempty, use $K(\mathbb{Z}, 1) \to K(R, 1)$ and the obstruction with $n = 0$. Its target is R-homology local by the same cohomology argument. Its R-homology is an isomorphism: R is the filtered union of its cyclic subgroups, the bar homology commutes with that union, and the only positive group homology of $\mathbb{Z}$ is in degree one. The map on π_1 is not surjective.

In positive characteristic use $K(\mathbb{Z}, 1) \to K(M, 1)$. The preceding locality proof works in degree one as well. The torsion-free abelian group M is a filtered union of finitely generated free abelian groups, whence

$$H_*(BM; \mathbb{F}_p) \simeq \bigwedge^*_{\mathbb{F}_p} (M/pM) \simeq \bigwedge^*_{\mathbb{F}_p} \mathbb{F}_p \quad (p \in P).$$

The diagonal $\mathbb{Z} \to M$ induces the displayed isomorphism in degree one, and hence is a k-homology equivalence. Again it is not onto on π_1. Lemma 9.4.5 with $n = 0$ proves nonproperness. Accessibility of these homology localizers is B.4, or Cisinski (2006, Theorem 9.4.2 and Corollary 9.4.3). Consequently (7F) fails in every nonzero-ring case.

The intermediate class $\mathcal{W}^1_{\mathbf{D}}$. The fundamental-localizer implications above must not be applied to this class, because it need not satisfy two-out-of-three. We instead check each assertion directly. For variance, put $\mathbf{D}^{\mathrm{op}}(A) = \mathbf{D}(A^{\mathrm{op}})^{\mathrm{op}}$. Then $f \in W^1_{\mathbf{D}}$ if and only if $f^{\mathrm{op}} \in W^1_{\mathbf{D}^{\mathrm{op}}}$. It therefore suffices to prove the assertion for Grothendieck opfibrations and apply that proof to $\mathbf{D}^{\mathrm{op}}$ when the manuscript uses Grothendieck fibrations. This dualizes both the shape and coefficient derivator; it does not assume shape-opposite invariance for a fixed bare derivator. For opfibrations, right Kan extensions in the presheaf convention have the strict-fibre formula of B.3. If $p : E \to S$ has transports in $\mathcal{W}^1_{\mathbf{D}}$, then p_* sends locally constant objects to locally constant objects, and does so after every base change. Indeed, each transport is in the cohomological-equivalence class by A.3; the comparison of the two fibre cohomology objects is an isomorphism for every locally constant coefficient. Pointwise Kan extension gives exactly this comparison on the base arrow. Consequently p^* and p_* restrict to an adjunction on locally constant categories.

If also $p \in \mathcal{W}^1_{\mathbf{D}}$, the unit of this adjunction is an isomorphism. Apply it to the constant object of value $M \in \mathbf{D}(e)$ and evaluate at $s \in S$. The strict-fibre formula gives

$$M \xrightarrow{\sim} (p_{E_s})_* p^*_{E_s} M. \tag{B.7.8}$$

Thus $E_s \to e$ is a constant-coefficient equivalence, equivalently a member of $\mathcal{W}^1_{\mathbf{D}}$ for this point target, by A.3. Every base change of p is again an opfibration with the same fibres. The fully faithful case of B.3 proves that each such base change belongs to $\mathcal{W}^1_{\mathbf{D}}$. This is literal Loc (7c), without any two-out-of-three argument.

Under the enhancement, C.2 identifies $\mathcal{W}^1_{\mathbf{D}}$ with the cohomological class. The all-local-coefficients direct-image property just proved and its Beck–Chevalley formula make p universally preserve that class, by the adjunction argument of D.2. Explicitly, for $f : T \to S$ in the cohomological class and $F \in \mathrm{LC}(E)$, cohomology on E equals cohomology on S with coefficient p_*F. This coefficient is locally constant, so f preserves it; base change identifies the result with the comparison on $T \times_S E$. This proves Loc (7a) for $\mathcal{W}^1_{\mathbf{D}}$ in the enhanced case. For the factorization assertion, factor any functor in the ordinary Thomason model structure. Its first map is in $W_\infty \subseteq \mathcal{W}^1_{\mathbf{D}}$. Its second map is an ordinary sharp fibration, whose strict base changes compute homotopy pullbacks. C.2 makes $\mathcal{W}^1_{\mathbf{D}}$ stable under homotopy pullback, so this second map is in $\mathrm{Fib}_{\mathcal{W}^1_{\mathbf{D}}}$. This proves the asserted factorization, without claiming the Brown axioms that require two-out-of-three.

Finally choose a pointed category Z with $Z \to e \in \mathcal{W}^1_{\mathbf{D}}$ and $e \to Z \notin \mathcal{W}^1_{\mathbf{D}}$. Let $R = Z \star \{t\}$ and $p : R \to [1]$ be the Grothendieck opfibration of $Z \to e$. The chosen point gives a section $s : [1] \to R$. Both structural maps p and $\mathrm{id}_{[1]}$ have transports in $\mathcal{W}^1_{\mathbf{D}}$. The section is a weak homotopy equivalence because both total categories have terminal objects, but its fibre at 0 is the excluded map $e \to Z$. Taking opposites gives the literal Grothendieck-fibration version. This refutes (7b), not (7c): the structural map p itself is universally in $\mathcal{W}^1_{\mathbf{D}}$ by (B.7.8). For r-types take a category model of $K(A, r+1)$ with $A \neq 0$; for groupoids use $r = 1$. For nonzero stable coefficients take $Z = BG$ for the nontrivial integrally acyclic group of B.10: its suspension is contractible, so constants satisfy the fibre criterion C.2, whereas $e \to BG$ fails full faithfulness by the regular-action argument in B.6. For a nontrivial thin coefficient category take $Z = e \amalg e$ and use (B.6.6). All these choices meet the two stated tests. For spaces $\mathcal{W}^1_{\mathbf{D}} = W_\infty$ and (7b) holds. □

Here is the resulting comparison for the computed theories. The zero coefficient theory is understood separately: all its arrows are equivalences and all listed axioms hold.

Class	Loc (6b,c)	Loc (7a,b,c)	Loc (7F)
W^{lc} for spaces, $D(\mathbb{Z})$, spectra	no	yes	yes
W^{lc} for sets, groupoids, finite r-types	no	no	yes
$W^{\mathrm{lc}}_{D(k)}$, $k \neq 0$, all primes nonunits	no	yes	yes
$W^{\mathrm{lc}}_{D(k)}$, $k \neq 0$, some prime a unit	no	no	no
Constant k-homology, $k \neq 0$	yes	no	no

For sets, groupoids and finite r-types the last column uses C.4 and the properness of every finite truncation localizer W_n, explicitly established in Cisinski (2006, §9.2.1). Thus (7F) does not imply (7a,b,c). For a general spectrum E, B.4 makes $W^{\mathrm{lc}}_{L_E\mathsf{Sp}}$ accessible, so the exact remaining test for (7F) is

$$W^{\mathrm{lc}}_{L_E\mathsf{Sp}} = (W^{\mathrm{lc}}_{L_E\mathsf{Sp}})_{\mathrm{asph}}, \tag{B.7.7}$$

equivalently properness. No assertion that (B.7.7) holds for every E is made; its evaluation beyond the coefficient theories computed here remains unresolved in this work. For an arbitrary derivator without an accessibility theorem, (7F) still implies aspherical generation by (B.7.4); the converse is not claimed without that hypothesis.

For an arbitrary unenhanced derivator, two intermediate-class questions remain unresolved here: whether every Grothendieck fibration with transports in $\mathcal{W}^1_{\mathbf{D}}$ belongs to $\mathrm{Fib}_{\mathcal{W}^1_{\mathbf{D}}}$ (literal Loc (7a)), and whether every functor factors as a member of $\mathcal{W}^1_{\mathbf{D}}$ followed by a member of $\mathrm{Fib}_{\mathcal{W}^1_{\mathbf{D}}}$ (the factorization assertion of Loc (7F)). The first argument above proves universal preservation of the cohomological class; identifying its output with $\mathcal{W}^1_{\mathbf{D}}$ used the enhancement in C.2. The second argument used the enhanced homotopy-fibre criterion to prove homotopy-pullback stability of $\mathcal{W}^1_{\mathbf{D}}$. Neither of these two extra steps is supplied just by the arbitrary-derivator

relative Theorem A of B.3. This precisely limits the two positive enhanced assertions; it does not leave (7c) unresolved.

Finally, the existence of the model-fibration class in D.6 is compatible with all the negative entries. An accessible fundamental localizer always has the Thomason model structure. Its model fibrations compose, are stable under strict base change, and detect fibre equivalences between fibrant objects of a slice. When the localizer is not proper, they need not belong to the manuscript's Fib_W. Replacing that distinction by an identification would incorrectly turn D.6 into Loc (7F).

2.3 A one-object test for truncation localizers

B.8. The sphere test, without accessibility. [Known]

For every fundamental localizer $\mathcal{W}$ and every integer $n \geq 0$,

$$\mathcal{W}_n \subseteq \mathcal{W} \quad \Longleftrightarrow \quad \big(\Delta/\partial\Delta^{n+2} \longrightarrow e\big) \in \mathcal{W}. \tag{B.8.1}$$

No accessibility hypothesis is needed. In particular, this applies to $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ for every derivator, by B.3. For $\mathcal{C} = \mathcal{S}_{\leq r}$ it holds exactly when $n \geq r+1$. For $\mathcal{C} = D(k)$ with $k \neq 0$, or for ordinary spectra, it holds for no finite n.

Proof. The equivalence is Cisinski (2006, Proposition 9.2.17), which we checked in the original text. The category of simplices on the right has classifying space S^{n+1}. C.4 identifies the truncated case with $\mathcal{W}_{r+1}$; inclusion $\mathcal{W}_n \subseteq \mathcal{W}_{r+1}$ is equivalent to $n \geq r+1$. In the linear cases C.7 and C.10 detect S^1 by its fundamental group, and detect every higher sphere by the nonzero top reduced homology of its universal cover with the stated coefficients. These are computations of the specified projection, so they prove the last assertion. TOTASPH F.28 is confirmed in precisely this form. □

2.4 Acyclic epimorphisms and cardinal-controlled cones

B.9. An acyclic map is an epimorphism in the unpointed homotopy category. [Proved]

Let $t : X \to Y$ be a map of small CW spaces whose homotopy fibres are integrally acyclic, where "acyclic" includes nonempty and connected. Then for every small space Z the map of unpointed homotopy classes

$$[Y, Z] \longrightarrow [X, Z] \tag{B.9.1}$$

induced by t is injective. The homotopy codiagonal $c : Y \cup_X^h Y \to Y$ is a weak homotopy equivalence.

Proof. First suppose Y is connected. The homotopy-fibre sequence makes X connected and $\pi_1(t)$ surjective. Put $P = Y \cup_X^h Y$, realized by the double mapping cylinder, with its two maps $j_1, j_2 : Y \to P$ and its fold $c : P \to Y$. Van Kampen identifies $\pi_1 P$ with $\pi_1 Y *_{\pi_1 X} \pi_1 Y$. Since $\pi_1(t)$ is onto, the fold of this amalgam is an isomorphism onto $\pi_1 Y$.

An acyclic map induces an isomorphism in homology with every local system on its target; this is the equivalence of (a) and (b) in Hausmann and Husemoller (1979, Definition/Proposition 1.2). For a local system L of abelian groups on Y, compare the Mayer–Vietoris sequence of P with that of the diagram $Y \leftarrow Y \to Y$, using t at the central term and identities at the two outer terms. The three component maps are homology isomorphisms with the pulled-back coefficients.

The induced map $H_*(P; c^*L) \to H_*(Y; L)$ is therefore an isomorphism. Take $L = \mathbb{Z}[\pi_1 Y]$. As $\pi_1 c$ is an isomorphism, this is the integral homology comparison between the universal covers. Those covers are simply connected CW spaces. Homological Whitehead, followed by the long exact sequences of the coverings, makes c a weak equivalence.

Since $cj_1 = cj_2 = \mathrm{id}_Y$, we have $j_1 = j_2$ in the unpointed homotopy category. If $a, b : Y \to Z$ and at is homotopic to bt, that actual homotopy defines $h : P \to Z$ on the double mapping cylinder. Consequently $[a] = [h][j_1] = [h][j_2] = [b]$. For disconnected Y, acyclicity gives a bijection on components, and the argument applies on each component. The product of these injective maps of sets is injective; this is a product of unpointed homotopy-class sets of spaces, not an interchange of an infinite product with components of ordinary functor categories. The empty case is immediate. □

B.10. Acyclic simple overgroups of controlled cardinality. [Proved]

Let H be a small group and let λ be an infinite small cardinal with $\lambda \geq |H|$. There is an embedding $H \hookrightarrow A$ into a nonabelian simple integrally acyclic group with $|A| = \lambda$. Consequently, if $\lambda > \mu$, every homomorphism $A \to L$ to a group of cardinality at most μ is trivial.

Proof. We give the two envelope constructions and then alternate them. For the simple envelope, start with $H * F_\lambda$, so that the cardinality is exactly λ. For each pair $g \neq 1, h$ in a current group L of cardinality λ, adjoin a separate infinite cyclic free generator $t_{g,h}$ and put, in $L * F(\{t_{g,h}\})$,

$$w_{g,h} = g t_{g,h} g^{-1} t_{g,h}^{-1} g t_{g,h}^{2} g^{-1} t_{g,h}^{-2}. \tag{B.10.1}$$

Both $w_{g,h}$ and $w_{g,h}h$ have infinite order. The first is cyclically reduced of length eight in the relevant free product. For the second, cyclic reduction combines its first and last L-syllables; the result has length eight if $hg \neq 1$ and length six if $hg = 1$, with every syllable nontrivial. Thus we may adjoin a stable letter $s_{g,h}$ satisfying

$$s_{g,h}^{-1} w_{g,h} s_{g,h} = w_{g,h} h. \tag{B.10.2}$$

The simultaneous HNN embedding theorem (Higman, Neumann, & Neumann, 1949, Theorem II) embeds the old group in this extension, because the identified cyclic subgroups are infinite cyclic. Equation (B.10.2) puts h in the normal closure of g. Iterate this step countably. In the union any nonidentity g and any h occur together at some stage; hence the normal closure of g contains h. The union is simple, is nonabelian because it contains F_λ, and has cardinality λ.

For clarity we also verify the needed acyclic envelope. For a group L, start with $L \times L$ and adjoin stable letters d, s such that, writing $j(h) = (h, 1)$ and $k(h) = (1, h)$,

$$d^{-1} j(h) d = j(h) k(h), \qquad s^{-1} j(h) s = k(h). \tag{B.10.3}$$

The three copies of L being identified are embedded copies, so HNN again embeds $L \times L$. Denote the resulting group by $m(L)$, with embedding $j : L \to m(L)$. This is the mitosis construction underlying Baumslag, Dyer, and Heller (1980, Proposition 4.1 and Theorem 4.2). Its countable iteration is acyclic, as follows. Over any field F, suppose $f : H_0 \to L$ induces zero in positive homology degrees below n. In the Künneth decomposition of $H_n(L \times L; F)$, the diagonal after f has only the two axis terms: the other terms contain a positive lower-degree factor killed by f. Under (B.10.3), its two axis maps are both conjugate to j, and its diagonal map is also conjugate to j. Hence

$$(jf)_* = (jf)_* + (jf)_* \quad \text{on } H_n(H_0; F),$$

so $(jf)_* = 0$. Induction shows that the map from any stage to its n-th successor kills positive homology through degree n. Every bar cycle has finite support, so the union has zero positive

homology over every field. Its positive integral homology groups are torsion by rational coefficients; the universal coefficient sequence in the next degree with $\mathbb{F}_p$ coefficients says they have no p-torsion, for every prime p. They are therefore zero. This construction preserves the bound $\max(|L|, \aleph_0)$.

Starting from $H * F_\lambda$, form a simple envelope, then an acyclic envelope, then a simple envelope, and so on, always using embeddings. Let A be the union. A nontrivial normal subgroup of A contains some $g \neq 1$; given any $h \in A$, both lie in a later simple stage, whose simplicity puts h in that normal subgroup. Thus A is simple. Every finite bar cycle lies in a later acyclic stage and bounds there, so A is acyclic. Its cardinality is exactly λ. Finally the kernel of a homomorphism from a simple group is trivial or the whole group. An injection into a group of size $\mu < \lambda$ is impossible, proving the last claim. □

B.11. The cardinal-controlled Kan–Thurston statement. [Proved]

Let Y be a connected $\mathbb{U}$-small CW complex and let μ be a cardinal belonging to $\mathbb{U}$. There are a $\mathbb{U}$-small group G and an acyclic map $t : BG \to Y$ such that $\phi = \pi_1(t) : G \twoheadrightarrow \pi_1 Y$ is onto, its kernel P is perfect, and every homomorphism $G \to L$ with $|L| \leq \mu$ vanishes on P. Base points are chosen only to write ϕ; acyclicity and the conclusion about the normal subgroup do not depend on their choice.

Proof. Choose a simplicial complex K and a homotopy equivalence $h : |K| \to Y$; the existence is Hatcher (2002, Theorem 2C.5). Choose an infinite cardinal $\lambda \in \mathbb{U}$ larger than μ and at least the number of simplices of K. Such a cardinal exists inside the fixed Grothendieck universe. We construct, simultaneously for all subcomplexes $L \subseteq K$, a CW space $T(L)$ and a map $t_L : T(L) \to |L|$, so that inclusions are actual CW inclusions, every component of $T(L)$ is aspherical, and every inclusion $T(L') \hookrightarrow T(L)$ is injective on the fundamental group of each component. All choices attached to a simplex are made once and used in every subcomplex containing it.

In dimension at most one, put $T(L) = |L|$ and $t_L = \mathrm{id}$. Graphs are aspherical, and inclusions of subgraphs induce injections on fundamental groups, as is seen from reduced edge paths after choosing maximal forests. Suppose the construction and its inclusion properties have been made through dimension $n-1$, with $n \geq 2$. For each n-simplex σ of K, the space $T(\partial\sigma)$ is connected and aspherical. Its fundamental group embeds, by B.10, in an acyclic nonabelian simple group C_σ of cardinality λ. Realize this embedding by a cellular map

$$a_\sigma : T(\partial\sigma) \longrightarrow BC_\sigma. \tag{B.11.1}$$

The realization is possible because both spaces are connected $K(\pi, 1)$ spaces; one can send their universal covers equivariantly into the contractible universal cover of BC_σ, extending over cells. Let Q_σ be its mapping cylinder, with $T(\partial\sigma)$ at the free end. Thus Q_σ deformation retracts onto BC_σ, $T(\partial\sigma) \hookrightarrow Q_\sigma$ is a CW cofibration, and this inclusion is injective on π_1. For a subcomplex L of dimension at most n, put

$$T(L) = T(L^{(n-1)}) \cup_{\coprod_{\sigma \in L_n} T(\partial\sigma)} \coprod_{\sigma \in L_n} Q_\sigma. \tag{B.11.2}$$

On the cylinder of (B.11.1), define t_L by moving $t_{\partial\sigma}(x)$ linearly inside σ to its barycentre as the cylinder coordinate runs from its free end to BC_σ. Send all of BC_σ to that barycentre. This defines an actual continuous map extending the one on the lower skeleton, and it commutes with all subcomplex inclusions. All these spaces have at most λ cells: the bar models of the groups do, and finite products and a countable union preserve this infinite cardinal bound.

We justify both induction assertions, including the subcomplex injectivity which is needed later. We use the following elementary graph-of-spaces argument. If connected aspherical vertex spaces are joined by mapping cylinders of aspherical edge spaces, and both edge homomorphisms are injective, the normal form for amalgamated products and HNN extensions embeds all vertex groups. In the universal cover the incidence graph is the corresponding tree of cosets. Vertex and edge inverse images are contractible. Every finite subtree of this lifted graph of spaces is contractible by successive attachment of a contractible vertex space along a contractible edge cylinder. A map from a sphere has compact image in a finite subcomplex, and hence lies over a finite subtree. It is nullhomotopic there. The universal cover is therefore weakly contractible, and, being a CW space, contractible. This proves asphericity as well as injectivity; disconnected initial spaces are handled component by component. It is the standard argument underlying the amalgamation statement in Baumslag et al. (1980, §1, before Proposition 1.1).

Apply this first to (B.11.2). Each edge group injects into the lower-skeleton group by the induction hypothesis and into C_σ by its construction. It follows that $T(L)$ is aspherical on every component and that its lower skeleton and all BC_σ inject on π_1. For the stronger assertion, let $L \subseteq K$ with both complexes of dimension at most n. Form first

$$V = T(K^{(n-1)}) \cup_{T(L^{(n-1)})} T(L). \tag{B.11.3}$$

This is a graph of spaces with one edge for each component of $T(L^{(n-1)})$. Its edge injections into $T(K^{(n-1)})$ are the dimension-$n-1$ induction assertion; its edge injections into $T(L)$ were proved in the preceding sentence. The argument just given embeds both vertex groups in each resulting component of V. Next attach the Q_σ for $\sigma \in K_n \setminus L_n$. Each attaching subgroup injects into $T(K^{(n-1)})$, which already injects into V and continues to inject at every attachment. The final space is exactly $T(K)$: both constructions use the same pushouts and the same chosen Q_σ. Thus $T(L) \to T(K)$ is injective on fundamental groups. This proves the missing subcomplex step without supposing that an arbitrary map between aspherical spaces is injective.

Take the union over dimensions. Compactness of spheres and homotopies in CW complexes proves the same assertions for arbitrary subcomplexes of an infinite-dimensional K. The construction does not introduce new components, so $T(K)$ is connected when K is. We now check the exact homological property, not merely constant homology. For every local system M of abelian groups on $|L|$, induction on dimension gives

$$H_*(T(L); t_L^* M) \xrightarrow{\cong} H_*(|L|; M). \tag{B.11.4}$$

Indeed, in the relative pair for an n-simplex the target system is constant because σ is contractible. The pair on the source is $(Q_\sigma, T(\partial\sigma))$. Its long exact homology sequence, acyclicity of $Q_\sigma \simeq BC_\sigma$, and the induction comparison for $T(\partial\sigma) \to \partial\sigma$ identify its relative homology with that of $(\sigma, \partial\sigma)$, with every constant coefficient group. Excision gives the direct sum of these relative comparisons for (B.11.2). The long exact sequence of the skeletal pair proves (B.11.4). Cellular chains with local coefficients commute with the filtered union over skeleta, which proves it in unbounded dimension as well.

Write $G = \pi_1 T(K)$ and let N be the normal closure in G of the embedded groups C_σ, for all simplices of dimension at least two. The map t_K kills each of these groups. Moreover

$$G/N \xrightarrow{\cong} \pi_1 |K|. \tag{B.11.5}$$

To verify the specified map, use van Kampen in (B.11.2). At dimension two, killing C_σ kills precisely the image of $\pi_1 T(\partial\sigma) = \pi_1 \partial\sigma$, the usual boundary relator of that two-simplex. At a dimension $n \geq 3$, the inductive version of (B.11.5) for $\partial\sigma$ says that its group is normally generated by its lower-dimensional cone groups, because $\pi_1 \partial\sigma = 0$. Those groups have already

been killed. No further relation in the quotient is therefore introduced. This proves (B.11.5) by induction and then by filtered union. In particular $\pi_1(t_K)$ is onto and its kernel is exactly N. Each C_σ is perfect; its conjugates generate N, and each such conjugate lies in $[N, N]$. Hence N is perfect.

Since $T(K)$ is a connected aspherical CW space, choose a based homotopy equivalence $b : BG \to T(K)$ inducing the identity on G, and put $t = h\, t_K\, b$. Equation (B.11.4) and Hausmann and Husemoller (1979, Definition/Proposition 1.2) show that this specified map is acyclic. Its kernel is $P = N$, transported by the chosen base-point identification. Every homomorphism $\rho : G \to L$ with $|L| \leq \mu$ restricts trivially to each C_σ, since that group is simple and has cardinality $\lambda > \mu$. Thus ρ kills their normal closure P. If K has dimension at most one, there are no cone groups, $P = 1$, and the same conclusion holds. This proves all the stated cases. □

B.12. Small ind-objects modulo natural transformations see at most one-types. [Proved]

Let $\mathcal{H}$ be the category of $\mathbb{U}$-small categories whose morphisms are functors modulo zigzags of natural transformations. Let I be a nonempty $\mathbb{U}$-small filtered category, and let $K : I \to \mathcal{H}$ be a diagram, represented by categories K_i. Suppose the presheaf

$$H_K(X) = \operatorname*{colim}_{i \in I} \operatorname{Hom}_{\mathcal{H}}(X, K_i) = \operatorname*{colim}_{i} \pi_0 \operatorname{Fun}(X, K_i) \tag{B.12.1}$$

sends every functor in $\mathcal{W}_\infty$ to a bijection. Then it sends every functor in $\mathcal{W}_1$ to a bijection. The specified unit of the reflector induces an isomorphism

$$(K_i)_{i \in I} \longrightarrow (\Pi_1 K_i)_{i \in I} \quad \text{in } \operatorname{Ind}_{\mathbb{U}}(\mathcal{H}). \tag{B.12.2}$$

In particular the small-index version of CATHOT E.17 is settled, with its original indexing category unchanged.

Proof. We first identify exactly the homotopy-invariant functor which (B.12.1) defines. The functor $q : \mathcal{H} \to \mathsf{Hot}$ is localization at nerve weak equivalences: a natural transformation is a functor on $X \times [1]$, and the two endpoint inclusions become equal after inverting its projection to X. Thus the localization of Cat factors through $\mathcal{H}$ and has the same universal property there. By hypothesis H_K is the pullback of a presheaf H' on Hot. There is a canonical identification

$$H'(Y) \cong \operatorname*{colim}_{i} [Y, BK_i]. \tag{B.12.3}$$

To justify this identification rather than assume it at each stage, use a further universe for presheaf categories. Restriction q^* is fully faithful because q is a localization. Its left adjoint, left Kan extension along q^{op}, sends the representable $\operatorname{Hom}_{\mathcal{H}}(-, K_i)$ to the representable $\operatorname{Hom}_{\mathsf{Hot}}(-, BK_i)$, as the Yoneda lemma verifies on every test presheaf. This left adjoint preserves colimits. Its unit is invertible on a presheaf already factoring through q. Applying these facts to (B.12.1) proves (B.12.3) with its natural comparison map.

Choose a small infinite cardinal μ bounding the cardinality of the arrows of every K_i. Such a bound exists because the index category and all its categories are small in the same universe. Let $Y = \coprod_{c \in \pi_0 Y} Y_c$ be any small CW space, choose a base point in each component, and apply B.11 to obtain acyclic maps $t_c : BG_c \to Y_c$. Put $T = \coprod_c BG_c$ and $t = \coprod_c t_c$. Write $\tau_1 Y = \coprod_c B\pi_1 Y_c$ and let $v : Y \to \tau_1 Y$ be its canonical first truncation in Hot. The composite $vt : T \to \tau_1 Y$ is represented by the quotient homomorphisms $G_c \twoheadrightarrow \pi_1 Y_c$.

For every i, restriction along that quotient gives an *isomorphism of ordinary functor categories*

$$\mathrm{Fun}\left(\coprod_c B\pi_1 Y_c, K_i\right) \xrightarrow{\cong} \mathrm{Fun}\left(\coprod_c BG_c, K_i\right). \tag{B.12.4}$$

A functor from BG_c chooses an object k and a homomorphism $G_c \to \mathrm{Aut}_{K_i}(k)$. The target group has cardinality at most μ, so this homomorphism kills $\ker \pi_1(t_c)$ by B.11 and factors uniquely through $\pi_1 Y_c$. A natural transformation is a family of arrows of K_i commuting with the group actions. Since the quotient is onto, commuting with all G_c actions is equivalent to commuting with all $\pi_1 Y_c$ actions. Thus objects and morphisms factor uniquely, proving (B.12.4) even with noninvertible endomorphisms in K_i and with infinitely many components. We take components only after proving this actual isomorphism; no invalid commutation of π_0 with an infinite product is used.

Using (B.12.3), the resulting bijection factors as

$$H'(\tau_1 Y) \xrightarrow{H'(v)} H'(Y) \xrightarrow{H'(t)} H'(T). \tag{B.12.5}$$

The second map is injective. Indeed B.9 says that $[Y, BK_i] \to [T, BK_i]$ is injective for every i. Filtered colimits of compatible injections of sets are injective: an equality in the target colimit holds at some common stage, where injectivity applies. The composite in (B.12.5) is bijective by (B.12.4). If $b \circ a$ is bijective and b is injective, then a is bijective: injectivity is immediate, and for y choose x with $ba(x) = b(y)$ and cancel b. Thus $H'(v)$ is bijective. For $Y = BX$, the checked CATHOT C.1–C.4 identify this comparison with $H_K(\Pi_1 X) \to H_K(X)$ induced by the specified unit. It follows that H_K inverts every $\mathcal{W}_1$ map, since Π_1 sends those maps to equivalences of groupoids.

We finish by checking that the conclusion is the canonical ind-isomorphism (B.12.2). The supplied reflector $\Pi_1 : \mathcal{H} \to \mathcal{G}$, where $\mathcal{G}$ is the full reflected subcategory of groupoids, induces an adjunction on small ind-categories. This follows directly from

$$\mathrm{Hom}_{\mathrm{Ind}(\mathcal{H})}((A_i), (B_j)) = \lim_i \operatorname*{colim}_j \mathrm{Hom}_{\mathcal{H}}(A_i, B_j) \tag{B.12.6}$$

and the reflector adjunction on its individual Hom sets. Write $L(K) = \mathrm{Ind}(\Pi_1)(K)$ and $\eta : K \to L(K)$ for its unit. The property just proved is orthogonality of K to every $X \to \Pi_1 X$. Taking the limit over i of these bijections produces a morphism $r : L(K) \to K$ with $r\eta = \mathrm{id}_K$. The object $L(K)$ has the same orthogonality property: each target $\Pi_1 K_j$ does by reflection, and filtered colimits preserve the resulting bijections of sets. Consequently precomposition with η is injective on maps $L(K) \to L(K)$. Since $(\eta r)\eta = \eta = \mathrm{id}_{L(K)}\eta$, it follows that $\eta r = \mathrm{id}_{L(K)}$. This proves (B.12.2).

If Y is empty, (B.12.4) is the isomorphism between two terminal functor categories, so the argument includes that case. No uniform bound on the sizes of arbitrary test categories X was used; only the one bound on the small family (K_i) was used. An index family unbounded in the original universe lacks this bound and is not covered by the statement. Thus the large-index examples retained in the CATHOT discussion do not contradict it. □

The construction in B.11 follows the simplicial-cone strategy of Kan–Thurston proofs, but the cardinal and subcomplex arguments are supplied above. We searched for Maunder's 1981 proof, for cardinal refinements of Kan–Thurston, and for acyclic simple envelopes. We verified the HNN and mitosis sources used in B.10 and the acyclic-map criterion used in B.9–B.11; we did not obtain Maunder's full text. Consequently no assertion about what his proof establishes is used as a premise, and no priority claim is attached to B.11 or B.12.

B.13. Accessibility with only the abstract derivator axioms. [Open]

For an arbitrary derivator $\mathbf{D} : \mathsf{Cat}_{\mathbb{U}}^{\mathrm{coop}} \to \mathrm{CAT}_{\mathbb{V}}$ satisfying Der 1–4, must $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ be generated by a $\mathbb{U}$-small set of functors as a fundamental localizer? Must its full subcategory of $\mathsf{Cat}_{\mathbb{U}}^{[1]}$ satisfy the literal Loc (9)? We likewise leave the corresponding accessibility assertion for $\mathcal{W}_{\mathbf{D}}$ and the literal accessibility assertion for $\mathcal{W}_{\mathbf{D}}^{1}$ unresolved in this full generality. B.4 answers the first two questions positively for every $\mathbb{U}$-presentable enhanced coefficient category, and answers all three for each concrete coefficient theory in Part C.

The obstruction to extending B.4 is the missing uniform small family of tests or a uniform bound on witnesses. Der 1–4 and B.3 give Kan extensions and relative descent, but do not in our proof supply such a generating family. For the constant and locally constant classes, the first untreated scope is an abstract derivator with no supplied presentable enhancement or small homotopical generating family. For $\mathcal{W}_{\mathbf{D}}^{1}$, accessibility is also unproved here for a general presentable coefficient category outside the concrete families of Part C: the accessible copower argument for equivalences has not been extended here to its fully faithful arrows. No explicit counterexample is supplied to either assertion. Under Vopěnka's principle, the set-generation questions for $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$ and $\mathcal{W}_{\mathbf{D}}$ are positive by Cisinski (2006, Scholie 4.2.17). We neither assume that principle nor infer an unconditional counterexample from the lack of a proof. The exact unresolved step is to produce a single $\lambda \in \mathbb{U}$ and a $\mathbb{U}$-small family of arrows generating the stated classes; merely knowing their filtered-colimit closure does not do so.

B.14. The remaining general fibration questions. [Open]

For an arbitrary Der 1–4 derivator without an enhancement, we leave unresolved whether every Grothendieck fibration whose transports belong to $\mathcal{W}_{\mathbf{D}}^{1}$ universally preserves $\mathcal{W}_{\mathbf{D}}^{1}$ under strict pullback, and whether every small functor factors as a member of $\mathcal{W}_{\mathbf{D}}^{1}$ followed by such a universally $\mathcal{W}_{\mathbf{D}}^{1}$-preserving map. These are respectively literal Loc (7a) and the factorization assertion of Loc (7F); Loc (7c) is proved in B.7. For the accessible localizer $W_{L_E\mathsf{Sp}}^{\mathrm{lc}}$ of a general nonzero spectrum E, we also leave its properness, equivalently its Loc (7F), unevaluated beyond the cases proved in B.7.

The first two gaps are exactly the steps isolated after B.7: the direct-image argument gives cohomological invariance, while the identification with full faithfulness and the homotopy-fibre criterion have only been proved here with the enhancement of C.2. The first untreated coefficient theories have no specified such enhancement; no counterexample derivator is supplied. For the spectrum question the exact necessary and sufficient condition is (B.7.7), namely generation by the aspherical projections already in the class. The loop-homology test of C.11 decides membership but does not prove that generation statement. This evaluation is not supplied here already for $E = K(1)$ at a fixed prime. We do not claim that these residual questions are open in the literature.

3 Coefficient calculations

3.1 The homotopy-category and enhancement tests

Throughout this part, $S = BX$, $T = BY$, and $v = Bf : S \to T$ are spaces in $\mathbb{U}$. A point means a map from the one-point space; all fibres of maps of spaces are homotopy fibres. The coefficient ∞-category $\mathcal{C}$ is locally $\mathbb{U}$-small, possibly $\mathbb{V}$-large in objects, and has $\mathbb{U}$-small limits and colimits. We use the derivator $I \mapsto \mathrm{Ho}\,\mathrm{Fun}(NI^{\mathrm{op}}, \mathcal{C})$. Inverting the arrows of NI identifies its locally constant objects with $\mathrm{Ho}\,\mathrm{Fun}(S, \mathcal{C})$; reversing paths accounts for the displayed contravariance. The results below concern the specified restriction functor, not an unspecified equivalence between abstract categories.

C.1. Ordinary full faithfulness detects the enhanced restriction. [PROVED]

For every map $v : S \to T$, the functor $\mathrm{Ho}(v^*) : \mathrm{Ho}\,\mathrm{Fun}(T, \mathcal{C}) \to \mathrm{Ho}\,\mathrm{Fun}(S, \mathcal{C})$ is fully faithful if and only if v^* is fully faithful as an ∞-functor. It is an equivalence if and only if v^* is an equivalence of ∞-categories.

Proof. For $M, N \in \mathrm{Fun}(T, \mathcal{C})$ and any $\mathbb{U}$-small space K, the pointwise cotensor N^K exists and $v^*(N^K) \simeq (v^*N)^K$. The tensor–cotensor adjunction identifies

$$\mathrm{Hom}_{\mathrm{Ho}\,\mathrm{Fun}(T,\mathcal{C})}(M, N^K) = [K, \mathrm{Map}_{\mathrm{Fun}(T,\mathcal{C})}(M, N)].$$

Full faithfulness of $\mathrm{Ho}(v^*)$ therefore says that the specified mapping-space map induces bijections on $[K, -]$ for every K. Yoneda in the homotopy category of spaces makes this map a weak equivalence. Essential surjectivity is the same condition before and after passing to the homotopy category. The converse implications follow directly. The mapping spaces are $\mathbb{U}$-small by our standing local-smallness convention. No assertion about arbitrary functors between arbitrary homotopy categories is being made. □

C.2. The middle three classes coincide for enhanced coefficients. [PROVED]

For the above coefficient theory, put $F_y = S \times^h_T \{y\}$ for each point y of T. The following conditions on v are equivalent:

$$v \in W^1_{\mathcal{C}}, \qquad v \in W^{\mathrm{coh}}_{\mathcal{C}}, \qquad v \in W^{\mathrm{hom}}_{\mathcal{C}}, \qquad F_y \otimes M \longrightarrow M \text{ is an equivalence for every } y, M.$$

Here $M \in \mathcal{C}$, $K \otimes M = \mathrm{colim}_K M$, and the displayed map is induced by $F_y \to *$. Equivalently, $M \to M^{F_y}$ is an equivalence for every M. The assertion includes empty spaces and does not assume a symmetric monoidal structure on $\mathcal{C}$.

Proof. The left Kan extension $v_!$ is left adjoint to v^*. At y, the counit on $N \in \mathrm{Fun}(T, \mathcal{C})$ is

$$(v_! v^* N)_y \simeq F_y \otimes N_y \longrightarrow N_y.$$

Indeed an object of the indexing fibre is $(x, \gamma : v(x) \to y)$, and transport along γ identifies the diagram being integrated with the constant diagram at N_y. Hence the last condition is equivalent to full faithfulness of v^*, by the counit criterion and C.1.

Write $p : S \to *$, $q : T \to *$. Full faithfulness gives, for every N and M, equivalences of mapping spaces

$$\mathrm{Map}(q^*M, N) \simeq \mathrm{Map}(p^*M, v^*N), \qquad \mathrm{Map}(N, q^*M) \simeq \mathrm{Map}(v^*N, p^*M).$$

Adjunction and Yoneda identify these with the cohomology comparison $q_*N \to p_*v^*N$ and homology comparison $p_!v^*N \to q_!N$.

Conversely let $i_y : * \to T$ denote the selected point. If all homology comparisons are equivalences, take $N = (i_y)_!M$. The homotopy pullback square defining F_y, together with the pointwise Kan formula, identifies that comparison with $F_y \otimes M \to M$. If all cohomology comparisons are equivalences, take $N = (i_y)_*M$; it becomes $M \to M^{F_y}$. Tensor and cotensor with F_y are adjoint, and a natural transformation from tensor to the identity is invertible exactly when its adjoint mate from the identity to cotensor is invertible. This proves the claimed equivalences. With $T = \varnothing$ the conditions are vacuous and necessarily $S = \varnothing$. If a nonempty component of T is missed, the corresponding fibre is empty and the condition is $0 \to M$ for all M, forcing a trivial coefficient category. □

C.3. Constant coefficients are tensor tests. [PROVED]

For every $v : S \to T$ one has $v \in W_{\mathcal{C}}$ if and only if the canonical map $S \otimes M \to T \otimes M$ is an equivalence in $\mathcal{C}$ for every $M \in \mathcal{C}$.

Proof. For constants M, N, the set of morphisms between them over S is $\mathrm{Hom}_{\mathrm{Ho}\,\mathcal{C}}(S \otimes M, N)$. Thus the definition of $W_{\mathcal{C}}$ asserts that the displayed map induces bijections into every N. Yoneda in $\mathrm{Ho}\,\mathcal{C}$ makes it an isomorphism, equivalently an equivalence in $\mathcal{C}$. This also proves the converse and handles all components simultaneously. □

3.2 Sets, groupoids, truncated spaces, and spaces

C.4. The truncation calculation. [PROVED]

Let $r \geq 0$ be an integer and let $\mathcal{C} = \mathcal{S}_{\leq r}$ be the ∞-category of r-truncated spaces. Then

$$W_{\mathcal{C}} = W_r, \qquad W^{\mathrm{lc}}_{\mathcal{C}} = W_{r+1}, \qquad W^{\mathrm{coh}}_{\mathcal{C}} = W^{\mathrm{hom}}_{\mathcal{C}} = W^1_{\mathcal{C}}.$$

Membership in the common middle class is equivalent to

$$\pi_0 v \text{ bijective}, \qquad \pi_i v \text{ isomorphic } (1 \leq i \leq r), \qquad \pi_{r+1} v \text{ onto}.$$

The conditions on homotopy groups hold at every source point. For $r = 0$ this includes sets, and for $r = 1$ groupoids. The set-valued $W^{\mathrm{lc}} = W_1$ agrees with CATHOT C.1–C.4, which we cite without reproving its categorical-reflector statement.

Proof. The tensor is $K \otimes M = \tau_{\leq r}(K \times M)$. In C.3 the test $M = *$ forces $\tau_{\leq r} v$ to be an equivalence, and conversely truncation preserves products. This gives W_r. C.2 says precisely that every fibre F_y has contractible r-truncation. The long exact homotopy sequence of a fibre identifies this condition with the displayed homotopy-group conditions, including the bijection on components.

For the 2-complete set-valued assertion we use the checked CATHOT C.1–C.4. For the remaining $r \geq 1$ we give the uniform higher-categorical argument. Every mapping space in $\mathcal{S}_{\leq r}$ is r-truncated. For each integer $m \geq 0$, mapping spaces in $\mathrm{Fun}([m], \mathcal{S}_{\leq r})$ are limits of r-types, hence are r-types; its maximal subgroupoid is therefore an $(r+1)$-type. The complete Segal nerve of $\mathrm{Fun}(S, \mathcal{S}_{\leq r})$ in degree m is the space of maps from S into that maximal subgroupoid. The universal property of truncation identifies it with the corresponding mapping space out of $\tau_{\leq r+1} S$. These levelwise identifications prove equivalence of the complete Segal nerves, hence of the functor ∞-categories. Thus $W_{r+1} \subset W^{\mathrm{lc}}$.

For the converse, C.2 gives the component bijection. Choose $x \in S$, $y = v(x)$, and let i_x, i_y be their inclusions. If v^* is an equivalence, the adjunction unit is an equivalence on $(i_x)_! *$ and $v_!(i_x)_! * \simeq (i_y)_! *$. Evaluating its unit at x gives the canonical map

$$\tau_{\leq r} \Omega_x S \longrightarrow \tau_{\leq r} \Omega_y T.$$

It is an equivalence, so v induces isomorphisms on π_i for $1 \leq i \leq r+1$. These arguments use chosen basepoints only component by component and require no global choice of an isomorphism of fundamental groupoids. □

C.5. Strictness and the failure of two-out-of-three. [PROVED]

For every finite $r \geq 0$, both inclusions $W_{r+1} \subset W^1_{\mathcal{S}_{\leq r}} \subset W_r$ are strict. The middle class does not satisfy two-out-of-three. For $\mathcal{C} = \mathcal{S}$, all five coefficient classes equal W_∞.

Proof. Take a nonzero abelian group A and the Eilenberg–Mac Lane space $K(A, r+1)$. Its projection to $*$ belongs to the middle class but not to W_{r+1}; the point inclusion belongs to W_r but not to the middle class. Their composite on $*$ is the identity, which proves the failure of two-out-of-three. For spaces, C.3 tested at $M = *$ gives W_∞, and C.2 tested at $M = *$ says that all fibres are contractible. The ordinary weak equivalences induce equivalences of functor ∞-categories, completing the calculation. All displayed spaces can be represented by categories of simplices of Kan models; the last-vertex map from the nerve of that category has the prescribed weak homotopy type, so these examples are examples in Cat as required. □

3.3 Derived modules and loop algebras

C.6. Monodromy, with the algebra and the restriction specified. [Known]

Let k be an associative unital ring and let (S, x) be a nonempty connected pointed space. Put $A_S = C_*(\Omega_x S; k)$, interpreted as the associative ring spectrum $Hk \wedge \Sigma^\infty_+ \Omega_x S$, or the corresponding dg ring $k \otimes_{\mathbb{Z}} C_*(\Omega_x S; \mathbb{Z})$; the loop operators commute with the k-action. There is an equivalence over Mod_k

$$\mathrm{Fun}(S, \mathrm{Mod}_k) \simeq \mathrm{Mod}_{A_S},$$

where evaluation at x corresponds to forgetting the A_S-action. For a pointed map $v : (S, x) \to (T, y)$ the restriction functor corresponds to restriction of scalars along the specified algebra map $a_v : A_S \to A_T$. No finiteness, nilpotence, or simple-connectivity hypothesis is needed. This is the monadicity theorem (Lurie, 2017, Theorem 4.7.3.5) applied to evaluation; the compatible chain model is Rivera and Zeinalian (2018, Corollary 9.2).

Proof. Evaluation i_x^* is conservative because every point of S is joined to x by a path, and it preserves all colimits, which are pointwise. Its left adjoint is $(i_x)_!$. At x the composite monad is $\Omega_x S \otimes (-)$; multiplication is path concatenation and the unit is the constant path. The two hypotheses of the cited Barr–Beck theorem therefore hold, and modules for this monad are precisely modules for A_S. For v, the equality $v_!(i_x)_! \simeq (i_y)_!$ identifies the monad comparison with the map induced by Ωv, so it identifies the actual restriction, not just the categories. With the presheaf convention, path reversal gives the same assertion using opposite loop multiplication and right modules; inversion canonically transports one convention to the other. The chain-complex realization of Mod_{Hk} is the ring case of Schwede and Shipley (2003, Theorem 5.1.6). □

C.7. All coefficient classes for an arbitrary ring. [Proved]

Suppose $k \neq 0$. Then the following are exact descriptions of the indicated classes, for every map $v : S \to T$ of spaces represented by small categories:

$$\begin{aligned}
v \in W_{D(k)} &\iff H_j(S; k) \xrightarrow{\sim} H_j(T; k) \quad (j \geq 0),\\
v \in W^{\mathrm{coh}}_{D(k)} = W^{\mathrm{hom}}_{D(k)} = W^{1}_{D(k)} &\iff H_0(F_y; k) \xrightarrow{\sim} k \text{ and } H_j(F_y; k) = 0 \quad (j > 0,\ y \in T),\\
v \in W^{\mathrm{lc}}_{D(k)} &\iff \begin{cases} \pi_0 v \text{ bijective}, \\ H_j(\Omega_x S; k) \xrightarrow{\sim} H_j(\Omega_{v(x)} T; k) \quad (j \geq 0,\ x \in S), \end{cases}\\
&\iff \begin{cases} \pi_0 v \text{ bijective}, \quad \pi_1(v, x) \text{ isomorphic}, \\ H_j(\widetilde{S}_x; k) \xrightarrow{\sim} H_j(\widetilde{T}_{v(x)}; k) \quad (j \geq 0,\ x \in S). \end{cases}
\end{aligned}$$

Here the last map is the lift determined by the chosen basepoints of the map on the corresponding connected components; it is equivariant for the specified isomorphism of fundamental groups. For $k = 0$, every category of coefficients is terminal and every functor belongs to all five classes.

Proof. C.3 can be tested on $M = k$: the map is $C_*(S;k) \to C_*(T;k)$, and its equivalence implies the test for every M by derived tensor product over k. C.2 gives the fibre condition in the same way. The augmented H_0 condition forces F_y to be nonempty and connected: the fold map from the free k-module on a set to k is an isomorphism only for a singleton, even when k does not have invariant basis number. Thus W^1 in particular forces a bijection on components and surjections on fundamental groups.

On each connected component C.6 identifies restriction with Res_{a_v}. This is an equivalence if and only if a_v is a quasi-isomorphism. One direction is invariance of module categories under an equivalence of algebras. Conversely its left adjoint is $A_T \otimes^{\mathbf{L}}_{A_S} (-)$, and if restriction is an equivalence, the unit at the free module A_S is the underlying map $A_S \to A_T$; it must be an equivalence. Components cannot be omitted or merged, by the preceding consequence of W^1. This proves the loop-homology description without any assumption on commutativity of k.

The degree-zero loop-homology map is the map $k[\pi_1 S] \to k[\pi_1 T]$ induced on free modules by $\pi_1 v$. Injectivity and surjectivity of this specified map force injectivity and surjectivity of the group map: two identified basis vectors give a nonzero difference, and a missing basis vector cannot lie in the image. Once $\pi_1 v$ is an isomorphism, the map on loop spaces preserves their component decompositions bijectively. The identity components identify with $\Omega\widetilde{S}$ and $\Omega\widetilde{T}$, and translation by a loop identifies every other component with the identity component. It remains to prove that a map $g : U \to V$ of simply connected CW spaces is a k-homology equivalence exactly when Ωg is one.

Here are the comparison and convergence details for this last assertion. Replace a pointed simply connected space by the simplicial set of its singular simplices whose one-skeleton is at the basepoint. This is a Kan simplicial set: singular horn fillers may be chosen with the required constant edges. It has one vertex and no nondegenerate edges, and hence trivial fundamental group. For $j \geq 2$, the usual based-simplex description of π_j and its based homotopies computes the same groups as the full singular complex, because those representatives and homotopies already have constant edges. Its inclusion is therefore a weak equivalence when the space is simply connected. This replacement is natural in pointed maps and has normalized integral chains C with $C_0 = \mathbb{Z}$, $C_1 = 0$, free in each degree. The cobar comparison identifies ΩC with chains on the based loop space; see Rivera and Zeinalian (2018, Corollary 9.2 and Remark 9.3). Use the decreasing filtration of its reduced tensor algebra by word length; the coproduct part of the cobar differential increases word length and thus preserves this filtration. The generators $s^{-1}\overline{C}$ have positive degree, so only finitely many word lengths occur in any fixed total degree. A k-homology equivalence of such C's induces equivalences on all tensor powers after tensoring with the abelian group underlying k: telescope the tensor powers one factor at a time, using that the other integral chain complexes are bounded below and degreewise free. The word-length filtration is degreewise finite, so the filtered comparison is a quasi-isomorphism. This proves the implication from g to Ωg.

Conversely the two-sided bar construction on integral chains of the loop monoid models chains of $B\Omega U \simeq U$. Its ascending length filtration is preserved by the bar differential and is again finite in each total degree, now because each bar factor is suspended once. The same tensor-power argument, followed by tensoring with k, proves that a k-homology equivalence of loop spaces induces a k-homology equivalence of their classifying spaces. These integral constructions followed by scalar extension explain why the argument also applies to an associative noncommutative k; no coalgebra over a noncommutative base is being assumed. Each replacement is connected by the displayed natural weak equivalence and the cobar or bar comparison, which preserve the exact homology test in both directions. □

C.8. The homological-epimorphism criterion and the point-target correction.

[Proved]

For a pointed map of connected spaces, write $A = A_S$ and $B = A_T$. Then

$$v \in W^1_{D(k)} \quad \Longleftrightarrow \quad B \otimes^{\mathbf{L}}_A B \xrightarrow{\mu} B \text{ is an equivalence of } B\text{-bimodules.}$$

For every prederivator, irrespective of enhancement, a map $X \to e$ lies in W_D if and only if it lies in W^1_D. Consequently the assertion in the CATHOT addendum that an acyclic-group map $BA \to e$ is not 1-complete is false.

Proof. The first map is the counit of extension–restriction evaluated at the free B-module. If it is an equivalence, then for every B-module N the counit is its derived tensor product with N over B, hence is an equivalence; thus restriction is fully faithful. The converse follows by evaluating the counit. In the second assertion every object of $D(e)$ is a constant coefficient, so the two definitions ask for exactly the same maps between exactly the same pairs of objects. In particular a nontrivial integrally acyclic group gives a map $BA \to e$ in $W^1_{D(k)}$ and outside $W^{\mathrm{lc}}_{D(k)}$ for every nonzero ring. Reversing that map gives a constant-coefficient equivalence $e \to BA$ outside W^1, since its homotopy fibre is the disconnected discrete group A. □

C.9. Integral detection and arbitrary-ring arithmetic. [Proved]

Let $P(k)$ be the set of rational primes which are not units in a nonzero unital ring k. Each k-homology test in C.7 is equivalent to the simultaneous $\mathbb{F}_p$-homology tests for $p \in P(k)$, together with the $\mathbb{Q}$-homology test when $\mathrm{char}(k) = 0$. In particular

$$W^{\mathrm{lc}}_{D(\mathbb{Z})} = W_\infty, \qquad W^{\mathrm{lc}}_{D(k)} = W_\infty \quad \Longleftrightarrow \quad P(k) \text{ contains every prime.}$$

Proof. For any bounded-below complex C of free abelian groups, consider $C \otimes^{\mathbf{L}}_{\mathbb{Z}} k$. A bounded-below integral complex is zero if and only if its derived tensors with $\mathbb{Q}$ and with every $\mathbb{F}_p$ vanish. Indeed, for its least nonzero homology group A, the universal coefficient sequences force $A/pA = A[p] = 0$ for every p; then A is a rational vector space, which the rational test detects. For $p \in P(k)$ the derived tensor $k \otimes^{\mathbf{L}} \mathbb{F}_p$ is a nonzero complex over a field, because $k/pk \neq 0$. Tensoring with it detects whether $C \otimes \mathbb{F}_p$ vanishes, by the field Künneth theorem. For $p \notin P(k)$ that derived tensor is zero. The rational tensor of k is nonzero exactly in characteristic zero. Applying the all-fields criterion to $C \otimes^{\mathbf{L}} k$ proves both directions of the asserted arithmetic description.

With $k = \mathbb{Z}$, the last condition of C.7 gives an integral homology equivalence of simply connected universal covers. Its mapping cylinder has zero relative homology and is simply connected, so relative Hurewicz, applied to a least nonzero relative homotopy group, gives a contradiction. Thus the lifted map is a weak equivalence; together with the specified π_1 isomorphism this proves v is a weak equivalence. This is the homology Whitehead argument of Hatcher (2002, Corollary 4.33). If no prime is invertible in k, the ring has characteristic zero (otherwise any prime coprime to its positive characteristic would be invertible), and the preceding arithmetic description makes its homology test the integral one. Conversely, if a prime p is invertible in k, the degree-p self-map of S^2 is a k-homology equivalence between simply connected spaces and belongs to $W^{\mathrm{lc}}_{D(k)}$ by C.7, but it is not a weak equivalence because multiplication by p on $\pi_2 S^2 = \mathbb{Z}$ is not an isomorphism. □

3.4 Spectra and local spectra

C.10. Spectra have the integral coefficient classes. [PROVED]

For spectra, W_{Sp} is the class of integral homology equivalences, $W^{\mathrm{coh}}_{\mathsf{Sp}} = W^{\mathrm{hom}}_{\mathsf{Sp}} = W^{1}_{\mathsf{Sp}}$ is the class of maps with integrally acyclic homotopy fibres, and $W^{\mathrm{lc}}_{\mathsf{Sp}} = W_{\infty}$.

Proof. The sphere generates spectra under colimits and shifts, so C.3 reduces the first test to $\Sigma^{\infty}_{+}v$. A map between connective spectra is an equivalence if and only if it is an integral homology equivalence: its connective cofiber, if nonzero, has a least nonzero homotopy group, which the stable Hurewicz theorem identifies with its least nonzero integral homology group. C.2 applied to the sphere gives the fibre assertion by the same argument. Evaluation on a connected space is monadic with monad $\Sigma^{\infty}_{+}\Omega S \wedge (-)$, exactly as in C.6. Therefore restriction is an equivalence exactly when the specified map of suspension ring spectra of loops is an equivalence. The connective argument turns this into integral loop-homology equivalence, and C.7–C.9 identify it with weak homotopy equivalence. All reductions are componentwise after C.2 has forced the component bijection. □

C.11. A complete generalized-homology description for local spectra. [PROVED]

Let E be any nonzero spectrum, let $L_E\mathsf{Sp}$ denote Bousfield localization at E-homology equivalences, and use the associated derivator. Here $E_j(Z) = \pi_j(E \wedge \Sigma^{\infty}_{+}Z)$ is unreduced generalized homology. Then

$$\begin{aligned} & W_{L_E\mathsf{Sp}} = \{v : E_j(S) \xrightarrow{\sim} E_j(T)\ (j \in \mathbb{Z})\}, \\ W^{\mathrm{coh}}_{L_E\mathsf{Sp}} = W^{\mathrm{hom}}_{L_E\mathsf{Sp}} = & W^{1}_{L_E\mathsf{Sp}} = \{v : E \wedge \Sigma^{\infty}_{+}F_y \xrightarrow{\sim} E\ (y \in T)\}, \\ & W^{\mathrm{lc}}_{L_E\mathsf{Sp}} = \{v : \pi_0 v \text{ bijective and } E_j(\Omega_x S) \xrightarrow{\sim} E_j(\Omega_{v(x)}T)\ (j \in \mathbb{Z},\ x \in S)\}. \end{aligned}$$

The last class forces isomorphisms of the discrete fundamental groups. These descriptions apply in particular to $E = K(n)$ at any prime and height $n \geq 1$. For such $K(n)$ the last class strictly contains W_{∞}. For $E = 0$ all classes consist of every map.

Proof. The E-acyclic spectra form a tensor ideal, so the localized category has the symmetric monoidal product $L_E(M \wedge N)$ and unit $U = L_E\mathbb{S}$. Its colimits are localized colimits of spectra. The unit generates under colimits and shifts: apply L_E to a sphere-cell presentation of any spectrum. Since $\mathbb{S} \to U$ is an E-equivalence, $E \wedge U \simeq E$. Consequently testing $S \otimes M \to T \otimes M$ or $F_y \otimes M \to M$ for all local M is equivalent to testing $M = U$, and the latter is precisely the indicated E-homology test. This proves the first two lines using C.2–C.3.

On a connected component, the monad of evaluation is tensoring with $L_E\Sigma^{\infty}_{+}\Omega S$. Barr–Beck applies as in C.6. An equivalence of the specified restriction forces the adjunction unit on the free object generated by U to be an equivalence. After evaluation it is the map

$$L_E\Sigma^{\infty}_{+}\Omega S \longrightarrow L_E\Sigma^{\infty}_{+}\Omega T.$$

Conversely this equivalence identifies the two monads and their categories of modules, proving sufficiency. Missing or merged components are excluded by C.2: each nonempty component of a fibre has a point whose sphere summand is split by collapse, and two such summands give a nonzero kernel for the augmented fold on E-homology. The same split-summand argument on loop components proves that the group map on π_1 is bijective. In detail, identified loop components give the difference of their point classes as a nonzero kernel element, and an omitted target component has a point class outside the image; use any nonzero group $\pi_j E$.

For $K(n)$ at p, choose a prime $\ell \neq p$. The degree-ℓ self-map of S^2 is a $\mathbb{Z}_{(p)}$-homology equivalence. C.7 for $k = \mathbb{Z}_{(p)}$ makes its loop map a $\mathbb{Z}_{(p)}$-homology equivalence. The cofiber of its localized suspension spectrum is bounded below, so the integral Hurewicz argument in the p-local category makes that loop map a stable p-local equivalence. Smashing with the p-local spectrum $K(n)$ preserves it. Thus the degree map belongs to $W^{\mathrm{lc}}_{L_{K(n)}\mathsf{Sp}}$, while multiplication by ℓ on $\pi_2 S^2$ is not invertible. For a general nonconnective E, we retain the exact loop-homology criterion; we do not replace it by an unproved criterion involving E-homology of universal covers. □

3.5 Strict inclusions and finite verification

C.12. Strictness uniform in every nonzero ring and in spectra. [Proved]

For every nonzero unital ring k, both inclusions

$$W^{\mathrm{lc}}_{D(k)} \subsetneq W^{1}_{D(k)} \subsetneq W_{D(k)}$$

are strict. The same is true for spectra. For $k = \mathbb{Q}$ or $k = \mathbb{F}_p$ one also has $W_\infty \subsetneq W^{\mathrm{lc}}_{D(k)}$. The first two strict inclusions hold for every nonzero coefficient $L_E\mathsf{Sp}$.

Proof. Let K be the connected presentation CW complex with one vertex, oriented edges x, y, and two oriented 2-cells attached by x^3y^{-5} and $(xy)^2y^{-5}$. There are no higher cells. Its cellular boundary $C_2(K;\mathbb{Z}) \to C_1(K;\mathbb{Z})$ is

$$\begin{pmatrix} 3 & 2 \\ -5 & -3 \end{pmatrix}, \qquad \det = 1.$$

Hence $\widetilde{H}_*(K;\mathbb{Z}) = 0$. Its fundamental group is nontrivial: on the set $\{0,1,2,3,4\}$ let

$$x = (0\ 1\ 2), \qquad y = (0\ 2\ 4\ 1\ 3).$$

With rightmost permutations applied first, x has order 3, y has order 5, and xy has order 2. These satisfy the relators and generate a quotient of order 60, so the edge x is not null-homotopic. The program `code/verify_coefficient_examples.py` enumerates this quotient and verifies the boundary determinant; its complete output is supplied.

Let $a : * \to K$ be the vertex and $b : K \to *$ the projection. Both are integral homology equivalences, hence lie in $W_{D(k)}$ for every k and in W_{Sp}. The map b lies in W^1 by C.8 and fails W^{lc} by its noninjective fundamental-group map. The fibre of a is ΩK, which has more than one component, so a is outside W^1 by C.7 or C.10. Also $ba = \mathrm{id}_*$, proving the corresponding failure of two-out-of-three for W^1. For local spectra $K \to *$ induces a stable equivalence of suspension spectra by C.10, so the same proof applies with C.11. For a literally specified indexing category take $\Delta/\operatorname{Sing} K$, whose objects are pairs $(n, \sigma : \Delta^n \to K)$ and whose morphisms $(m,\tau) \to (n,\sigma)$ are monotone maps $\alpha : [m] \to [n]$ with $\tau = \sigma\alpha$; composition is composition of these monotone maps. The selected vertex gives a functor $e \to \Delta/\operatorname{Sing} K$, and the unique functor to e realizes b. The usual realization map of its nerve to K is a weak equivalence, so the five coefficient properties transfer by restriction equivalences.

The promised group-theoretic examples are obtained from the acyclic overgroup A in B.10. The one-object category BA has endomorphism set A and multiplication as composition. The projection $BA \to e$ supplies the first strict inclusion, and the unit functor $e \to BA$ supplies the second. This is the corrected use of CATHOT B.6–B.7. Finally, a degree-2 map of S^2 gives strictness over $\mathbb{Q}$, and a degree-ℓ map for a prime $\ell \neq p$ does so over $\mathbb{F}_p$, by C.7. For $\mathbb{Z}$ and spectra the smallest class is already W_∞, so that last inclusion is equality. □

3.6 Rational homotopy and schematization

C.13. The rational class is the established fibrewise rational class. [KNOWN]

For a map of connected pointed CW spaces,

$$v \in W^{\mathrm{lc}}_{D(\mathbb{Q})} \quad \Longleftrightarrow \quad \pi_1 v \text{ is an isomorphism and } \pi_j v \otimes \mathbb{Q} \text{ is an isomorphism for every } j \geq 2.$$

Thus the class is that of π_1-rational homotopy equivalences, equivalently the maps inverted by fibrewise rationalization over $B\pi_1$. This characterization, including the cobar and local-coefficient formulations, is Rivera, Wierstra, and Zeinalian (2021, Theorem 3). For spaces whose universal covers have finite-dimensional rational homology in each degree, the algebraic models of Gómez-Tato, Halperin, and Tanré (2000, Theorems 5.10, 5.11, 6.2, and 6.4) model that fibrewise rational homotopy category. Their Definition 6.1 leaves the discrete fundamental group unchanged.

Proof. C.7 identifies the class with a specified π_1 isomorphism and a rational homology equivalence of the simply connected universal covers. The rational Whitehead theorem identifies the latter with isomorphisms of rationalized higher homotopy groups; alternatively Theorem 3 of the cited 2021 paper applies directly to the equivalent loop-chain test. Fibrewise rationalization changes the simply connected fibre of $S \to B\pi_1 S$ and keeps its base and action data. A map of these fibrations is a weak equivalence after that operation precisely under the displayed conditions. The finite-type assumption is needed for the stated minimal algebraic models, not for the equivalence class computed here. No assertion identifies the class with rationalization that also changes π_1: for example $B\mathbb{Z} \to B\mathbb{Q}$ is an ordinary rational homology equivalence but fails the displayed discrete-group test. □

C.14. Schematization is a strictly weaker rational test. [PROVED]

On connected pointed spaces, every map in $W^{\mathrm{lc}}_{D(\mathbb{Q})}$ induces an equivalence of Toën's schematizations over $\mathbb{Q}$, and the converse is false. The latter test requires an isomorphism of proalgebraic fundamental groups and cohomology isomorphisms for all finite-dimensional rational local systems, as in Toën (2006, Lemma 3.3.2 and Theorem 3.3.4); its fundamental group is the proalgebraic completion, by Corollary 3.3.7 there. These assertions concern schematization of the specified map.

Proof. A map in $W^{\mathrm{lc}}_{D(\mathbb{Q})}$ induces the actual isomorphism of discrete fundamental groups by C.7, hence the required isomorphism of proalgebraic completions. C.2 gives its cohomology comparison for every rational local system, including all finite-dimensional ones. The cited criterion and existence theorem therefore give the asserted equivalence of schematizations.

For strictness choose, by B.10, a nontrivial integrally acyclic simple group A of cardinality greater than $\aleph_0$ in $\mathbb{U}$. Every homomorphism $A \to \mathrm{GL}_n(\mathbb{Q})$ is trivial: simplicity makes a nontrivial homomorphism injective, while the target is countable. Hence its proalgebraic completion over $\mathbb{Q}$ is trivial and every finite-dimensional rational local system on BA is constant. Integral acyclicity, followed by scalar extension and vector-space duality, gives exactly the required rational cohomology comparisons for $BA \to *$. Thus this projection is a schematization equivalence. It is outside $W^{\mathrm{lc}}_{D(\mathbb{Q})}$, since $A \to 1$ is not an isomorphism. The one-object indexing category, its arrows and composition, and the explicit construction of A are specified in B.10 and C.12.

The supplied Toën source has a reversed implication in the sentence immediately following Definition 3.1.1, where it says that an $\mathcal{O}$-equivalence is a P-equivalence. Lemma 3.3.2 gives

P-equivalence $\Rightarrow$ $\mathcal{O}$-equivalence. Over $\mathbb{Q}$, $BC_2 \to *$ disproves the written converse: it preserves ordinary rational cohomology but changes the nontrivial finite proalgebraic group C_2. The proof above uses the actual criterion of Lemma 3.3.2. For simply connected spaces of finite type, Corollary 3.3.9 identifies the schematic higher homotopy sheaves in characteristic zero with the additive-group versions of the rational homotopy groups; this agrees with C.13 in that restricted case. □

C.15. The requested values, including their edge cases. [Proved]

The following table records the exact calculations. In the column headed "middle", the cohomological, homological, and 1-complete classes coincide. "Fibre k-acyclic" means the augmented condition in C.7, "cover k-equivalence" includes bijectivity on π_0 and isomorphisms on π_1, and "loop E-equivalence" includes bijectivity on π_0.

Coefficient	Constant class	Middle class	2-complete class
Sets	W_0	π_0 bijective, π_1 onto	W_1
Groupoids	W_1	$\pi_{\leq 1}$ isomorphic, π_2 onto	W_2
r-types	W_r	$\pi_{\leq r}$ isomorphic, π_{r+1} onto	W_{r+1}
Spaces	W_∞	W_∞	W_∞
$D(\mathbb{Z})$	Integral homology equivalences	Fibre $\mathbb{Z}$-acyclic	W_∞
$D(\mathbb{Q})$	Rational homology equivalences	Fibre $\mathbb{Q}$-acyclic	Cover $\mathbb{Q}$-equivalence
$D(\mathbb{F}_p)$	Mod-p homology equivalences	Fibre $\mathbb{F}_p$-acyclic	Cover $\mathbb{F}_p$-equivalence
$D(k)$, $k \neq 0$	k-homology equivalences	Fibre k-acyclic	Cover k-equivalence
Spectra	Integral homology equivalences	Fibre $\mathbb{Z}$-acyclic	W_∞
$L_E\mathsf{Sp}$, $E \neq 0$	E-homology equivalences	Fibre E-acyclic	Loop E-equivalence
Zero coefficients	Every map	Every map	Every map

Proof. C.4, C.5, C.7, C.10, and C.11 prove the rows. Infinite component sets cause no additional restriction: the augmented free-module argument and the pointwise Kan formulas apply to arbitrary $\mathbb{U}$-small sets, and each component is then treated separately. For nontrivial coefficients the map $\varnothing \to *$ fails already the constant class in every row, whereas the map $\varnothing \to \varnothing$ belongs to every class. Idempotents or noninvertible endomorphisms in an indexing category introduce no additional proviso: localization of its nerve at all arrows is the space BX, and the coefficient diagrams considered here are exactly the functors factoring through this localization. □

4 The manuscript's fibration assertions

D.1. The two pullback tests are different. [Proved]

Fix a class W of functors between small categories. A *universal W-preserving map* is a functor $p : E \to B$ such that, for every functor $v : T' \to T$ over B belonging to W, the induced functor

$$p^*v : E \times_B T' \longrightarrow E \times_B T$$

belongs to W. All pullbacks in this definition are strict pullbacks in Cat. This class of maps is closed under composition and arbitrary strict base change. It differs in general from the class obtained by testing only T', T with weakly contractible nerves.

Proof. For composable $E \xrightarrow{p} B \xrightarrow{q} S$, the pullback of $v : T' \to T$ over S first lies in W after pullback by q, and then lies in W after pullback by p. The canonical isomorphism of the iterated strict pullbacks with the pullbacks along qp proves composition. A pullback of p along $B' \to B$ acts on Cat/B' by the same strict pullback as p after forgetting to B; this proves base-change stability. These arguments do not require two-out-of-three for W. The difference from the restricted test is demonstrated in D.4 below. □

D.2. What the local-coefficient repair actually proves. [Proved]

Let $\mathbf{D}$ satisfy the derivator axioms of Part A on all small categories, with the manuscript's presheaf variance. Suppose $p : E \to B$ has the following property after every strict base change: for each $M \in \mathbf{D}(e)$, $p_* M_E$ is locally constant and its evaluation at b is canonically $H^*_{\mathbf{D}}(E_b, M)$. If $v : T' \to T$ is a 1-complete equivalence over B, then

$$w : E \times_B T' \longrightarrow E \times_B T$$

is a constant-coefficient $\mathbf{D}$-equivalence. The analogous statement with left Kan extensions holds. This does not assert that w is 1-complete or 2-complete.

If the direct-image hypothesis instead holds for every locally constant coefficient on every base-changed total category, then $v \in W^{\mathrm{coh}}_{\mathbf{D}}$ implies $w \in W^{\mathrm{coh}}_{\mathbf{D}}$. Such cohomological fibrations are closed under composition and base change; the dual assertion holds for homological fibrations. Under the enhancement hypotheses of C.2 these are universal $W^1_{\mathbf{D}}$-preserving fibrations.

Proof. Put $E_T = E \times_B T$, $E_{T'} = E \times_B T'$, and denote their projections by $p_T, p_{T'}$. Universality and pointwise detection identify the canonical mate

$$v^*(p_T)_* M_{E_T} \longrightarrow (p_{T'})_* M_{E_{T'}}$$

with an isomorphism: at $t' \in T'$ both sides are the cohomology of the same strict fibre $E_{v(t')}$. The object $K = (p_T)_* M_{E_T}$ is locally constant. Full faithfulness of v^* on local systems, applied to $\mathrm{Hom}(N_T, K)$ for every $N \in \mathbf{D}(e)$, and the two adjunctions to the point identify the canonical comparison $H^*_{\mathbf{D}}(T, K) \to H^*_{\mathbf{D}}(T', v^*K)$ as an isomorphism by Yoneda. Composition of right Kan extensions identifies this map with the constant-coefficient cohomology comparison induced by w. Testing every M proves the first assertion. The left-Kan proof uses $\mathrm{Hom}(K, N_T)$ instead.

For the strengthened assertion fix $L \in \mathrm{LC}_{\mathbf{D}}(E_T)$. The same mate identifies $v^*(p_T)_* L$ with $(p_{T'})_* w^* L$, and $(p_T)_* L$ is locally constant. The defining property of $W^{\mathrm{coh}}_{\mathbf{D}}$ applied to this coefficient proves the comparison for L, as required. For $E \xrightarrow{p} B \xrightarrow{q} S$, first $p_* L$, then $q_* p_* L$ is locally constant. The canonical composition isomorphism identifies the latter with $(qp)_* L$. Pasting the two universally invertible base-change mates gives the base-change mate for qp. This proves composition with all quantifiers on coefficients retained. A preliminary pullback proves base-change stability. Finally C.2 identifies $W^{\mathrm{coh}}_{\mathbf{D}}$ and $W^1_{\mathbf{D}}$ in the stated enhanced setting. □

D.3. Composition for the manuscript's weak rational fibrations. [Refuted]

For $\mathbf{D} = D(\mathbb{Q})$ there are finite categories and functors $E \xrightarrow{p} B \xrightarrow{q} [1]$ such that p and q are the universal weak fibrations of I ms. 117, while qp is not one. Thus I-b-B081-04 is false, as is the unqualified composition claim at I ms. 118.

Proof. Write $G = C_2$ additively. The category B has objects a, b, one identity at a, automorphism group G at b, two arrows $a \to b$ labelled by G, and no arrows $b \to a$. Postcomposition adds labels. The functor q sends a to 0, b to 1, and is the Grothendieck opfibration of $e \to BG$. Its strict fibres are e and BG.

An opfibration has a left adjoint from its over-comma category to its strict fibre, given by cocartesian transport. For presheaf coefficients this computes the right Kan extension on the fibre. Its constant direct image is locally constant precisely when the transport functors are constant-coefficient equivalences, and the same computation applies after arbitrary base change. Here the only nonidentity transport is $e \to BG$. For an arbitrary unbounded complex M of rational vector spaces, the trivial $\mathbb{Q}[G]$-module $\mathbb{Q}$ is projective, since the averaging idempotent $(1+\sigma)/2$ splits its augmentation. Hence both its derived invariants and derived coinvariants with trivial coefficient M are canonically M. Therefore q has the required universal weak property.

Define $L : B \to \mathsf{Set}$ by $L(a) = L(b) = G$ and let the arrow with label g act by $z \mapsto g + z$. Let $E = \int_B L$ and $p : E \to B$ be the projection. There is a unique lift of a base arrow with any prescribed source, and also with any prescribed target. Thus p is a covering, its transports are isomorphisms of discrete categories, and it satisfies the universal weak property even for spaces. The composite qp has fibre G_{disc} at 0 and the regular action groupoid $G \ltimes G$ at 1. The latter has one arrow between every ordered pair of objects and is contractible. The transition induces

$$\mathbb{Q} \longrightarrow \mathbb{Q}^2, \qquad r \longmapsto (r, r)$$

on degree-zero cohomology. It is not an isomorphism, so $(qp)_*\mathbb{Q}_E$ is not locally constant. The finite-category laws, all associative triples, and the component counts 2 and 1 are independently checked in `code/D_verify_finite.py` and its output. □

D.4. The restricted 2-complete fibration condition still fails composition. [Refuted]

Let $\mathbf{D}(A) = \mathrm{Fun}(A^{\mathrm{op}}, \mathsf{Set})$. Call $p : E \to B$ a restricted 2-complete fibration if, for every $v : T' \to T$ over B with BT' and BT weakly contractible, $E \times_B T' \to E \times_B T$ belongs to $W^{\mathrm{lc}}_{\mathbf{D}} = W_1$. There are finite $E \xrightarrow{p} B \xrightarrow{q} [1]$ for which both factors have this property and the composite does not. The first factor is a Hot-fibration; the second is even a universal W_1-equivalence. Consequently this is a counterexample to the replacement "2-complete throughout" in Lead 3, including its composition assertion.

Proof. We use distinct symbols K, H, B, E for the four categories in the construction. Let K be the poset of nonempty proper subsets of $\{0,1,2,3\}$. Its nerve is the barycentric subdivision of a tetrahedral boundary. Its objects, arrows and compositions are respectively those subsets, inclusions and composition of inclusions. Fix the maximal chain

$$\sigma = (\{0\} < \{0,1\} < \{0,1,2\}).$$

For composable inclusions $u : x \to y$, $v : y \to z$, let $c(u,v) \in C_2$ equal 1 precisely on σ and 0 otherwise. It is normalized: a pair containing an identity has value zero. It is a cocycle, since there is no chain of three strict inclusions, while the cocycle identity for a triple containing an identity cancels directly. Its evaluation on the mod-2 fundamental cycle is 1.

Let H have the objects of K and arrows (u, m) with $u \in K(x,y)$ and $m \in C_2$. Define

$$(v,n)(u,m) = (vu, m + n + c(u,v)), \qquad 1_x = (1_x, 0).$$

The normalized cocycle identity proves both unit laws and associativity. The projection $H \to K$ is a fibration and an opfibration: once a lift of an arrow is fixed, any required factorization is obtained by solving uniquely for one C_2-label. Its fibre is BC_2 and each transport acts identically on C_2.

We calculate $\Pi_1 H$. Put $t_x = (1_x, 1)$ and $h_{xy} = (x \le y, 0)$. In the groupoid completion the defining relations are

$$t_x^2 = 1, \quad t_y h_{xy} = h_{xy} t_x, \quad h_{yz} h_{xy} = h_{xz} t_x^{c(x,y,z)}.$$

Collapse a spanning tree of the one-skeleton. All t_x become a central element t of order dividing two. Delete the interior of the triangle σ from the triangulated sphere. The remaining disk has the same one-skeleton; its untwisted triangle relations kill the edge generators. The omitted triangle relation is then $t = 1$. Thus H is connected and simply connected. This calculation uses the full presentation, and not merely vanishing of its first homology.

Put $x_0 = \{0\}$ and let B be the Grothendieck construction of $e \xrightarrow{x_0} K$ over $[1]$. It has the objects of K and a new object a, with $B(a, x) = K(x_0, x)$, no arrows $x \to a$, and the evident compositions. The projection $q : B \to [1]$ sends a to 0 and K to 1. There is a retraction $r : B \to K$, $r(a) = x_0$, and the arrow $a \to x_0$ defines $1_B \Rightarrow ir$, where $i : K \hookrightarrow B$. Set $E = B \times_K H$ and $p : E \to B$. The retraction and natural transformation lift to $E \to H$: the lift over $a \to x_0$ has label zero, and normalization of c proves naturality. Thus $BE \simeq BH$, so E is connected and simply connected. Its strict fibre over $0 \in [1]$ is BC_2.

We verify the universal restricted tests. For a weakly contractible category $T \to B$, the pullback cocycle is a coboundary because $H^2(BT; C_2) = 0$. Choose a normalized one-cochain d with $\delta d = c|_T$. Relabelling an arrow by $m \mapsto m+d(u)$ gives an isomorphism between the pullback of p and $T \times BC_2$. This isomorphism preserves the vertical C_2-loops. A map between two weakly contractible test bases therefore induces a bijection on components and an isomorphism on fundamental groups of the total categories, because it preserves their common vertical generator. This proves the exact proposed property for p. Moreover opfibration transport is an equivalence of groupoids; the fibre-to-comma adjunction and Quillen's Theorem B, after every base change, show that p is a Hot-fibration (Quillen, 1973, §1, Theorem B and its corollary).

For any category $T \to [1]$, with no contractibility assumption, the projection $q_T : B \times_{[1]} T \to T$ induces an equivalence of fundamental groupoids. Its fibres are e or K, and it has a section choosing a or x_0. Each fibre is connected and simply connected. Consequently its fundamental groupoid has a unique path from the section object to any given fibre object. These paths are natural: factor an arrow of the Grothendieck construction into its cocartesian lift followed by a vertical arrow, and use uniqueness in the target fibre. They give a natural isomorphism from section followed by projection to the identity in the groupoid completion. Thus $q_T \in W_1$ for every T. Two-out-of-three now proves the restricted test for q.

Apply the test $\{0\} \to [1]$ to the composite. Its upstairs comparison is $BC_2 \to E$. The target is simply connected, whereas the source has fundamental group C_2. Restriction sends a set to the trivial C_2-action. The two-point transitive action is absent from its essential image. Thus this comparison is 1-complete but not 2-complete, proving failure of composition. The checker verifies all 1,552 associative triples in H and 2,064 in E; an independent finite presentation calculation eliminates all 73 and 87 generators, respectively. Its fibre presentation remains $\langle t \mid t^2 = 1 \rangle$. □

D.5. Total equivalence need not detect fibres after either blanket substitution. [Refuted]

The following two assertions are false. First, a 2-complete equivalence between Hot-fibrations over a completely **D**-aspherical base must be 2-complete on every strict fibre. Second, a 1-complete equivalence between universal $W^1_{\mathbf{D}}$-preserving fibrations must be 1-complete on every fibre. Both failures occur for set-valued coefficients. In contrast, under the enhancement hypotheses of C.2, a 1-complete equivalence between Hot-fibrations over the same base is 1-complete on every fibre.

Proof. For the first assertion use $H \to K$ from D.4 as a map over K to $1_K : K \to K$. Both structural maps are Hot-fibrations. The categories H and K are connected and simply connected, so the total map and the map $K \to e$ belong to $W_1 = W^{\mathsf{lc}}_{\mathsf{Set}}$. Its fibre map is $BC_2 \to e$, which

does not belong to W_1. This also refutes the qualification in I-b-B084-07 after replacing ordinary asphericity by complete asphericity and ordinary equivalence by 2-complete equivalence.

For the second assertion let R have two objects b, a, automorphism group C_2 at b, only the identity at a, one arrow $b \to a$, no arrows $a \to b$, and the relation that precomposing the arrow $b \to a$ by either automorphism leaves it fixed. Thus a is terminal. The opfibration $q : R \to [1]$ has transition $BC_2 \to e$. The section $s : [1] \to R$ chooses the unique object in each fibre. Its total map is a weak homotopy equivalence because both categories have terminal objects. At 0 its fibre map is $e \to BC_2$, which is not 1-complete: an underlying function from a singleton to a regular C_2-set need not be equivariant.

We verify the stronger structural hypothesis on q. After an arbitrary base change $T \to [1]$, it is the Grothendieck opfibration with fibres BC_2 over the objects of colour 0 and e over colour 1. For a locally constant set-valued presheaf L on the total category, transport over an arrow from colour 0 to colour 1 is invertible and its compatibility with vertical loops forces the C_2-action at its source to be trivial. Right Kan extension is calculated on the strict fibres by the opfibration adjunction. Its values are invariants at colour 0 and the original set at colour 1. Transport is invertible within each colour, and over a cross-colour arrow it is invertible because the source action is trivial. Thus the direct image is locally constant. The same fibre formula proves the base-change mate for every further pullback and every local system on that pullback. D.2 and the set case of C.2 show that q universally preserves W^1_{Set}. The identity of $[1]$ does too. This proves the second refutation, with all four arrows of R checked by `D_verify_more.py`.

For the last assertion, write $f : E \to F$ over S. The strict fibres of both structural Hot-fibrations compute their homotopy fibres. The map $BE_s \to BF_s$ is therefore the homotopy pullback of Bf along $BF_s \to BF$. C.2 characterizes $W^1_{\mathbf{D}}$ by a property of all homotopy fibres; that property is preserved by homotopy pullback. Hence every f_s is 1-complete. No analogous argument applies to $W^{\mathrm{lc}}_{\mathbf{D}}$, as the first example proves. □

D.6. A fibration class with the requested stability and fibre detection. [Known]

Assume $W = W^{\mathrm{lc}}_{\mathbf{D}}$ is an accessible fundamental localizer. Equip Cat with the Thomason model structure associated to W, whose existence is Cisinski (2006, Theorem 5.2.15). Its fibrations are closed under arbitrary strict base change and composition. If $p : E \to S$ and $q : F \to S$ are such fibrations and $f : E \to F$ is a functor over S belonging to W, then $f_s : E_s \to F_s$ belongs to W for every object $s \in S$. Right properness of the model structure is not needed for this assertion.

Proof. The cited theorem gives a cofibrantly generated, left proper model structure with weak equivalences exactly W and the usual Thomason cofibrations. Fibrations have the right lifting property against its trivial cofibrations; this directly implies stability under base change and composition. For $s : e \to S$, the pullback

$$s^* : \mathsf{Cat}/S \longrightarrow \mathsf{Cat}/e$$

is right Quillen: its left adjoint sends $T \to e$ to the composite $T \to e \to S$ and preserves cofibrations and trivial cofibrations because their underlying functors do not change. The objects p, q are fibrant in Cat/S, since a terminal object there is 1_S. A right Quillen functor preserves weak equivalences between fibrant objects, so $s^* f = f_s \in W$. For completeness this last model-category fact follows by factoring a map between fibrant objects through a trivial cofibration and a fibration, then using a path-object factorization to express its image as a retract, in the homotopy category, of the images of trivial fibrations; a right Quillen functor preserves these trivial fibrations. Equivalently it is the dual form of Ken Brown's lemma.

These are the model fibrations of the specified localized Thomason structure. We do not identify them with either the restricted fibrations of D.4 or every universal W-preserving functor of D.1. In particular, absent right properness, a model fibration is not asserted to preserve an arbitrary W-equivalence between arbitrary objects of a slice. This distinction is why the preceding counterexamples do not contradict the theorem. □

D.7. Universal constant equivalences control arbitrary coefficients. [PROVED]

Let $\mathbf{D}$ be a derivator on all small categories with both pointwise homotopy Kan extensions, pointwise detection, and the disjoint-union axiom. If every strict pullback of $u : A \to B$ is a constant-coefficient $\mathbf{D}$-equivalence, then $u^* : \mathbf{D}(B) \to \mathbf{D}(A)$ is fully faithful. In particular this holds if every strict pullback of u is 1-complete or 2-complete. Thus I-b-B084-01 is affirmative under these hypotheses, without an enhancement assumption.

Proof. Universal equivalence implies that each strict fibre A_b is $\mathbf{D}$-aspherical. Pulling back over B/b shows that A/b is $\mathbf{D}$-aspherical as well. Their inclusion and every transition between the comma categories are constant-coefficient equivalences by two-out-of-three. The same reasoning applies after every base change. Hence u is universally a weak cohomological fibration.

We prove the needed fibre formula for arbitrary inverse-image coefficients. Fix $b \in B$, put $H = A/b$, $S = A_b$, and let $v : H \to B$ remember $u(a)$, with $h : H \to e$, $t : S \to e$. The two functors

$$h_* v^*, \quad t_* t^* b^* : \mathbf{D}(B) \longrightarrow \mathbf{D}(e)$$

are the right adjoints in the desired comparison $(u_* u^* F)_b \to H^*_{\mathbf{D}}(A_b, F_b)$. Their left adjoints are $v_! h^*$ and $b_! t_! t^*$, respectively. We check that the canonical left mate $b_! t_! t^* \to v_! h^*$ is invertible.

For $z \in B$ and $\gamma : z \to b$, let J_γ have factorizations $z \xrightarrow{\beta} c \xrightarrow{\alpha} b$ with $\alpha\beta = \gamma$, and morphisms $k : c \to c'$ satisfying $k\beta = \beta'$ and $\alpha' k = \alpha$. It has initial object $(z, 1_z, \gamma)$ and terminal object $(b, \gamma, 1_b)$. Put $K_\gamma = A \times_B J_\gamma$ using the middle-object functor. The universal weak fibration property over J_γ implies $A_b \to K_\gamma$ is a constant-coefficient equivalence. The comma category $z\backslash v$ is the disjoint union of the K_γ indexed by the set $B(z, b)$. Pointwise left Kan extension therefore identifies the mate, evaluated at $M \in \mathbf{D}(e)$ and z, with

$$\coprod_{\gamma \in B(z,b)} H^{\mathbf{D}}_*(A_b, M) \longrightarrow \coprod_{\gamma \in B(z,b)} H^{\mathbf{D}}_*(K_\gamma, M).$$

Each summand is invertible. Pointwise detection proves that the mate is an isomorphism, so the right-adjoint comparison is an isomorphism as well. Every indexing category used is small, including an empty hom-set or an infinite hom-set; the disjoint-union axiom covers both cases.

Now evaluate the unit $F \to u_* u^* F$ at b and follow it by the fibre comparison. Its composite is the constant unit $F_b \to H^*_{\mathbf{D}}(A_b, F_b)$, which is invertible because A_b is $\mathbf{D}$-aspherical. Therefore the unit is pointwise invertible and hence invertible. This is exactly full faithfulness of u^*. The second adjunction then also has invertible counit $u_! u^* F \to F$. All maps are canonical mates; no abstract identification of endpoints is used. □

D.8. The HOT conclusions and the weak-contractible-base equivalence. [KNOWN]

For the homotopy derivator of spaces, all five equivalence classes of Part A equal W_∞. The universal weak cohomological fibration condition, its homological counterpart, the test on every map between weakly contractible bases, and universal preservation of W_∞ are equivalent. This settles I-b-B096-04 for the usual derivator axioms and settles Ib-C021, Ib-C026, Ib-C048,

Ib-C069, Ib-C092 and Ib-C096 in the modern HOT setting. For a general derivator, the weak-contractible-base test is equivalent to the two universal weak conditions under Der 3 a'; the latter axiom follows from Cisinski (2009, Corollary 3.8).

Proof. For spaces, a constant-coefficient equivalence $u : A \to B$ induces bijections $[BB, Z] \to [BA, Z]$ for every small space Z, by choosing the source coefficient to be a point. Yoneda in the homotopy category implies that the specified map Bu is a weak equivalence. Conversely Cisinski's theorem gives equivalence of local-system categories under a weak equivalence of nerves; full faithfulness then gives the local homology and cohomology comparisons by adjunction. The chain of Part A proves equality of the five classes. Thus in the HOT case changing any one of these five adjectives changes none of the assertions.

For the general weak-condition equivalence, suppose first that right direct images of constants are locally constant and satisfy the universal strict-fibre formula. For $v : T' \to T$ over the base, both BT' and BT weakly contractible, the canonical base-change map on such a direct image is invertible by pointwise detection, since it compares the same strict fibres. Der 3 a' makes that direct image constant on T. Its cohomology is its constant value, and the same holds on T'. The comparison of total cohomology is therefore invertible. The left-Kan proof is dual. Conversely the restricted test includes every inclusion of a chosen point into a category with a terminal object. Apply these tests over the slices B/b: their pullbacks are the fibre inclusions $E_b \to E/b$. Selecting any object $(b, u : b \to b')$ of B/b' and using two-out-of-three gives all transition comparisons $E/b \to E/b'$. Repeating after every base change yields the universal weak cohomological condition. Initial objects and undercategories give the dual condition. The point is arbitrary, not necessarily the distinguished terminal or initial object.

For HOT, these fibre and comma comparisons are weak equivalences. Quillen's Theorem B identifies the strict fibres with homotopy fibres, after every base change. Consequently each strict base-change square is homotopy Cartesian: compare its homotopy fibres over every component of the new base, using the identical strict fibre on both sides. It follows that pullback preserves weak equivalences. The reverse implication follows by applying universal preservation to the contractible test bases. Equivalently, these are the sharp maps of the right proper Thomason model structure; Rezk (1998, Proposition 2.7) gives precisely this homotopy-Cartesian characterization. This is the intrinsic class appearing in Chapter XI, and is not asserted to coincide with its smaller class of model fibrations. □

D.9. Errors not removed by changing the equivalence class. [Refuted]

The uniqueness assertion I-b-B087-05, the two implications I-b-B093-04 and I-b-B093-05, the orbit relations I-b-B095-01, the properness assertion Ib-C003, and the internal-Hom assertion Ib-C082 are false as stated. Their corrections are independent of a wholesale replacement by $W^{\mathrm{lc}}_{\mathbf{D}}$.

Proof. A trivial two-point bundle over a circle has prescribed identity monodromy but admits the nonidentity automorphism exchanging its two sheets. Hence there is no unique isomorphism realizing its classification. This can be modeled by the product of the four-object circle poset with a two-point discrete category. Every arrow in the product preserves the sheet label, and exchanging labels defines a nonidentity automorphism over the base. The classification is of an isomorphism class, with a framing if one wishes to prescribe rather than conjugate monodromy. No change of weak-equivalence class removes these automorphisms.

For I-b-B093-04 take $G = C_2$, $E = G$ as a discrete regular G-set, and Z the four-object circle poset with an orientation-reversing involution. All are finite posets, $E/G = e$, and for a

field of characteristic different from two, $H^*(E;k)^G = k$. The involution acts as minus one on $H^1(Z;k)$, so $H^*(Z;k)^G \to H^*(Z;k)$ is not an isomorphism. Yet

$$(G \times Z)/G \longrightarrow Z, \qquad [g,z] \longmapsto g^{-1}z$$

is an isomorphism of categories. The inclusion of the fibre over the unique base object is the identity of Z. It is therefore an equivalence for every one of the classes under discussion. Conditions (a)–(c) in the manuscript do not force the asserted failure on a fibre. The same example also has $H^*(Z;\mathbb{Q})^{C_2} = \mathbb{Q}$ but total category Z, so it refutes the proposed implication to asphericity even when the base is completely aspherical. In condition (b) the transcription reproduces a printed map from the invariants to themselves; the only interpretation consistent with the preceding page is the inclusion into $H^*(Z;k)$.

For I-b-B093-05 use $E = S^2$ with the antipodal action and $Z = S^1$ with reflection, over $k = \mathbb{Q}$. The actions on their reduced cohomology in degrees 2 and 1 are both multiplication by -1. Since the diagonal action is free and finite, averaging identifies rational cohomology of its quotient with invariants of the rational cohomology of $S^2 \times S^1$. The tensor of the two negative eigenspaces is a positive eigenspace in degree 3. Thus

$$H^*((S^2 \times S^1)/C_2;\mathbb{Q}) \cong \mathbb{Q}[0] \oplus \mathbb{Q}[3]$$

as graded vector spaces. It is not rationally acyclic, despite $H^*(S^1;\mathbb{Q})^{C_2} = \mathbb{Q}$. The omitted cross term is the error. The program `D_verify_more.py` constructs the quotient of the product of the six-object sphere poset and the reflected four-object circle poset. Its normalized rational chain dimensions are $(12, 60, 96, 48)$, its boundary ranks are $(11, 49, 47)$, and its Betti numbers are $(1, 0, 0, 1)$. The categories are given there by all objects, arrows and the orbit composition rule. This independently checks the top-degree class in a finite categorical model. As failure of constant equivalence implies failure of 1- and 2-complete equivalence, strengthening the conclusion does not repair this implication under its original cohomological hypotheses.

For I-b-B095-01 write the six-object sphere poset as pairs (i, ϵ) with $i = 0, 1, 2$ and $\epsilon \in \{\pm 1\}$, ordered by increasing i. The antipodal involution changes both endpoint signs. Label an arrow orbit by the product of its endpoint signs. Composition then multiplies labels. Therefore the correct relations are

$$p_1^- p_0^- = p_1^+ p_0^+, \qquad p_1^- p_0^+ = p_1^+ p_0^- .$$

The displayed relations in the manuscript equate arrows of different signs and are false. This is a categorical composition calculation, unaffected by the choice of coefficients. The program checks all four sign products. The count of the later poset model should not be inferred from these incorrect relations.

For Ib-C003 take the inclusion $\{1\} \hookrightarrow [1]$. It is a Grothendieck opfibration: the only arrow issuing from its image object is an identity. The strict-fibre-to-over-comma comparisons are identities, after all base changes. Nevertheless its comma categories at 0 and 1 are empty and terminal, so the transition is $\varnothing \to e$. It belongs to none of W_∞, W^1_{Set} or $W^{\mathrm{lc}}_{\mathsf{Set}}$. Properness supplies a fibre formula, but does not supply local constancy of the fibres. An additional transport hypothesis is required.

For Ib-C082 take $e \to BC_2$ with $\mathbf{D} = D(\mathbb{Q})$. Averaging proves the constant-equivalence assertion exactly as in D.3. The trivial representation and the sign representation, both in degree zero, have no nonzero equivariant map between them, whereas their restrictions have a one-dimensional space of linear maps. Thus restriction is not full. Derived rational local systems have compatible tensor products and internal Homs, so the stated hypothesis is present. The missing adjective is "cohomological": with locally constant internal Homs preserved by pullback, cohomological invariance applied to $\underline{\mathrm{Hom}}(L, M)$ compares $\mathrm{Hom}(L, M)$ and proves 1-completeness. In this argument the unit must be the actual monoidal unit. For a stable theory

it cannot be replaced by the terminal zero object. Replacing "equivalence" in the conclusion by "2-complete" only strengthens an already false conclusion; replacing the hypothesis by 2-completeness makes full faithfulness tautological and does not validate the manuscript's inference. □

D.10. The doubled-arrow test and the scope of the repair. [PROVED]

For the space derivator, the doubled-arrow construction of I-b-A069 cannot produce the proposed failure of weak cohomological fibration. The same gluing step is valid for $W^{\mathrm{lc}}_{\mathbf{D}}$ and $W^{1}_{\mathbf{D}}$ when $\mathbf{D}$ has the complete enhanced coefficient category of C.1. This statement concerns that specified base change only.

Proof. A functor $X \to [1]$ decomposes X into a sieve X_0 and a cosieve X_1. Pullback along the fork $0 \to 1 \leftarrow 0'$ is $P = X \amalg_{X_1} X$. Every composable chain in this category lies in a single copy of X, because once a chain enters the common cosieve it cannot leave it. Hence

$$NP = NX \amalg_{NX_1} NX.$$

The two inclusions of NX_1 are monomorphisms of simplicial sets. Its realization is therefore the homotopy pushout of $BX \leftarrow BX_1 \to BX$. If $X \to [1]$ is a weak cohomological HOT-fibration, both inclusions $X_i \to X$ are weak equivalences. Thus either copy of X maps by a weak equivalence into P, and $X_1 \to P$ is one too. The other two comma categories for the fork are copies of X_0, so their fibre and transport comparisons are also weak equivalences.

Under the enhanced hypotheses, apply $\mathrm{Fun}(-, \mathcal{C})$ to this homotopy pushout. It becomes the pullback of infinity categories

$$\mathrm{Fun}(BP, \mathcal{C}) \simeq \mathrm{Fun}(BX, \mathcal{C}) \underset{\mathrm{Fun}(BX_1, \mathcal{C})}{\times} \mathrm{Fun}(BX, \mathcal{C}).$$

If restriction to BX_1 is an equivalence, its pullback is an equivalence. If restriction is only fully faithful, the diagonal into this self-pullback is still an equivalence: full faithfulness identifies a specified equivalence between the two restricted objects with a unique equivalence between the two original objects, together with its contractible space of choices. Thus the gluing step retains the required full faithfulness, and the remaining comparisons follow by composition. C.1 transfers the exact fully faithful or equivalence condition to the homotopy categories used in the question. □

D.11. The doubled-arrow generalization for every derivator. [PROVED]

Let $\mathbf{D}$ satisfy Der 1–4 on all small categories. If a cosieve inclusion $i : X_1 \to X$ belongs to $W^{\mathrm{lc}}_{\mathbf{D}}$, then both copy inclusions $X \to X \amalg_{X_1} X$ belong to $W^{\mathrm{lc}}_{\mathbf{D}}$. Consequently the strengthened 2-complete gluing step in the specified doubled-arrow construction is valid without an enhancement. The corresponding 1-complete gluing step is also valid, by B.5.

Proof. B.3 proves that $W^{\mathrm{lc}}_{\mathbf{D}}$ is a fundamental localizer, including relative Theorem A. Let V be the fundamental localizer generated by the single functor i. It is accessible and is contained in $W^{\mathrm{lc}}_{\mathbf{D}}$. The nerve correspondence and its associated simplicial model structure, whose cofibrations are all monomorphisms, are Cisinski (2006, Theorems 1.4.3 and 4.2.15). In that model structure Ni is a trivial cofibration. As in D.10, the nerve of this categorical pushout is the simplicial pushout, since a composable chain cannot leave the common cosieve after entering it. Each inclusion $NX \to NX \amalg_{NX_1} NX$ is therefore a pushout of the trivial cofibration Ni, and is a V-equivalence. Passing back to categories proves the stated $W^{\mathrm{lc}}_{\mathbf{D}}$ comparison. This argument does not assume that the whole localizer $W^{\mathrm{lc}}_{\mathbf{D}}$ is accessible.

For $W^1_{\mathbf{D}}$ apply Loc (6a), as proved in B.5 from the fully faithful relative-Theorem-A assertion of B.3, to the map of cosieve Mayer–Vietoris squares

$$(X_1, X_1, X, X) \longrightarrow (X_1, X, X, X \amalg_{X_1} X).$$

The first three maps are id_{X_1}, i, and id_X, so the fourth is 1-complete. Interchanging the two copies gives the other inclusion. The remaining fibre and transport comparisons for this specified fork are the ones already identified in D.10. Thus the previous general descent gap is resolved by B.3; no conclusion about all possible base changes of the original weak condition is added. □

The file `research/D_audit.csv` gives one disposition for each of the twenty-one rows in `DER_I-b_claims.csv` whose printed status is exactly `false` or `conjectured` and whose first manuscript page is between 76 and 126. It also records the stronger hypotheses needed when the answer differs for 1- and 2-complete equivalences. Rows classified `stated` or `sketched` are not silently counted as conjectures; the companion's separate ms. 118 and ms. 122 composition developments are verified in D.3 and D.4.

D.12. The original claim already fails for 0-type coefficients. [Refuted]

For $\mathbf{D}(A) = \mathrm{Fun}(A^{\mathrm{op}}, \mathsf{Set})$, a universal weak cohomological and homological fibration need not preserve constant-coefficient equivalences. Thus the truncated-type qualification I-b-B079-07, as well as I-b-B079-03 and I-b-B079-04, is false in degree zero.

Proof. Let $G = C_2$. The category $EG = G \ltimes G$ has object set G, a unique arrow between any ordered pair of objects, and the unique possible composition. The functor $EG \to BG$ sends the arrow $x \to y$ to yx^{-1}. It is a covering: an arrow of BG has exactly one lift from any prescribed object and exactly one lift ending at any prescribed object. A presheaf on a strict fibre is just a family indexed by a discrete two-point set. Right and left Kan extensions along the covering are respectively fibrewise products and coproducts, and transport permutes their factors. Hence constants have locally constant direct images, the strict fibre formulas hold, and all these assertions persist under arbitrary base change.

The functor $e \to BG$ is a constant-coefficient equivalence for sets since both indexing categories are connected, and constants have the same Hom sets over every connected indexing category. Its pullback comparison is $G_{\mathrm{disc}} \to EG$. These categories have two and one connected components. For a two-element set M, the restriction of sections of the constant presheaf is the nonbijective diagonal $M \to M^2$. This proves failure of the constant-equivalence conclusion. The input map $e \to BG$ is neither 1-complete nor 2-complete; consequently it does not contradict the weaker repaired statement D.2. The groupoid and its two-point fibre occur explicitly in the finite checker. □

5 Realization and higher coefficients

E.1. What "one categorical level up" means. [Proved]

Let $\mathcal{C}$ be an ∞-category, small in a universe $\mathbb{V}$ larger than the universe $\mathbb{U}$ of indexing spaces. Put

$$K_0(\mathcal{C}) = \mathcal{C}^{\simeq}, \qquad K_1(\mathcal{C}) = \mathrm{Fun}(\Delta^1, \mathcal{C})^{\simeq}.$$

For a map $f : X \to Y$ of $\mathbb{U}$-small spaces, restriction $f^* : \mathrm{Fun}(Y, \mathcal{C}) \to \mathrm{Fun}(X, \mathcal{C})$ is an equivalence if and only if both canonical maps

$$\mathrm{Map}(Y, K_i(\mathcal{C})) \longrightarrow \mathrm{Map}(X, K_i(\mathcal{C})), \qquad i = 0, 1,$$

are equivalences of spaces. This is a two-object orthogonality description, not by itself the constant-coefficient class of a derivator. If $\mathcal{C}$ is locally $\mathbb{U}$-small and admits the required $\mathbb{U}$-small cotensors, equivalence of the ordinary homotopy categories under the specified restriction functor is equivalent to the stated ∞-categorical condition. The same conclusion holds without local $\mathbb{U}$-smallness whenever that restriction has a Kan-extension adjoint.

Proof. Since X is a groupoid, the maximal subgroupoid of $\mathrm{Fun}(X,\mathcal{C})$ is canonically $\mathrm{Map}(X,\mathcal{C}^{\simeq})$. Exponential adjunction identifies the other displayed map with the map on maximal subgroupoids of the arrow categories of the two functor categories. For any ∞-category $\mathcal{A}$ and objects a,b, its mapping space $\mathrm{Map}_{\mathcal{A}}(a,b)$ is the homotopy fibre at (a,b) of

$$\mathrm{Fun}(\Delta^1,\mathcal{A})^{\simeq} \longrightarrow \mathcal{A}^{\simeq}\times\mathcal{A}^{\simeq}.$$

An equivalence on the two displayed cores therefore gives essential surjectivity and equivalences on every mapping space. Conversely an equivalence of categories gives both equivalences of cores.

For the last assertion, restriction commutes with cotensors. Full faithfulness on homotopy categories, applied to M and N^T for every $\mathbb{U}$-small space T, identifies $[T,\mathrm{Map}(M,N)]$ with $[T,\mathrm{Map}(f^*M,f^*N)]$. Yoneda in the homotopy category of spaces identifies the specified map of mapping spaces as an equivalence. Essential surjectivity on the homotopy category is the same as essential surjectivity on objects. All constructions are canonical up to the contractible choices inherent in mapping objects. For the adjoint formulation, an adjunction of ∞-categories induces an adjunction of homotopy categories. Full faithfulness of a right adjoint is equivalent to invertibility of its counit, and full faithfulness of a left adjoint to invertibility of its unit. Invertibility is detected in the homotopy category, so this criterion upgrades full faithfulness directly; essential surjectivity is unchanged. This applies in particular to restriction into $\mathcal{P}_R$, which has the Kan adjoints although its entire mapping spaces need not be $\mathbb{U}$-small. □

E.2. An ordinary derivator realization for linear coefficients. [Proved]

Fix universes $\mathbb{U}\subset\mathbb{V}\subset\mathcal{W}$, and an $\mathbb{U}$-small E_∞-ring spectrum R. Let $\mathcal{P}_R$ denote the ∞-category of $\mathbb{U}$-presentable R-linear stable ∞-categories and R-linear colimit-preserving functors. We regard it as a category in the larger universes; all limits and colimits required below have $\mathbb{U}$-small shapes. Let

$$\mathbf{D}_R(A)=\mathrm{Ho}\,\mathrm{Fun}(NA^{\mathrm{op}},\mathrm{Mod}_R),\qquad \mathbf{D}'_R(A)=\mathrm{Ho}\,\mathrm{Fun}(NA^{\mathrm{op}},\mathcal{P}_R).$$

Both are derivators on $\mathsf{Cat}_{\mathbb{U}}$, and their classes of functors satisfy

$$W_{\mathbf{D}'_R}=W^{\mathrm{lc}}_{\mathbf{D}_R}. \tag{E.2.1}$$

Thus the equality in Part E has an ordinary-derivator realization for commutative rings, by taking $R=Hk$. It uses the specified symmetric monoidal enhancement of $\mathbf{D}_R$. It is functorial in maps of E_∞-rings by extension of scalars; it is not an assignment constructed from an arbitrary unadorned derivator.

Proof. We first check the size and existence assertions. The category of presentable categories admits small limits, computed on underlying categories, and small colimits, computed by taking limits of right adjoints (Lurie, 2009, Proposition 5.5.3.13 and Theorem 5.5.3.18). It is symmetric monoidal, with tensor preserving colimits separately (Lurie, 2017, Proposition 4.8.1.15 and Remark 4.8.1.24). We may identify $\mathcal{P}_R$ with $\mathrm{Mod}_{\mathrm{Mod}_R}(\mathrm{Pr}^L)$. Limits and colimits of these modules exist by Lurie (2017, Corollaries 4.2.3.3 and 4.2.3.5). The Mod_R-action makes their

underlying categories stable; the corresponding assertion for spectra is Lurie (2017, Proposition 4.8.2.18). The standard pointwise Kan-extension construction consequently gives both displayed derivators. We use $\mathcal{W}$ for their values and do not assert that $\mathcal{P}_R$ itself is $\mathbb{U}$-presentable. For a map $R \to S$ of E_∞-rings, the coefficient functor is

$$\mathcal{P}_R \longrightarrow \mathcal{P}_S, \qquad \mathcal{A} \longmapsto \mathrm{Mod}_S \otimes_{\mathrm{Mod}_R} \mathcal{A}.$$

Relative tensor products are characterized by balanced colimit-preserving functors. Their universal property supplies the associative change-of-scalars comparisons for $R \to S \to T$ and the identity comparisons coherently, so this gives functoriality in the enhanced ring datum and induced morphisms of the displayed derivators. No choice of a ring presenting an arbitrary derivator is implicit in this assertion.

Write $X = BA$ and $Y = BB$ for the source and target of the map of spaces induced by a functor of indexing categories. For $\mathcal{A}, \mathcal{B} \in \mathcal{P}_R$, the constant-coefficient mapping space is

$$\mathrm{Map}\big(Y, \mathrm{Fun}^L_R(\mathcal{A}, \mathcal{B})^{\simeq}\big). \tag{E.2.2}$$

Here Fun^L_R denotes the category of R-linear left adjoints and all R-linear natural transformations. Constant diagrams factor through groupoid completion; this explains the use of BB rather than NB in (E.2.2). A fixed choice of inversion identifies the presheaf variance with the covariant groupoid formula, naturally up to equivalence.

Suppose first that f is in $W_{\mathbf{D}'_R}$. For every $\mathcal{B}$ and space T, its cotensor $\mathcal{B}^T = \mathrm{Fun}(T, \mathcal{B})$ is again an object of $\mathcal{P}_R$. Applying the constant tests with $\mathcal{A} = \mathrm{Mod}_R$ and this cotensor, and using

$$\mathrm{Fun}^L_R(\mathrm{Mod}_R, \mathcal{B}) \simeq \mathcal{B}, \tag{E.2.3}$$

we obtain bijections $[T, \mathrm{Map}(Y, \mathcal{B}^{\simeq})] \to [T, \mathrm{Map}(X, \mathcal{B}^{\simeq})]$ for every T. Formula (E.2.3) is evaluation at R; its inverse sends b to $M \mapsto M \otimes_R b$. It is the unit case of the linear functor theorem (Lurie, 2017, Theorem 4.8.4.1). Yoneda gives equivalence on these mapping spaces. Although the cores may have a larger set of components, each component is $\mathbb{U}$-small: its loop spaces are the equivalences in mapping spaces of a locally $\mathbb{U}$-small presentable category, and the same is true after a $\mathbb{U}$-small functor shape. The test $T = *$ gives the component bijection; Yoneda with connected $\mathbb{U}$-small T then applies component by component. Taking successively $\mathcal{B} = \mathrm{Mod}_R$ and $\mathcal{B} = \mathrm{Fun}(\Delta^1, \mathrm{Mod}_R)$, which is presentable, stable, and R-linear, proves by E.1 that $\mathrm{Fun}(Y, \mathrm{Mod}_R) \to \mathrm{Fun}(X, \mathrm{Mod}_R)$ is an equivalence. This proves one inclusion in (E.2.1).

For the converse we spell out the monodromy argument needed uniformly for all coefficient categories. For a pointed connected space (Z, z) and a cocomplete R-linear category $\mathcal{M}$, evaluation $U_z : \mathrm{Fun}(Z, \mathcal{M}) \to \mathcal{M}$ has a left adjoint F_z, given by left Kan extension. Its monad is

$$U_z F_z(m) = R[\Omega_z Z] \otimes_R m, \qquad R[\Omega_z Z] = R \otimes \Sigma^\infty_+ \Omega_z Z. \tag{E.2.4}$$

Composition in the monad is path concatenation. Evaluation is conservative because Z is connected, and preserves geometric realizations because colimits of functors are pointwise. The ∞-categorical Barr–Beck theorem (Lurie, 2017, Theorem 4.7.3.5) therefore identifies $\mathrm{Fun}(Z, \mathcal{M})$, over $\mathcal{M}$, with modules for the monad in (E.2.4). This proof also applies when $\mathcal{M}$ is cocomplete without being presentable. A current systematic treatment is Pascaleff, Pavia, and Sibilla (2025, Proposition 1.2.7 and Corollary 1.2.8).

If $R \neq 0$, equivalence of Mod_R-valued local systems forces a bijection on components. An omitted target component makes the identity and zero endomorphisms of the constant R on that component indistinguishable after restriction. Two source components lying over one target component prevent the source system with fibres 0 and R on those components from being in the essential image. These arguments also handle empty source and target. Choose a basepoint in

each source component. Restriction, being an equivalence compatible with evaluation, identifies the left adjoints of evaluation. The canonical mate $F_x \to f^* F_{f(x)}$ is consequently an equivalence. Evaluating it at R and then at x shows that the specified algebra map

$$R[\Omega_x X] \longrightarrow R[\Omega_{f(x)} Y] \tag{E.2.5}$$

is an equivalence of E_1-algebras, not merely a Morita equivalence. By (E.2.4), restriction on local systems in every cocomplete R-linear $\mathcal{M}$ is therefore an equivalence on each component.

Apply this with $\mathcal{M} = \mathrm{Fun}^L_R(\mathcal{A}, \mathcal{B})$. Its colimits are computed pointwise: colimits commute with colimits and with the R-action. It is a cocomplete R-linear category. Taking maximal subgroupoids and then components proves invariance of every constant-coefficient hom-set (E.2.2), as required. If $R = 0$, both Mod_R and $\mathcal{P}_R$ are terminal categories, and both classes contain every map. This proves the remaining edge case. □

E.3. The two-categorical version of the linear realization. [Proved]

Retain the hypotheses of E.2, and retain the full ∞-categories of natural transformations between R-linear left adjoints. The class of maps $f : X \to Y$ for which all restrictions

$$\mathrm{Fun}\big(Y, \mathrm{Fun}^L_R(\mathcal{A}, \mathcal{B})\big) \longrightarrow \mathrm{Fun}\big(X, \mathrm{Fun}^L_R(\mathcal{A}, \mathcal{B})\big) \tag{E.3.1}$$

are equivalences is also $W^{\mathrm{lc}}_{\mathbf{D}_R}$. The same is true if equivalence in (E.3.1) is tested on ordinary homotopy categories. These are the Hom-category constant tests in the coefficient $(\infty, 2)$-category $\mathcal{P}_R$. No assertion that its diagram assignment satisfies every axiom of Di Vittorio's 2-derivators is needed for this equality.

Proof. The forward implication in E.2 gave equivalence of (E.3.1), before taking its core. Conversely the single pair $\mathcal{A} = \mathcal{B} = \mathrm{Mod}_R$ gives $\mathrm{Fun}^L_R(\mathrm{Mod}_R, \mathrm{Mod}_R) \simeq \mathrm{Mod}_R$, so (E.3.1) already includes the defining test for $W^{\mathrm{lc}}_{\mathbf{D}_R}$. For the homotopy-category formulation the same converse is immediate, and the forward implication follows from the equivalence already proved. Thus the ordinary realization and the Hom-category realization agree in this case. Passing to homotopy categories at one shape loses information, but the cotensor and arrow tests in E.2 explain exactly why that observation does not obstruct the ordinary realization. □

E.4. A specified second linear stage detects degree-two torsion. [Proved]

For $R = H\mathbb{Q}$ use the particular coefficient category $\mathcal{P}_{\mathbb{Q}}$ of E.2, and define

$$W^{(2)}_{\mathbb{Q}} = W^{\mathrm{lc}}_{\mathbf{D}'_{\mathbb{Q}}}, \qquad W^{(1)}_{\mathbb{Q}} = W^{\mathrm{lc}}_{\mathbf{D}_{\mathbb{Q}}} = W_{\mathbf{D}'_{\mathbb{Q}}}.$$

For every prime p, the projection $K(\mathbb{Z}/p, 2) \to *$ is not in $W^{(2)}_{\mathbb{Q}}$. The projection $K(\mathbb{Z}, 2) \to *$ is not in that class either. The first projection is in $W^{(1)}_{\mathbb{Q}}$, so the inclusion $W^{(2)}_{\mathbb{Q}} \subset W^{(1)}_{\mathbb{Q}}$ is strict.

Proof. For a field extension $L/\mathbb{Q}$, multiplication by a unit of L is an invertible $\mathbb{Q}$-linear natural transformation of the identity of Mod_L. Scalar multiplication and composition give a coherent map

$$B^2(L^\times) \longrightarrow \mathcal{P}^{\simeq}_{\mathbb{Q}} \tag{E.4.1}$$

based at Mod_L: the single 1-morphism acts by the identity functor and each 2-morphism acts by its scalar natural automorphism. This may be constructed with complexes of L-vector spaces, where all the coherence equations are literal identities, and then localized. On π_2, (E.4.1) sends

a unit to its class as an automorphism of the identity functor. This homomorphism is injective: evaluating at the free module L sends it to that scalar in $\pi_0 \operatorname{Aut}_{\mathrm{Mod}_L}(L) = L^\times$.

For $A = \mathbb{Z}/p$ choose $L = \mathbb{Q}(\zeta_p)$ and the homomorphism $A \to L^\times$ sending 1 to a primitive pth root of unity. Composing $K(A,2) = B^2A \to B^2L^\times$ with (E.4.1) gives a local system of $\mathbb{Q}$-linear categories which is not constant: its induced map on π_2 is nonzero, whereas every constant system has zero map on π_2. For $A = \mathbb{Z}$, use $L = \mathbb{Q}$ and $n \mapsto 2^n$. An equivalence to a constant system could change the identification of its fibre by an autoequivalence, but conjugation acts by an isomorphism on π_2 and cannot turn a nonzero homomorphism into zero. Thus the constant-system functor is not essentially surjective in either case.

Finally $K(\mathbb{Z}/p, 2)$ is simply connected and rationally acyclic. One direct verification uses the path fibration with fibre $K(\mathbb{Z}/p, 1)$: transfer makes the latter rationally acyclic, and the Serre spectral sequence for the contractible total space then gives the assertion. The rational universal-cover criterion of Part C therefore puts its projection in $W^{(1)}_{\mathbb{Q}}$. The inclusion of the two stages follows in general because constants are locally constant. The character mechanism above agrees with the algebraically closed-field calculation of Pascaleff et al. (2025, Proposition 2.21); our use of the coefficient category $\mathrm{Mod}_{\mathbb{Q}(\zeta_p)}$ permits every prime over $\mathbb{Q}$. □

E.5. The explicitly specified tower starting with sets. [Known]

For $r \geq 0$ put $\mathcal{C}_r = \mathcal{S}_{\leq r}$ and let $\mathbf{D}_r$ be its derivator. Then

$$W_{\mathbf{D}_r} = W_r, \qquad W^{\mathrm{lc}}_{\mathbf{D}_r} = W_{r+1}, \qquad \bigcap_{r \geq 0} W_r = W_\infty. \tag{E.5.1}$$

Consequently the stipulated choice $\mathcal{C}_r \mapsto \mathcal{C}_{r+1}$ realizes the descending tower in Lead 4 for sets. It does not characterize a unique coefficient-raising operation.

Identification and intersection. The first two identities are the truncation computations in Part C. At $r = 0$, the identification of the reflector with Π_1 is the checked CATHOT C.1–C.4; we do not repeat that proof. A map in all W_r is bijective on components and induces isomorphisms on every homotopy group at each basepoint. It is therefore a weak equivalence, by the definition of weak equivalence of spaces. The reverse inclusion is immediate. Empty spaces cause no exception because W_0 already detects components. CATHOT C.2 and B.11 also show why retaining only categories modulo zigzags of natural transformations cannot produce the subsequent stages: that representability theory stops at 1-types. □

E.6. What the attached 2-derivator paper actually proves. [Known]

In arXiv:2309.05216v2, Di Vittorio's shapes are small simplicially enriched categories, and a 2-prederivator is a trihomomorphism $\mathrm{Dia}^{\mathrm{op}} \to \mathrm{GRAY}$, not a prederivator on ordinary Cat alone. For the 2-prederivator of a combinatorial $\mathrm{sSet}_{\mathrm{Joyal}}$-enriched model category, axioms (HDer 1)–(HDer 4) are proved, whereas (HDer 5) is Conjecture 3.2.23. All five axioms, when assumed, are preserved by shift.

Exact source check. The shapes and 2-prederivators are Definitions 3.1.1 and 3.1.5; the model construction is Theorem 3.1.14. The four model verifications are Propositions 3.2.3, 3.2.7, 3.2.11, and 3.2.18. (HDer 1) concerns coproducts; (HDer 2) detection of equivalences of 1-cells; (HDer 3) left and right biadjoints; (HDer 4) exactness of the weighted-collage squares. (HDer 5) requires smothering underlying diagram 2-functors for the walking arrow, walking isomorphism, and walking adjunction. The paper explicitly leaves the needed fullness and conservativity statements conjectural, again in Section 4.2(2). Shift invariance is Propositions 3.2.2, 3.2.6,

3.2.10, 3.2.16, and 3.2.22 (Di Vittorio, 2026). The checked arXiv record identifies v2 as the accepted version; we do not silently replace its conjecture by an asserted theorem.

Our ordinary derivator $\mathbf{D}'_R$ satisfies ordinary Der 1–4 because its enhanced coefficient category is complete and cocomplete. E.3 concerns Hom-categories on groupoid shapes. We have not constructed an extension of it to Di Vittorio's enriched shapes, nor verified his (HDer 5). It would therefore be incorrect to call E.2 or E.3 a construction of a 2-derivator in his sense without additional work. □

E.7. A conditional realization by localized spaces. [Proved]

Let $\mathcal{L} \subset \mathcal{S}$ be an accessible reflective localization with reflector L, and let W be the maps inverted by L. Assume W is preserved by product with every small space. For the derivator $\mathbf{D}_{\mathcal{L}}$ of $\mathcal{L}$, its constant-coefficient class is W. In particular, if $W^{\mathrm{lc}}_{\mathbf{D}}$ is accessible as a fundamental localizer, there exists a derivator of localized spaces $\mathbf{D}^{\mathrm{loc}}$ with

$$W_{\mathbf{D}^{\mathrm{loc}}} = W^{\mathrm{lc}}_{\mathbf{D}}.$$

The construction is functorial for specified inclusions between accessible fundamental localizers. By B.4, the accessibility hypothesis holds whenever $\mathbf{D}$ is the derivator of a $\mathbb{U}$-presentable infinity-category.

Proof. For a local space B and any space A, the space $\mathrm{Map}(A, B)$ is local: testing a map $f \in W$ against it is the same as testing $f \times A$ against B. Thus each mapping space between local objects is local, and a map in W preserves all the required constant-coefficient hom-sets. Conversely the terminal space is local. Testing constants $*$ and all $B \in \mathcal{L}$ gives bijections $\mathrm{Hom}_{\mathrm{Ho}\,\mathcal{L}}(LY, B) \to \mathrm{Hom}_{\mathrm{Ho}\,\mathcal{L}}(LX, B)$. Yoneda in $\mathrm{Ho}\,\mathcal{L}$ says that the specified Lf is an isomorphism there, and hence an equivalence. This is precisely $f \in W$. The accessible localization is presentable and complete, so its associated derivator exists.

For the additional assertion, B.3 supplies the fundamental-localizer axioms for $W^{\mathrm{lc}}_{\mathbf{D}}$. Every fundamental localizer is stable under products: for $f : A \to B$ in the localizer, apply the relative Theorem A axiom to $f \times C$ over C. Its slices are $f \times (C/c)$, and their projections to f are ordinary weak homotopy equivalences since C/c has a terminal object. Two-out-of-three gives the required slice equivalences. The correspondence for the test category Δ in Cisinski (2006, Theorems 4.2.15 and 1.4.3) sends an accessible fundamental localizer to an accessible localization of spaces with exactly the same specified weak equivalences under categorical realization. Its class is product stable by the preceding argument. The first part of this proof therefore applies. If $W \subseteq W'$ are specified accessible fundamental localizers, restriction of the W' reflector to W-local spaces is the canonical colimit-preserving functor $\mathcal{S}_W \to \mathcal{S}_{W'}$. Uniqueness in the universal property of localization supplies coherent composition. This is the asserted functoriality on the poset of localizers; an arbitrary morphism of derivators need not supply such an inclusion. □

E.8. The complete second-stage loop criterion. [Proved]

Let $R \neq 0$ be an E_∞-ring spectrum, and put $W^{(2)}_R = W^{\mathrm{lc}}_{\mathbf{D}'_R}$ for the realization in E.2. A map $f : X \to Y$ of $\mathbb{U}$-small spaces is in $W^{(2)}_R$ if and only if it is bijective on components and, for one basepoint x in every component, the specified map

$$\Omega_x f : \Omega_x X \longrightarrow \Omega_{f(x)} Y$$

belongs to $W^{\mathrm{lc}}_{\mathbf{D}_R}$. In particular, for a nonzero commutative ordinary ring k,

$$f \in W_k^{(2)} \quad \Longleftrightarrow \quad \begin{cases} \pi_0(f) \text{ is bijective,} \\ \pi_1(f), \pi_2(f) \text{ are isomorphisms at every basepoint,} \\ H_*(X\langle 2\rangle; k) \longrightarrow H_*(Y\langle 2\rangle; k) \text{ is an isomorphism on each component.} \end{cases} \tag{E.8.1}$$

Here $X\langle 2\rangle$ is the homotopy fibre at the chosen basepoint of $X \to \tau_{\leq 2}X$; all maps of covers are the ones induced by f and the chosen basepoints. This is a full computation of the specified rational second stage on taking $k = \mathbb{Q}$, with no nilpotence or finite-type assumption on X or Y. In that case the last condition is equivalently that $\pi_j(f) \otimes \mathbb{Q}$ is an isomorphism for every $j \geq 3$; the 2-connected covers to which rational Whitehead applies are automatically simply connected.

Proof. First suppose X, Y are connected and pointed. The symmetric monoidal functor

$$\mathcal{S}_{\mathbb{U}} \longrightarrow \mathcal{P}_R, \qquad T \longmapsto \mathcal{P}_R[T] := \operatorname*{colim}_{t \in T} \mathrm{Mod}_R$$

sends Cartesian products to relative tensor products, by the universal property of colimits and the separate preservation of colimits by the tensor product. Its unit is Mod_R. The free R-linear presheaf universal property identifies

$$\mathcal{P}_R[T] \simeq \mathrm{Fun}(T^{\mathrm{op}}, \mathrm{Mod}_R). \tag{E.8.2}$$

To check this identification, an R-linear left adjoint from either side to $\mathcal{A}$ is precisely a functor $T \to \mathcal{A}$: on the right it is the colimit-preserving extension of its values on the R-linear representables; on the left it is the compatible family of functors $\mathrm{Mod}_R \to \mathcal{A}$ determined by those same values. These universal properties identify both functors and natural transformations. For a map $g : T \to T'$, the induced map in (E.8.2) is left Kan extension $g_!$; its right adjoint is the specified restriction g^*. Thus the comparison is natural in the direction required here, not just an abstract equivalence of categories. For groupoids the inversion equivalence may remove the opposite.

Evaluation $U_x : \mathrm{Fun}(X, \mathcal{P}_R) \to \mathcal{P}_R$ is conservative and preserves geometric realizations. It has a left adjoint F_x by pointwise Kan extension. Barr–Beck therefore applies even though $\mathcal{P}_R$ need not itself be presentable in the indexing universe. The free-evaluation monad is

$$U_x F_x(\mathcal{A}) = \mathcal{P}_R[\Omega_x X] \otimes_{\mathrm{Mod}_R} \mathcal{A}. \tag{E.8.3}$$

The E_1-algebra structure of $\mathcal{P}_R[\Omega_x X]$ comes from loop multiplication, and (E.8.3) is compatible with the full monad structure. For f the canonical comparison of these monads is tensoring with the specified algebra map

$$a_f : \mathcal{P}_R[\Omega_x X] \longrightarrow \mathcal{P}_R[\Omega_{f(x)} Y]. \tag{E.8.4}$$

If restriction of $\mathcal{P}_R$-valued local systems is an equivalence, its compatibility with evaluation identifies the left adjoints of evaluation, so the comparison of monads is an equivalence. Evaluate it at the unit Mod_R to see that (E.8.4) is an equivalence of underlying categories, and hence of E_1-algebra objects. Conversely equivalence of (E.8.4) makes the monads equivalent, and the monadic comparison identifies their restriction functor with f^*; consequently that specified restriction is an equivalence. E.1's adjunction argument identifies this with the ordinary-derivator definition.

By (E.8.2), (E.8.4) is left Kan extension along $\Omega_x f$. A left adjoint is an equivalence exactly when its right adjoint is, so (E.8.4) is an equivalence exactly when $\Omega_x f$ belongs to $W^{\mathrm{lc}}_{\mathbf{D}_R}$. For

disconnected spaces the same zero-versus-unit argument as in E.2 forces a component bijection. All functor categories then decompose as products over the corresponding components, and the connected proof applies to each factor. It also shows that an empty source is permitted precisely when the target is empty.

For $R = Hk$, apply the loop-algebra computation of Part C to $\Omega_x f$. Its component bijection is exactly the isomorphism on $\pi_1(X, x)$, and its fundamental-group isomorphism is the isomorphism on $\pi_2(X, x)$. The universal cover of the identity component of $\Omega_x X$ is $\Omega(X\langle 2\rangle)$: loop the fibre sequence $X\langle 2\rangle \to X \to \tau_{\leq 2}X$, then take the identity component. Other components are identified by translations, and $\Omega_x f$ commutes with these translations coherently. Finally, on simply connected spaces, a specified map is a k-homology equivalence if and only if its based loop map is a k-homology equivalence, as proved by the connected bar–cobar comparison in Part C. Apply this to the 2-connected covers to obtain (E.8.1). □

E.9. Degree-three torsion survives the second rational stage. [PROVED]

For every prime p, the projection $K(\mathbb{Z}/p, 3) \to *$ belongs to the specific $W_{\mathbb{Q}}^{(2)}$ of E.4. More generally this holds for every nonempty, 2-connected, rationally acyclic space in place of $K(\mathbb{Z}/p, 3)$. In particular

$$W_\infty \subsetneq W_{\mathbb{Q}}^{(2)} \subsetneq W_{\mathbb{Q}}^{(1)}.$$

Proof. Let $\mathcal{A} \in \mathcal{P}_{\mathbb{Q}}$. For $i \geq 3$ the homotopy groups of $\mathcal{P}_{\mathbb{Q}}^{\simeq}$ at $\mathcal{A}$ are

$$\pi_i(\mathcal{P}_{\mathbb{Q}}^{\simeq}, \mathcal{A}) \cong \pi_{i-2} \operatorname{End}_{\operatorname{Fun}_{\mathbb{Q}}^{L}(\mathcal{A},\mathcal{A})}(\mathrm{id}_{\mathcal{A}}), \tag{E.9.1}$$

where End denotes the mapping spectrum. Indeed the loop space at $\mathcal{A}$ is the space of its autoequivalences; the loop space there at the identity is the space of invertible natural endomorphisms of that identity. In positive homotopy degrees this is the same as the underlying space of its endomorphism spectrum. The functor category is $\mathbb{Q}$-linear and stable, so the groups in (E.9.1) are $\mathbb{Q}$-vector spaces.

The endpoint map

$$\operatorname{Fun}(\Delta^1, \mathcal{P}_{\mathbb{Q}})^{\simeq} \longrightarrow (\mathcal{P}_{\mathbb{Q}}^{\simeq})^2$$

has homotopy fibre $\operatorname{Fun}_{\mathbb{Q}}^{L}(\mathcal{A}, \mathcal{B})^{\simeq}$ at $(\mathcal{A}, \mathcal{B})$. Its homotopy groups in degrees at least two are homotopy groups of rational mapping spectra, hence are $\mathbb{Q}$-vector spaces. The long exact homotopy sequence shows that the total space has $\mathbb{Q}$-vector-space homotopy groups in every degree at least three as well: kernels and cokernels of maps between uniquely divisible abelian groups are uniquely divisible, and an extension of two such groups is uniquely divisible.

We use the following elementary Postnikov observation. If T is 2-connected and rationally acyclic, and K is any space with $\pi_i(K, k)$ a $\mathbb{Q}$-vector space for $i \geq 3$, then evaluation $\operatorname{Map}(T, K) \to K$ is an equivalence. Its fibre at k is $\operatorname{Map}_*(T, K)$, which agrees with the pointed mapping space into the 2-connected cover of (K, k), since $\operatorname{Map}_*(T, \tau_{\leq 2}K)$ is contractible. For a $\mathbb{Q}$-vector space V all reduced cohomology groups of T with coefficients in V vanish. Hence $\operatorname{Map}_*(T, K(V, n))$ is contractible for every n. Induction over the Postnikov fibrations proves the same for every finite Postnikov section of the 2-connected cover; taking their inverse limit proves the observation. This argument uses Postnikov completeness of spaces, and does not replace a limit by a colimit or assume finite type. If K is empty both sides are empty; otherwise the argument applies independently to all components.

Apply the observation to the two spaces K_0, K_1 in E.1 for $\mathcal{C} = \mathcal{P}_{\mathbb{Q}}$. It proves that the constant functor $\mathcal{P}_{\mathbb{Q}} \to \operatorname{Fun}(T, \mathcal{P}_{\mathbb{Q}})$ is an equivalence. Finally $K(\mathbb{Z}/p, 3)$ is 2-connected and rationally acyclic. Transfer gives rational acyclicity of $K(\mathbb{Z}/p, 1)$, and two applications of the

path-fibration Serre spectral sequence give it for degrees two and three. Its nonzero π_3 shows that its projection is not a weak equivalence. Strictness of the other inclusion is E.4. □

E.10. A rational loop tower and its ordinary coefficient realization. [PROVED]

Define the following specified tower of classes of maps of spaces:

$$V_0 = W_{\mathbf{D}_{\mathbb{Q}}}, \qquad V_{r+1} = \{f : X \to Y \mid \pi_0(f) \text{ is bijective and } \Omega_x f \in V_r \text{ for every component}\}. \tag{E.10.1}$$

For $r \geq 1$ its exact description is

$$f \in V_r \iff \begin{cases} \pi_0(f) \text{ is bijective,} \\ \pi_i(f) \text{ is an isomorphism } (1 \leq i \leq r), \\ \pi_j(f) \otimes \mathbb{Q} \text{ is an isomorphism } (j > r). \end{cases} \tag{E.10.2}$$

The rationalization condition concerns the higher abelian homotopy groups and their specified maps. We have $V_1 = W^{(1)}_{\mathbb{Q}}$, $V_2 = W^{(2)}_{\mathbb{Q}}$ for the actual coefficient categories constructed above, strict inclusions $V_{r+1} \subsetneq V_r$ for every $r \geq 0$, and $\bigcap_{r\geq 0} V_r = W_\infty$. Every V_r is an accessible fundamental localizer. For $r \geq 1$ let $\mathcal{C}_r = \mathcal{S}_{V_r}$ be its category of local spaces, and write $\mathbf{D}_r = \mathbf{D}_{\mathcal{C}_r}$, with $\mathbf{D}_0 = \mathbf{D}_{\mathbb{Q}}$. These are a specified tower of ordinary derivators satisfying

$$W_{\mathbf{D}_r} = V_r, \qquad W^{\mathrm{lc}}_{\mathbf{D}_r} = V_{r+1} \quad (r \geq 0). \tag{E.10.3}$$

One may instead use $\mathbf{D}_1 = \mathbf{D}'_{\mathbb{Q}}$ from E.2 and retain $\mathbf{D}_r = \mathbf{D}_{\mathcal{C}_r}$ for $r \geq 2$; the same equalities hold. Thus the tower, including its intersection, has a construction by ordinary coefficient categories. The stages $\mathcal{C}_r$ are localized spaces; they are not identified here with categories of higher $\mathbb{Q}$-linear categories. More generally, if an initial derivator $\mathbb{E}$ has accessible $W^{\mathrm{lc}}_{\mathbb{E}} \subseteq W_0$, the same repeated localized-space realization constructs an ordinary coefficient tower whose intersection is W_∞. In particular this applies to every nonterminal $\mathbb{U}$-presentable infinity-category coefficient.

Proof. For a group-like loop space, rational homology equivalence detects its component group by H_0, and translation identifies its components. The loop-chain criterion in Part C consequently identifies V_1 with $W^{\mathrm{lc}}_{\mathbf{D}_{\mathbb{Q}}}$. Rational Whitehead on the simply connected covers gives (E.10.2) at $r = 1$. E.8 identifies the next loop step with the actual second categorical stage. In general $\pi_i(\Omega_x X) = \pi_{i+1}(X, x)$ for $i \geq 1$, while $\pi_0(\Omega_x X) = \pi_1(X, x)$ as a group. Apply the inductive criterion to all components of the loop map; translations make this equivalent to the criterion at its identity component. This proves (E.10.2) for $r + 1$ and proves all the inclusions. The inclusion $V_1 \subset V_0$ also follows from the constant-versus-locally-constant inclusion.

For any prime p, the projection $K(\mathbb{Z}/p, r + 1) \to *$ is in V_r but not V_{r+1}. For $r = 0$ use finite-group transfer; for $r \geq 1$ use (E.10.2), since its only nonzero homotopy group is the torsion group in degree $r + 1$. Thus every inclusion is strict. A map in every V_r is bijective on components and is an isomorphism on each homotopy group by taking r large enough for that degree. It is therefore a weak equivalence. The reverse inclusion is immediate from (E.10.1).

We now construct (E.10.3). We first record the coefficient calculation which makes the induction possible. Let $L : \mathcal{S} \to \mathcal{C} \subseteq \mathcal{S}$ be an accessible reflective localization whose inverted maps W are product stable, and suppose $W \subseteq W_0$. The point is always local, and the empty space is local because the localization unit belongs to W and therefore preserves its empty set of components. These two local spaces are distinct. For any connected pointed space (X, x), evaluation

$$U_x : \mathrm{Fun}(X, \mathcal{C}) \longrightarrow \mathcal{C}$$

has a left adjoint, is conservative, and preserves geometric realizations. Its monad is

$$M \longmapsto L(\Omega_x X \times M). \tag{E.10.4}$$

Indeed the free local system has value the copower by the path space, and colimits of local spaces are obtained by applying L to the corresponding colimits of spaces. Barr–Beck applies as in E.8. For a pointed map $f : X \to Y$ between connected spaces, the canonical comparison of the monads (E.10.4) is an equivalence if and only if $\Omega_x f \in W$. Necessity follows by evaluating the comparison at $M = *$. For sufficiency, product stability makes $\Omega_x f \times M$ a member of W for every local M, so the comparison is an equivalence on every object, compatibly with its monad structure. Monadicity identifies its restriction functor with the specified f^*, exactly as in E.8.

For arbitrary spaces, an equivalence of local-system categories forces a component bijection. If a target component is missed, restriction cannot detect the difference between the empty and point-valued constant local systems on that component. If two source components map into one target component, the local system which has empty value on one and point value on the other cannot be a restriction: the fibers of a local system on a connected space are equivalent. Once components correspond, apply the connected calculation componentwise. This proves

$$W^{\mathrm{lc}}_{\mathbf{D}_{\mathcal{C}}} = \{f : \pi_0(f) \text{ is bijective and } \Omega_x f \in W \text{ on every component}\}. \tag{E.10.5}$$

E.1 identifies the enhanced equivalence used in this argument with the ordinary homotopy-category definition.

The class $V_1 = W^{\mathrm{lc}}_{\mathbf{D}_{\mathbb{Q}}}$ is an accessible fundamental localizer by B.3–B.4. Apply E.7 to form $\mathcal{C}_1$. It is presentable, its constant class is V_1, and (E.10.5) identifies its locally constant class with V_2. B.3–B.4 now make V_2 accessible and fundamental, so E.7 forms $\mathcal{C}_2$. Inductively, the same argument constructs every $\mathcal{C}_r$ and proves (E.10.3). The base V_0 is the constant-coefficient class of $\mathbf{D}_{\mathbb{Q}}$ and is accessible by B.4. The component condition needed in (E.10.5) follows at every stage from (E.10.2). Finally E.2 and E.8 say respectively $W_{\mathbf{D}'_{\mathbb{Q}}} = V_1$ and $W^{\mathrm{lc}}_{\mathbf{D}'_{\mathbb{Q}}} = V_2$, which proves the assertion about the alternative first stage.

For the final assertion start with $U_1 = W^{\mathrm{lc}}_{\mathbb{E}}$ and set $U_{r+1} = \{f : \pi_0(f) \text{ bijective, } \Omega_x f \in U_r\}$ for $r \geq 1$. The same induction using (E.10.5) constructs the coefficient categories and preserves accessibility. Since $U_1 \subseteq W_0$, induction on r gives $U_r \subseteq W_{r-1}$, where W_{r-1} is equivalence on $(r-1)$-types. Hence a map in every U_r is an isomorphism on every homotopy group and on components. Conversely every weak homotopy equivalence belongs to every stage by A.2, so the intersection is W_∞. B.4 supplies accessibility for a presentable initial coefficient, and G.1 supplies component detection when that coefficient is nonterminal. □

E.11. Arbitrary bare-derivator functoriality and higher linear coefficients. [Open]

For an arbitrary derivator on $\mathsf{Cat}_{\mathbb{U}}$ we have not constructed a functorial assignment $\mathbf{D} \mapsto \mathbf{D}'$ with $W_{\mathbf{D}'} = W^{\mathrm{lc}}_{\mathbf{D}}$. A phrase such as "$\mathbf{D}$-linear categories" does not specify such an assignment without a monoidal enhancement and a choice of morphisms. E.2 settles a specified enhanced linear case, and B.4 together with E.7 gives a realizing derivator of localized spaces for every presentable infinity-category coefficient. Neither proves a functorial assignment on all bare derivators and all their morphisms. The full second linear stage is computed in E.8, and E.10 constructs an entire ordinary coefficient tower with its stated intersection. A realization of its further stages by higher $\mathbb{Q}$-linear categories, with a specified higher monodromy construction and morphism class, remains unresolved here.

Remaining mathematical task. The unenhanced problem asks for a construction on derivators and their morphisms, not merely for a representative of an equivalence class of localizers. The

failed step in extending E.2 is that neither the symmetric monoidal category of modules nor its category of linear categories is part of the data of an arbitrary derivator. The next linear-categorical task is to construct an operation on $\mathbb{Q}$-linear categories whose third weak-equivalence class can be compared with maps inducing an equivalence on 3-types and rational homology equivalences of 3-connected covers. The ordinary category $\mathcal{P}_{\mathbb{Q}}$ is not a presentable coefficient category in its original universe, so repeating the notation "modules in presentable categories" is not by itself a construction of that higher linear categorical level. E.10's localized-space coefficients already give an ordinary realization of the same third class and of every subsequent loop class. The higher monodromy equivalences of Pascaleff et al. (2025, Theorem 3.2.24) provide a precise framework, but their general higher-Hochschild statement is conditional on their Conjecture 3.3.1. We do not import that conditional statement as a theorem or assert that arbitrary iteration has been defined. □

6 Reconstruction

F.1. All-set fibre functors recover the fundamental groupoid. [Proved]

Let X be a $\mathbb{U}$-small space, let $G = \Pi_1 X$, and retain the ordinary category $\mathcal{L}_0(X) = \mathrm{Fun}(G, \mathsf{Set}_{\mathbb{U}})$ together with its evaluation functors $\omega_x : \mathcal{L}_0(X) \to \mathsf{Set}_{\mathbb{U}}$ for a set of points meeting every component. The category whose objects are these functors and whose arrows are all natural transformations between them is a groupoid equivalent to G. With every point retained, the comparison is the specified fully faithful functor

$$x \longmapsto \omega_x, \qquad \mathrm{Hom}_G(x, y) \longrightarrow \mathrm{Nat}(\omega_x, \omega_y). \tag{F.1.1}$$

Restricting instead to finite sets recovers the profinite Galois invariant; it must not be identified with the full groupoid in (F.1.1).

Proof. The covariant representable $h_x = \mathrm{Hom}_G(x, -)$ belongs to $\mathcal{L}_0(X)$ and represents ω_x: the ordinary Yoneda isomorphism is $\mathrm{Hom}(h_x, L) \cong L(x)$. Applying Yoneda once more gives

$$\mathrm{Nat}(\omega_x, \omega_y) \cong \mathrm{Hom}_{\mathcal{L}_0(X)}(h_y, h_x) \cong \mathrm{Hom}_G(x, y).$$

Under these isomorphisms (F.1.1) sends an arrow to transport along that arrow; composition is preserved. Every such transformation is invertible because every arrow of G is invertible. Points in different components have no arrows between their evaluations, exactly as the displayed formula states. A subset of points meeting every component gives an equivalent full subgroupoid. If X is empty the family is empty and the reconstructed groupoid is empty, even though $\mathcal{L}_0(X)$ is terminal.

The identification of $\mathcal{L}_0(X)$ as the set-valued local system category and the corresponding reflector is the checked CATHOT C.1–C.4. The above argument reconstructs that groupoid from its fibres; it does not reprove the CATHOT reflection theorem. In contrast, SGA 1, Exposé V, Theorem 4.1 explicitly takes a fibre functor with values in *finite* sets and obtains a topological automorphism group represented by a pro-object (Grothendieck, 2003). For a connected discrete group G this is its profinite completion, since finite continuous actions are precisely the finite actions factoring through finite quotients. Thus the finite and all-set statements have different quantifiers and different conclusions. □

F.2. Enhanced linear reconstruction with a specified fibre. [Proved]

Let R be an E_∞-ring spectrum and (X, x) a pointed connected $\mathbb{U}$-small space. Retain the R-linear stable ∞-category

$$\mathcal{L}_R(X) = \mathrm{Fun}(X, \mathrm{Mod}_R)$$

and its R-linear evaluation functor $\omega_x : \mathcal{L}_R(X) \to \mathrm{Mod}_R$, including its coherent functor structure. This pair determines the E_1-algebra $R[\Omega_x X]$ up to equivalence of E_1-algebras. For an ordinary commutative ring k and $R = Hk$, it determines $C_*(\Omega_x X; k)$ as an associative derived k-algebra, with the prescribed variance convention. Retaining fibres in every component gives the corresponding many-object linear path category.

Proof. The functor ω_x has a left adjoint F_x, unique with its adjunction data up to contractible choice. Left Kan extension gives the explicit formula

$$F_x(M)(y) = R[\mathrm{Map}_X(x, y)] \otimes_R M. \tag{F.2.1}$$

Here $\mathrm{Map}_X(x, y)$ is the space of paths from x to y, and $R[T] = R \otimes \Sigma^\infty_+ T$. Thus the monad $T_x = \omega_x F_x$ is tensoring with $R[\Omega_x X]$. Its unit comes from the constant path and its multiplication from concatenation, with the order fixed by the unit and counit of this adjunction. Evaluating the R-linear monad at the free rank-one module recovers

$$T_x(R), \qquad R \longrightarrow T_x(R), \qquad T_x(R) \otimes_R T_x(R) \longrightarrow T_x(R), \tag{F.2.2}$$

together with their coherent associativity and unit data. The linear functor theorem (Lurie, 2017, Theorem 4.8.4.1) identifies R-linear colimit-preserving endofunctors of Mod_R with tensoring by R-modules, so (F.2.2) is a reconstruction of the algebra itself. It does not reconstruct merely a Morita class.

The equivalence $\mathcal{L}_R(X) \simeq \mathrm{Mod}_{R[\Omega_x X]}$ over Mod_R is precisely the monadic equivalence proved in E.2: evaluation is conservative on a connected groupoid and preserves geometric realizations. Its proof therefore includes the hypotheses of the Barr–Beck theorem, rather than inferring the assertion from a comparison of homotopy categories. For $R = Hk$, the symmetric monoidal chain construction identifies $Hk \otimes \Sigma^\infty_+ \Omega_x X$ with the chain algebra of the based loop space. In the presheaf convention the opposite action is transported by path inversion; with right modules one instead obtains the opposite algebra. Either convention is valid, but the comparison map is fixed after the convention is chosen.

With more than one point, enriched adjunction gives the mapping R-module

$$\underline{\mathrm{Hom}}_{\mathcal{L}_R(X)}(F_y(R), F_x(R)) \simeq R[\mathrm{Map}_X(x, y)]. \tag{F.2.3}$$

Composition on the left, with this reversal of the representables, is path composition on the right. Formula (F.2.3), component by component, reconstructs the R-linearized path category. Between different components the path space is empty and its linearization is zero. No basepoint is chosen in an empty component. □

F.3. Spaces are recovered from enhanced space-valued local systems. [Proved]

For every $\mathbb{U}$-small space X, the ∞-topos $\mathcal{L}_{\mathcal{S}}(X) = \mathrm{Fun}(X, \mathcal{S}_{\mathbb{U}})$ has shape the constant pro-space X. Retaining its evaluation functors ω_x and their full spaces of natural transformations recovers X as an ∞-groupoid. Retaining only the homotopy category still suffices if, in addition, the specified constant functor $\mathrm{Ho}\,\mathcal{S} \to \mathrm{Ho}\,\mathcal{L}_{\mathcal{S}}(X)$ is retained: in that case one recovers the object X of Hot by Yoneda.

Proof. Straightening identifies $\mathrm{Fun}(X, \mathcal{S})$ with the slice ∞-topos $\mathcal{S}_{/X}$. The inverse-image part $\Delta_X : \mathcal{S} \to \mathcal{L}_{\mathcal{S}}(X)$ of its geometric morphism to spaces is the constant functor. Its global sections satisfy the canonical equivalence

$$\mathrm{Map}_{\mathcal{L}_{\mathcal{S}}(X)}(1, \Delta_X K) \simeq \mathrm{Map}_{\mathcal{S}}(X, K). \tag{F.3.1}$$

By definition the left side, as a functor of K, is corepresented by the shape. Equation (F.3.1) identifies it with the constant pro-object represented by X. This is also the classifying-topos case of the shape construction in Hoyois (2018, Section 2); it is a known reconstruction, not a priority claim.

For the fibre-functor formulation, the covariant representable $r_x(y) = \mathrm{Map}_X(x, y)$ corepresents ω_x. The enriched Yoneda equivalence yields

$$\mathrm{Map}_{\mathrm{Fun}(\mathcal{L}_{\mathcal{S}}(X),\mathcal{S})}(\omega_x, \omega_y) \simeq \mathrm{Map}_{\mathcal{L}_{\mathcal{S}}(X)}(r_y, r_x) \simeq \mathrm{Map}_X(x, y). \tag{F.3.2}$$

These equivalences respect composition coherently. They identify X with the full ∞-subcategory on its specified evaluation functors; every morphism there is an equivalence. If only a family meeting every component is retained, the resulting full subgroupoid is equivalent to X.

Finally taking components of (F.3.1) gives $\mathrm{Hom}(1, \Delta_X K) = [X, K]$, naturally in $K \in \mathsf{Hot}$. The ordinary Yoneda lemma recovers the representing object X in Hot from these hom-sets. The extra datum of the constant functor is essential to this particular proof: it supplies the specified functor whose representability is being used. None of these arguments identifies an arbitrary equivalence of bare homotopy categories with a geometric equivalence. For $X = \varnothing$ the topos is degenerate, (F.3.1) is the contractible space $\mathrm{Map}(\varnothing, K)$, and the same representability assertion recovers the empty space correctly. □

F.4. The form applicable to a locally connected topos. [KNOWN]

Let $\mathcal{E}$ be a locally connected Grothendieck topos, and let $\Pi_1\mathcal{E}$ be its fundamental pro-groupoid, equivalently the 1-truncation of the shape of its associated ∞-topos. There is a canonical equivalence

$$\mathrm{Fun}(\Pi_1\mathcal{E}, \mathsf{Set}) \simeq \mathrm{LC}_{\mathsf{Set}}(\mathcal{E}). \tag{F.4.1}$$

For a representing cofiltered diagram (G_i), the left side means the filtered colimit of the categories $\mathrm{Fun}(G_i, \mathsf{Set})$ in the sense of representations of a pro-groupoid; it is not the category of actions of the abstract inverse-limit group with its topology forgotten. The full subcategory of filtered colimits of locally constant objects is the corresponding classifying topos. In particular, if $\mathcal{E}$ is generated under colimits by its locally constant objects, the canonical map $\mathcal{E} \to \beta(\Pi_1\mathcal{E})$ is an equivalence.

Source, hypotheses, and specialization. Equation (F.4.1) is Hoyois (2018, Theorem 3.13, case $n = 0$), with 0-truncated coefficients. The proof uses local connectedness to make the constant-object functor preserve products on sets, then descends the resulting mapping-space identities along torsors. The description of the filtered-colimit completion is part of that same theorem. The final assertion is Hoyois (2018, Corollary 3.20, case $n = 1$), which permits several components and does not assume a point. For an ∞-topos locally n-connected, the same theorem identifies locally constant n-truncated objects with $\mathrm{Fun}(\Pi_\infty\mathcal{E}, \mathcal{S}_{\leq n})$. Without any connectedness hypothesis, its finite version is Hoyois (2018, Theorem 4.3), with finite homotopy types as coefficients.

This also specifies the reconstruction data needed by the multi-Galois application. Retain the category of locally constant objects with finite limits, and the families which are jointly effective epimorphic in $\mathcal{E}$. For each small groupoid G these data determine its principal G-torsors: they consist of objects indexed by the objects of G, transport isomorphisms, the principal-action identities, and the joint covering condition. The resulting groupoid-valued functor of G is the torsor functor corepresented by $\Pi_1\mathcal{E}$. Yoneda in pro-groupoids determines this pro-object up to equivalence. If a family of points is jointly conservative on $\mathcal{E}$, their fibre functors identify the covering families as those which are jointly surjective on every fibre. Without this conservativity assumption one must retain the covering condition itself; a statement about unspecified point functors would have omitted data.

The attached *Longue Marche*, Sections 1–4, uses the stronger locally simply connected setting when asserting that locally constant objects already form a multigaloisian topos (Grothendieck, 1981, Section 2, Theorem 2.1 and Definition 2.2). For merely locally connected $\mathcal{E}$, the distinction in (F.4.1) between split representations and their filtered-colimit completion is retained. We do not replace it by an unqualified assertion that $\mathrm{LC}(\mathcal{E})$ is always a topos. □

F.5. Kaledin's groupoid enhancement already reconstructs spaces. [Known]

For a small space X consider the family

$$J \longmapsto \Pi_1 \operatorname{Map}(BJ, X). \tag{F.5.1}$$

Kaledin's enhanced-groupoid representability theorem identifies the appropriate small families satisfying his axioms with homotopy types. More precisely, in the attached arXiv:2409.17489v3, Definition 6.1.1.3 and Corollary 6.1.2.5 give the groupoid case; Theorem 7.3.1.3 is the more general representability theorem for enhanced categories. Thus the reconstruction of a space from (F.5.1), with all the structure required there, is already recorded in the literature.

Exact comparison with the stated theorem. Kaledin's Pos is the category of partially ordered sets of finite chain dimension, not the category of finite posets. Its objects may have arbitrary permitted cardinality. His Pos^+ consists of left-bounded partially ordered sets. A small enhanced groupoid is a family of groupoids over Pos satisfying additivity, semiexactness, and excision as specified in Definition 6.1.1.1. A semicanonical extension to Pos^+ also has the stated interval-invariance and semicontinuity properties. The small or κ-bounded conditions, for an uncountable regular κ, are part of the claim and cannot be omitted.

The comparison of these families with families on simplicial sets is Proposition 6.1.2.3. Corollary 6.1.2.5 identifies their category with the localization of simplicial sets at weak equivalences, and supplies uniqueness of extension with the precise equivalence and isomorphism ambiguities stated there. In the representable case its nerve pullback is exactly (F.5.1), with derived mapping spaces. Theorem 7.3.1.3 proves that every restricted enhanced category is equivalent over Pos to one obtained from a complete fibrant Segal space; it also represents enhanced functors by maps and identifies homotopic maps with isomorphic enhanced functors (Kaledin, 2026).

These are statements of the opened preprint. Its arXiv record specifies that Chapters 6 and 7 remain preliminary, whereas Chapters 1–5 are intended for publication. We therefore do not describe the enhanced-groupoid theorem as a checked consequence of a refereed 2026 journal article. The space reconstruction F.3 was proved independently above and does not depend on the status of that preprint. □

F.6. The relation between the coefficient tower and varying shapes. [Proved]

The finite-type tower in E.5 and the family (F.5.1) both determine X up to equivalence, but their parameters and their truncations are different. The tower raises the coefficient category at fixed X. Kaledin fixes the target of the reported values to ordinary groupoids and varies the source shape J, retaining all coherence between those values. In general

$$\Pi_1 \operatorname{Map}(BJ, X) \not\simeq \operatorname{Map}(BJ, \Pi_1 X), \tag{F.6.1}$$

even on components. The explicit comparison from enhanced space-valued local systems is obtained by first forming the full ∞-groupoid of designated fibre functors in F.3 and then taking $\Pi_1 \operatorname{Fun}(BJ, -)$ of that groupoid.

Proof. For the coefficient tower, E.5 says that a specified map inducing equivalences at all stages induces isomorphisms on all homotopy groups. For varying shapes, F.5 gives the known representability statement. Formula (F.3.2) identifies the full fibre-functor ∞-groupoid with X, compatibly with composition, so the last construction gives (F.5.1) naturally in J.

For (F.6.1), take $X = K(\mathbb{Z}, 2)$ and take J to be the face poset of a finite triangulation of S^2. Then $BJ \simeq S^2$ and

$$\pi_0\Pi_1 \operatorname{Map}(BJ, X) = [S^2, K(\mathbb{Z}, 2)] \cong \mathbb{Z}, \qquad \operatorname{Map}(BJ, \Pi_1 X) \simeq *.$$

The left side contains degree-two information even though each reported value is an ordinary groupoid. Thus one cannot identify Kaledin's diagram with a diagram built only from the unenhanced groupoid $\Pi_1 X$. This example involves an infinite homotopy type as target; the finite poset itself is the nonempty proper subsets of a four-element set ordered by inclusion, whose nerve is the barycentric subdivision of the boundary of a tetrahedron. Its exact face counts and homology are verified in `code/ef_checks.py` and its output. □

F.7. A precise recognition question for a coefficient theory. [Proved]

For a complete coefficient ∞-category $\mathcal{C}$, the condition $W_{\mathcal{C}}^{\mathrm{lc}} = W_\infty$ means that the contravariant functor $X \mapsto \operatorname{Fun}(X, \mathcal{C})$ reflects equivalences of *specified maps* of spaces. Equivalently, the pair of test spaces $K_0(\mathcal{C}), K_1(\mathcal{C})$ of E.1 detects all weak equivalences. In the cases computed in Part C, this holds for spaces, integral derived coefficients, and spectra; it fails for sets, groupoids, every fixed finite truncation level, rational derived coefficients, and mod-p derived coefficients.

Proof. The equivalence of the two formulations is E.1. The positive and negative computations are those of Part C; F.1–F.3 describe their reconstructed invariants with fibres retained. For finite truncations the projection of a nontrivial $K(A, r+2)$ to a point passes the $\mathcal{S}_{\le r}$ local system test but is not a weak equivalence. For rational coefficients use $K(\mathbb{Z}/p, 2) \to *$. For mod-p coefficients use $K(\mathbb{Z}/q, 2) \to *$ with a different prime q; transfer and the path fibration give the needed mod-p acyclicity. All spaces in these examples are connected, so no ambiguity about component detection is involved. For a general ring or an E-local stable coefficient theory, the exact loop-algebra criteria and the qualifications in Part C remain in force. □

F.8. The reconstruction problem with only ordinary fibre functors. [Open]

If the input in Part F is restricted literally to the ordinary category $\mathrm{LC}_{\mathbf{D}_R}(X)$ and ordinary functors $\mathrm{LC}_{\mathbf{D}_R}(X) \to \mathbf{D}_R(e)$, with no enhanced mapping objects, no coherent adjunction data, and no specified family of shifted categories, F.2 does not establish reconstruction of the E_1-algebra $C_*(\Omega X; k)$. We leave that unenhanced assertion unresolved. Likewise F.3 without either its ∞-topos structure or its specified constant functor does not establish the corresponding bare-category assertion.

The precise missing comparison. The proof of F.2 constructs an algebra from a coherent monad. The ordinary adjunction on homotopy categories supplies a monad in ordinary categories, but no comparison reconstructing all of the higher multiplication coherences has been supplied. An invalid passage from the latter monad to an E_1-algebra would be a gap, not a counterexample. A smallest useful test is the class of pointed simply connected finite CW complexes with rational coefficients: does an equivalence of their ordinary local-system categories, compatible with the specified ordinary evaluation functors, imply equivalence of the rational loop-chain E_1-algebras? Neither a proof nor a counterexample to this exact implication is given here. The enhanced assertion and the constant-functor space assertion have been proved with their extra data explicitly named. □

7 Chapter X revisited

7.1 Component detection and the meaning of "useful"

G.1. Locally constant coefficients detect components for every nontrivial derivator. [Proved]

Let $\mathbf{D}$ be a derivator on all $\mathbb{U}$-small categories. If $\mathbf{D}(e)$ is not equivalent to the terminal category, then

$$\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}} \subseteq \mathcal{W}_0. \tag{G.1.1}$$

If $\mathbf{D}(e) \simeq e$, then $\mathbf{D}(A) \simeq e$ for every A, and all five classes consist of every functor. Thus the hypothesis is exact.

Proof. There are two nonisomorphic objects $M, N \in \mathbf{D}(e)$: a category with an initial object and only one isomorphism class of objects is equivalent to e. The initial object exists by Der 1 and Der 3. Suppose first that Bf misses a component of BB. Use Der 1 to choose two coefficients on B constant with value M on all other components and respectively M, N on the missed component. They are locally constant, and their pullbacks are isomorphic. An equivalence f^* would make them isomorphic on B, contradicting evaluation at a point of the missed component.

If two distinct components of A map into the same component of B, choose a locally constant coefficient on A taking the values M, N on those components, and M elsewhere. It cannot be a restriction: any locally constant object on the connected component of B has isomorphic values at all its objects, along a finite zigzag of arrows. This contradicts essential surjectivity. These arguments also exclude $\varnothing \to B$ for nonempty B and handle arbitrarily many components without a finiteness assumption.

If $\mathbf{D}(e) \simeq e$, all arrows of $\mathbf{D}(A)$ are isomorphisms by Der 2. The category $\mathbf{D}(A)$ has an initial object, from the left Kan extension along $\varnothing \to A$; a groupoid with an initial object is equivalent to e. Every restriction is then an equivalence and every comparison is invertible. □

G.2. The exact obstruction for constant coefficients. [Proved]

For a derivator, $\mathcal{W}_{\mathbf{D}} \subseteq \mathcal{W}_0$ holds if and only if $\mathbf{D}(e)$ is not a preorder up to equivalence. More explicitly, if there exist two distinct parallel morphisms $\alpha, \beta : M \to N$ in $\mathbf{D}(e)$, then every constant-coefficient equivalence is bijective on components. If $\mathbf{D}(e)$ is a nontrivial preorder up to equivalence, then

$$\mathcal{W}_{\mathbf{D}} = \{f : A \to B : \ A \text{ and } B \text{ are both empty or both nonempty}\}. \tag{G.2.1}$$

In particular, Chapter X's component-detection expectation needs a coefficient-separation hypothesis for $\mathcal{W}_{\mathbf{D}}$, but only nontriviality for $\mathcal{W}_{\mathbf{D}}^{\mathrm{lc}}$.

Proof. Suppose $\alpha \neq \beta$. If a target component is missed, define two morphisms $q^*M \to q^*N$ using α everywhere and changing it to β just on that component. Der 1 supplies these maps; evaluation distinguishes them, while restriction identifies them. If two source components coalesce, define a morphism $p^*M \to p^*N$ using different choices on them. Every morphism between constants over B induces, under the underlying diagram functor, a natural transformation between constant presheaves. Its component morphisms are equal along every arrow of B, hence equal throughout each connected component. The chosen source map is therefore not in the image of (A.1.1). Full bijectivity excludes both failures.

Now assume $\mathbf{D}(e)$ is thin. For nonempty A, choose $a : e \to A$. The adjunction unit $N \to p_*p^*N$ and evaluation at a give maps in both directions between N and p_*p^*N. In a thin category these are mutually inverse. Thus

$$\mathrm{Hom}_{\mathbf{D}(A)}(p^*M, p^*N) \cong \mathrm{Hom}_{\mathbf{D}(e)}(M, p_*p^*N) \cong \mathrm{Hom}_{\mathbf{D}(e)}(M, N).$$

Under these identifications every restriction between nonempty shapes is the identity. On the empty shape every Hom set is a singleton. In a nontrivial complete preorder the morphism from terminal to initial is absent; that pair distinguishes $\varnothing \to B$ for nonempty B. This proves (G.2.1). The represented complete lattice $0 < 1$ is an explicit example: constants see only emptiness, while its locally constant coefficient category is $(0 < 1)^{\pi_0 A}$ and detects $\mathcal{W}_0$ under equivalence. □

7.2 Functoriality in the coefficient theory

G.3. What an arbitrary morphism of derivators supplies. [Proved]

A pseudonatural morphism $\Phi : \mathbf{D} \to \mathbf{E}$ restricts to functors $\Phi_A : \mathrm{LC}_{\mathbf{D}}(A) \to \mathrm{LC}_{\mathbf{E}}(A)$, coherently commuting with restriction. This alone imposes no inclusion in either direction between $\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$ and $\mathcal{W}^{\mathrm{lc}}_{\mathbf{E}}$. Equivalence of derivators gives equality of the five classes. Products give intersections of the five classes.

Proof. Pseudonaturality at objects and arrows identifies the underlying presheaf of $\Phi_A F$ with the image under Φ_e of that of F. Functors preserve isomorphisms, proving the first assertion. There are morphisms in both directions between any derivator and the terminal derivator: the unique functor in one direction and the terminal-object section in the other. Restrictions preserve terminal objects because they have left adjoints. The terminal derivator has every functor in its class, while the derivator of sets has $\mathcal{W}^{\mathrm{lc}} = \mathcal{W}_1$. Thus no monotonicity direction is forced by the existence of an arbitrary morphism. An equivalence of derivators identifies each of the specified comparisons. For a product derivator, restrictions and all adjunction units and counits are products; their invertibility and full faithfulness are componentwise. Essential surjectivity is componentwise as well. □

G.4. Retracts and fully faithful comparisons. [Proved]

If $\mathbf{E}$ is a pseudoretract of $\mathbf{D}$, namely if there are morphisms $\Phi : \mathbf{D} \to \mathbf{E}$ and $\Psi : \mathbf{E} \to \mathbf{D}$ with $\Phi\Psi \cong \mathrm{id}$, then

$$\mathcal{W}^{\mathrm{lc}}_{\mathbf{D}} \subseteq \mathcal{W}^{\mathrm{lc}}_{\mathbf{E}}, \qquad \mathcal{W}^{1}_{\mathbf{D}} \subseteq \mathcal{W}^{1}_{\mathbf{E}}, \qquad \mathcal{W}_{\mathbf{D}} \subseteq \mathcal{W}_{\mathbf{E}}. \tag{G.4.1}$$

If instead every Φ_A is fully faithful, then $\mathcal{W}^{1}_{\mathbf{E}} \subseteq \mathcal{W}^{1}_{\mathbf{D}}$ and $\mathcal{W}_{\mathbf{E}} \subseteq \mathcal{W}_{\mathbf{D}}$. To infer the analogous assertion for 2-completeness one additionally needs descent of the essential image: whenever $F \in \mathrm{LC}_{\mathbf{D}}(A)$ and $G \in \mathrm{LC}_{\mathbf{E}}(B)$ satisfy $f^*G \cong \Phi_A F$, an object G with this property must be choosable in the essential image of Φ_B.

Proof. The restriction functor for $\mathbf{E}$ is a retract, up to the specified coherent isomorphism, of that for $\mathbf{D}$. Equivalences and fully faithful functors are closed under these retracts by the Hom-set and essential-surjectivity argument of A.4. Constants are preserved by both morphisms, and (A.1.1) for $\mathbf{E}$ is a retract of the corresponding map for $\mathbf{D}$ with coefficients $\Psi_e M, \Psi_e N$. This proves (G.4.1). For a fully faithful Φ, the two vertical maps in every Hom comparison square are bijections, so the asserted implications follow. Essential surjectivity has a separate quantifier over source coefficients; the last displayed condition supplies exactly the missing lift of the descended coefficient. A preimage under Φ_B is locally constant because the fully faithful Φ_e

reflects isomorphisms on all underlying transition arrows. No conservativity claim is substituted for that existence condition. □

G.5. Change of scalars and the Chapter X assignment. [PROVED]

Let $k \to l$ be a homomorphism of associative unital rings. Derived extension of scalars gives

$$\mathcal{W}^{\mathrm{lc}}_{D(k)} \subseteq \mathcal{W}^{\mathrm{lc}}_{D(l)}, \qquad \mathcal{W}_{D(k)} \subseteq \mathcal{W}_{D(l)}. \tag{G.5.1}$$

The first inclusion is equality if derived extension of scalars reflects equivalences and the canonical loop-algebra comparison can be reflected by it; in particular this holds for a faithfully flat map of nonzero commutative rings. Thus $\mathbf{D} \mapsto \mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$ is well defined on equivalence classes of derivators and has precise functoriality under additional hypotheses, but is not monotone on all morphisms of derivators.

Proof. The comparison of loop chains for l is obtained from the comparison for k by derived scalar extension; the same statement holds for ordinary singular chains. C.7 therefore proves (G.5.1) when the rings are nonzero. A zero target ring gives the trivial coefficient theory, and a unital map from the zero ring has zero target, so the exceptional cases satisfy the same assertion. For faithfully flat commutative extension, a complex is acyclic if and only if its scalar extension is acyclic. The loop criterion then reflects membership, including its component and fundamental-group conditions. The final assertions follow from G.3 and G.4. □

G.6. A functorial higher-coefficient construction on bare derivators. [OPEN]

For an arbitrary derivator $\mathbf{D}$ with no specified monoidal enhancement, this work does not construct a functorially assigned derivator $\mathbf{D}'$ with $\mathcal{W}_{\mathbf{D}'} = \mathcal{W}^{\mathrm{lc}}_{\mathbf{D}}$, nor an iteratable 2-derivator construction of that kind. The requested general statement is retained with those quantifiers. A chosen presentable linear enhancement has the positive realization in E.2; a specified accessible space localization has the conditional realization in E.7. The missing data in the general proposal are a category of $\mathbf{D}$-linear categories, its morphisms and coherent 2-morphisms, and Kan extensions on the enriched shapes required by Di Vittorio. The full relative Theorem A for arbitrary derivators is proved in B.3. That localizer theorem by itself does not specify an iteratable coefficient category or the required enriched shapes.

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